

Oracle AI for Fusion Applications

How do I use AI Agent Studio?



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Get Help

There are a number of ways to learn more about your product and interact with Oracle and other users.

Get Help in the Applications

Some application pages have help icons  to give you access to contextual help. If you don't see any help icons on your page, click your user image or name in the global header and select Show Help Icons. If the page has contextual help, help icons will appear.

Get Training

Increase your knowledge of Oracle Cloud by taking courses at [Oracle University](#).

Join Our Community

Use [Cloud Customer Connect](#) to get information from industry experts at Oracle and in the partner community. You can join forums to connect with other customers, post questions, suggest [ideas](#) for product enhancements, and watch events.

Share Your Feedback

We welcome your feedback about Oracle Applications user assistance. If you need clarification, find an error, or just want to tell us what you found helpful, we'd like to hear from you.

You can email your feedback to oracle_fusion_applications_help_ww_grp@oracle.com.

Thanks for helping us improve our user assistance!

1 Overview

Overview of AI Agent Studio

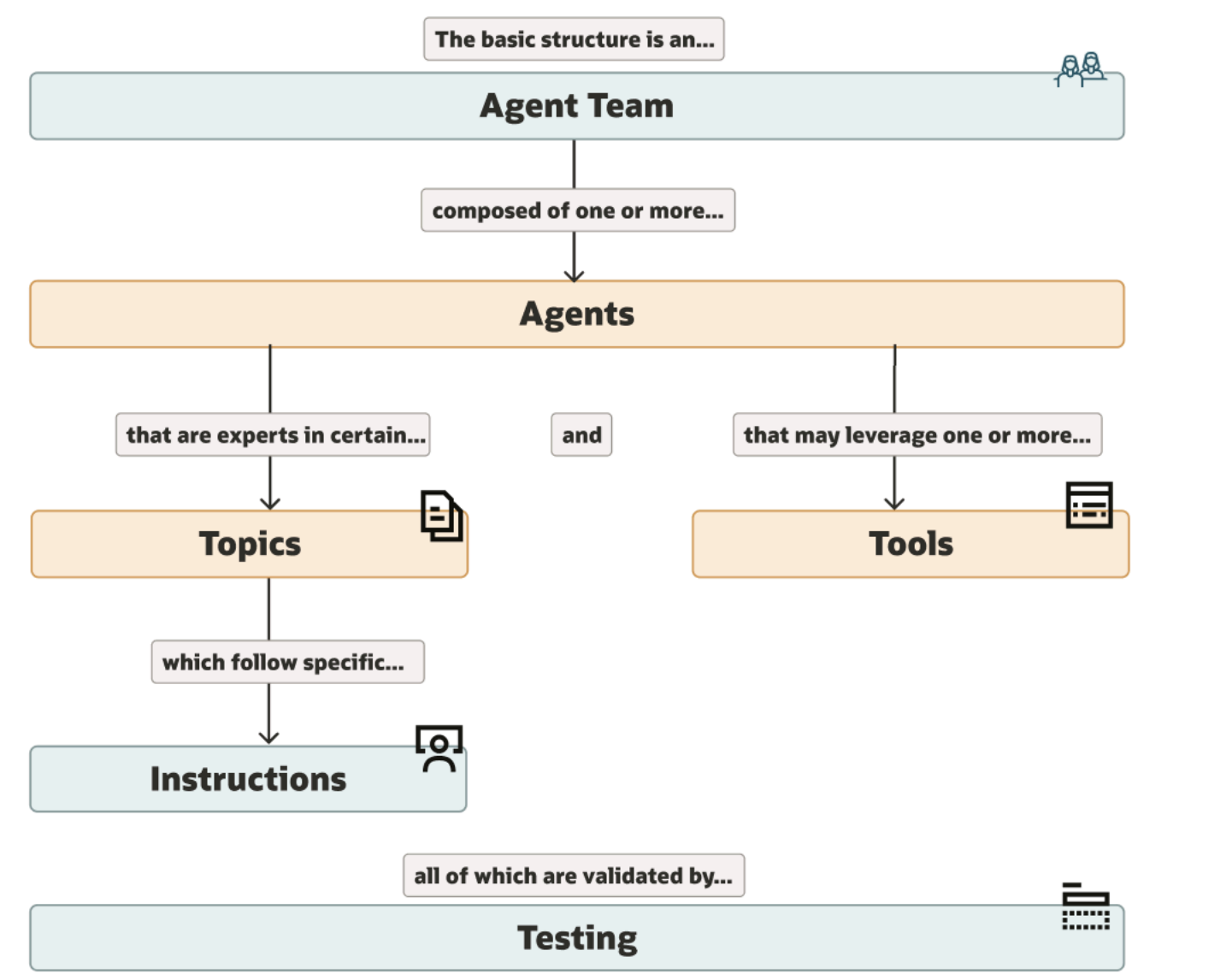
AI Agent Studio is a design-time environment that empowers you to create, configure, validate, and deploy AI agents to meet your organization's needs.

With AI Agent Studio, you can easily extend preconfigured agent templates, and even build new agents and multiagent flows. AI Agent Studio is fully integrated into Oracle Fusion Cloud Applications, providing secure and seamless access to the knowledge stores, tools, and APIs of Fusion Applications. This integration enables agents to be deployed directly into the flow, ensuring an efficient process.

Key Capabilities of AI Agent Studio

Feature	Description
Agent template libraries	Use templates and natural language prompts to create or fine tune agents for common business scenarios, such as opportunity-to-quote processing and shift scheduling.
Agent team orchestration	Configure multiple agents to collaborate on multistep processes, integrating user approvals where necessary.
Agent extensibility	Change and extend existing agents in Oracle Fusion Cloud Applications by incorporating new data sources, prompts, and APIs to fit specific business needs or industry requirements.
Native integration with Fusion Applications	Directly access APIs and tools in Fusion Applications, ensuring seamless deployment of agents without complex modifications.
Third-party system integration	Connect with external systems and collaborate with third-party agents for end-to-end automation with secure API support.
Trust and security framework	Automatically applies the security configurations, policies, and access controls of Fusion Applications, ensuring compliance with enterprise security standards.
Validation and testing tools	Use built-in tools to make your agents reliable, repeatable, easy to explain, and secure by verifying AI-driven flows before deployment.

Components of AI Agent Studio



Component	What It Does	Example
Agent Team	Consists of a structured sequence of steps or actions that a single agent or a group of agents follows to accomplish a specific business task or answer a user query. The agent team is the component that can be deployed for use. - Consists of conversation logic, system integration, and user support flow. - Defines how the agent acts for a particular use case — what to do, when to do it, and how to respond based on user inputs or back-end system data.	A recruitment agent (that might consist of multiple agents) schedules interviews, screens resumes, calculates salary, and generates offers, based on policies and approvals.

Component	What It Does	Example
Agents	Leverages a large language model to reason, create action plans, and interact with users to gather information and take direction. On behalf of users, the AI agent can do tasks that enhance productivity, efficiency, and the overall user experience. An agent must be added to an agent team, so that it can be deployed for use. AI agents can be categorized into various types. • User-proxy agent: Acts on behalf of a business user to provide input to another agent or group of agents. It's sometimes referred to as a conversational agent. • Supervisor agent: Orchestrates the use of agents within an agentic flow. • Specialist or utility agent: Focuses on a specific role or expertise and can be skilled in using a particular tool. AI agents can also have one or more of these characteristics: • Persona-based agent: Represents a specific role, such as benefits administrator, customer service representative, and finance administrator. • Tool user: Uses technology-related tools, such as calculators, web search queries, document embedding, and calendar schedulers. • Task-oriented agent: Understands their assignment or task, as a single agent or as part of a multiagent flow.	A scheduling or calendar agent that manages your workday by following your instructions. It can accept new calendar invites and propose alternative times when needed.
Topics	Defines the areas of expertise through instructions that set the boundaries and constraints for agent conversations and abilities.	An employee benefits agent can contain topics such as Health Savings Account (HSA), retirement benefits, and stock plans.
Tools	Defines the additional utilities an agent can use to accomplish a task. One or more tools are assigned to agents, and they're reusable among agents.	• Calculator tool • Connector tool • Email tool • Business object tool • User query tool • Document retrieval tool for retrieval-augmented generation (RAG)
Instructions	Specifies the natural language rules that define the rules or conditions applied to a given topic. Instructions are part of the prompts that are sent to the underlying large language model. They can also contain guidelines and guardrails that set the parameters of an agent response.	Instructions for the payroll deduction topic: Make sure you've information regarding the number of dependents either by asking the user or querying the system. If you don't know the answer, don't make up a response.
Testing	Enables administrators to test the agent team design, ensure correct tool, topic, and instruction configurations, and validate reasoning and sources cited by the agent.	Provide example responses to a series of questions a user would likely ask an agent and details about how the agent arrived at its response.

2 Access Requirements for AI Agent Studio

Access Requirements for AI Agent Studio

You can give access to AI Agent Studio by assigning predefined duty roles to job roles. Also, make sure to complete these prerequisites:

- Enable security console to work with permission groups
- Run scheduled processes to import security data
- Run scheduled processes to get help from the integrated AI agent
- (Optional) Assign privilege to use external REST API tools
- (Optional) Assign permission groups for channels
- (Optional) Assign role to schedule workflow agent teams
- Give users access to AI agents



Enable Security Console to Work with Permission Groups

For the Security Console to work with permission groups and related objects, set the **Enable Security Console External Application Integration** (ORA_ASE_SAS_INTEGRATION_ENABLED) profile option at the site level.

1. In the Setup and Maintenance work area, search for the **Manage Administrator Profile Values** task using the search link in the  panel.
2. Search for the profile option and set the value for the **Site** profile level to **Yes**.

Run Scheduled Processes to Import Security Data

To import resources from LDAP, and transfer the necessary information into the security tables of Fusion Applications, run these two scheduled processes sequentially.

1. Import Resource Application Security Data
2. Import User and Role Application Security Data

You must run the processes one after the other.

1. Go to **Navigator > Tools > Scheduled Processes**.
2. Click **Schedule New Process**.
3. Leave the type as **Job**.
4. Search for and select the process.
5. Submit the process.

Run Scheduled Processes to Get Help from the Integrated AI Agent

You can get answers to questions about developing AI agents from the AI help agent integrated into AI Agent Studio. Using this conversational agent, you can get answers to questions about existing agents, search for agents, tools, and topics using natural language, and receive AI powered suggestions for relevant resources to use in your agents.

To use the integrated AI help agent, run these two scheduled processes daily.

- Index AI Agent Studio Assistant Documents
- Index AI Agent Studio Assistant Objects and Attributes

Run each process using these steps:

1. Go to Scheduled Processes, and click **Schedule New Process**.
2. Leave the type as **Job**.
3. Search for and select the process.
4. Submit the process.

Assign Privilege to Use External REST API Tools

To create External REST API tools in AI Agent Studio, the Create and Edit Backends for Visual Builder Studio (ORA_FND_TRAP_PRIV) privilege must be added to the custom role assigned to the user. You can add this privilege while creating or editing a custom role.

1. Go to the Security Console.
2. To use a new custom role, create it. To use an existing custom role, search for the custom role and edit it.
3. Go to the Function Security Policies page and select **Add Function Security Policy**.
4. Add the Create and Edit Backends for Visual Builder Studio (ORA_FND_TRAP_PRIV) privilege to the role and save it.
5. Save the custom role and assign to the user.

Assign Permission Groups for Channels

To create channels from Credentials tab in AI Agent Studio, additional permission groups must be added to the custom role assigned to the user. You can add these permission groups to a duty role and assign the duty role while creating or editing a custom role.

1. Go to the Security Console and create a new duty role for permission groups.
2. Open the **Permission Groups** page and select **Add Permission Groups**.
3. Search for and add these permission groups:
 - create:ChannelManifest
 - create:ExternalChatCorrelation
 - delete:ChannelManifest
 - delete:ExternalChatCorrelation
 - read:ChannelManifest
 - read:ExternalChatCorrelation
 - update:ChannelManifest
 - update:ExternalChatCorrelation
4. Add security view for each permission group.

- a. Select the permission group added.
 - b. In Details section, open Security Views tab.
 - c. Select **Add Security Views** and add the **AllRowsAllFields** security view.
 - d. Add the security view for all the permission groups.
 5. Save the duty role and assign this duty role to the custom job role.
 6. To use a new custom job role, create it. To use an existing custom job role, search for the custom job role and edit it.
- Note:** Make sure to enable permission groups.
7. Go to the **Role Hierarchy** page. From the Roles and Permission Groups tab select **Add Role**.
 8. Search for and add the duty role you've created.
 9. Save the custom role and assign to the user.

Assign Role to Schedule Workflow Agent Teams

You can call a workflow agent team using a scheduled trigger. To schedule the workflow, the Fai Batch Job Manager Duty (ORA_DR_FAI_BATCH_JOB_MANAGER_DUTY) role must be added to the custom role assigned to the user.

1. Go to the Security Console.
 2. To use a new custom role, create it. To use an existing custom role, search for the custom role and edit it.
- Note:** Make sure that permission groups are enabled for the custom role.
3. Go to the Role Hierarchy page and open the Roles and Permission Groups tab.
 4. Add the Fai Batch Job Manager Duty (ORA_DR_FAI_BATCH_JOB_MANAGER_DUTY) role.
 5. Save the custom role and assign to the user.

Give Users Access to AI Agents

After your agents are created and ready for use, provide access for users to interact with the AI agents. For information, see [How can I give users access to AI agents?](#).

Assign Predefined Duty Roles to Job Roles

Assign predefined product-specific duty roles to the appropriate job roles, and make sure permission groups are enabled. You can give people access to configure AI agents in all or specific products.

CAUTION: Using predefined roles might account for subscription consumption irrespective of whether you purchased the cloud service or not. See [Guidance for Assigning Predefined Roles](#).

- [Provide Access to Configure AI Agents in all Products](#)
- [Provide Access to Configure AI Agents in Oracle Fusion Cloud Human Capital Management](#)
- [Provide Access to Configure AI Agents in Oracle Fusion Cloud Supply Chain & Manufacturing](#)
- [Provide Access to Configure AI Agents in Oracle Fusion Cloud Procurement](#)
- [Provide Access to Configure AI Agents in Oracle Permitting and Licensing](#)

Provide Access to Configure AI Agents in all Products

1. Go to the Security Console and create a new custom job role.

Note: Make sure to enable permission groups.

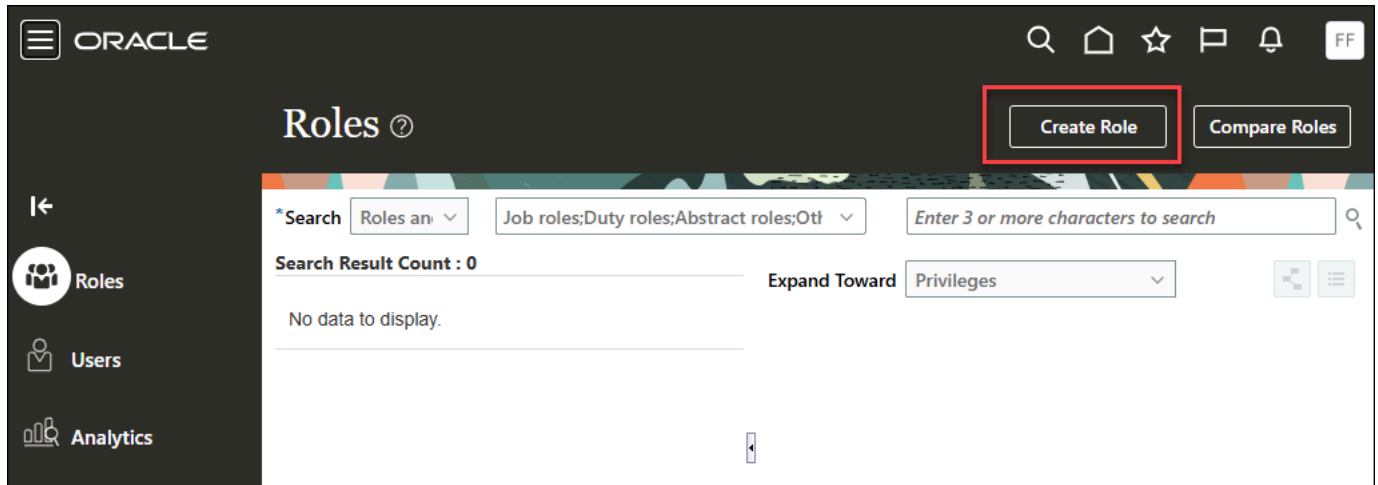
2. On the Role Hierarchy page, open the Roles and Permission Groups tab and add these duty roles:
 - Fai Genai Agent CX Administrator Duty
(ORA_DR_FAI_GENERATIVE_AI_AGENT_CX_ADMINISTRATOR_DUTY)
 - Fai Genai Agent FIN Administrator Duty
(ORA_DR_FAI_GENERATIVE_AI_AGENT_FIN_ADMINISTRATOR_DUTY)
 - Fai Genai Agent GRC Administrator Duty
(ORA_DR_FAI_GENERATIVE_AI_AGENT_GRC_ADMINISTRATOR_DUTY)
 - Fai Genai Agent HCM Administrator Duty
(ORA_DR_FAI_GENERATIVE_AI_AGENT_HCM_ADMINISTRATOR_DUTY)
 - Fai Genai Agent PRC Administrator Duty
(ORA_DR_FAI_GENERATIVE_AI_AGENT_PRC_ADMINISTRATOR_DUTY)
 - Fai Genai Agent PRJ Administrator Duty
(ORA_DR_FAI_GENERATIVE_AI_AGENT_PRJ_ADMINISTRATOR_DUTY)
 - Fai Genai Agent PSC Administrator Duty
(ORA_DR_FAI_GENERATIVE_AI_AGENT_PSC_ADMINISTRATOR_DUTY)
 - Fai Genai Agent SCM Administrator Duty
(ORA_DR_FAI_GENERATIVE_AI_AGENT_SCM_ADMINISTRATOR_DUTY)
3. Open the Roles and Privileges tab and add the Manage All Intelligent Agents (ORA_FAI_MANAGE_ALL_AI_AGENTS) role.
4. Save the custom role and assign it to users who want access.

Provide Access to Configure AI Agents in Oracle Fusion Cloud Human Capital Management

To give access to users without the Human Capital Management Application Administrator Job Role:

1. Go to the Security Console and create a new custom job role.

Note: Make sure to enable permission groups.



*Role Name

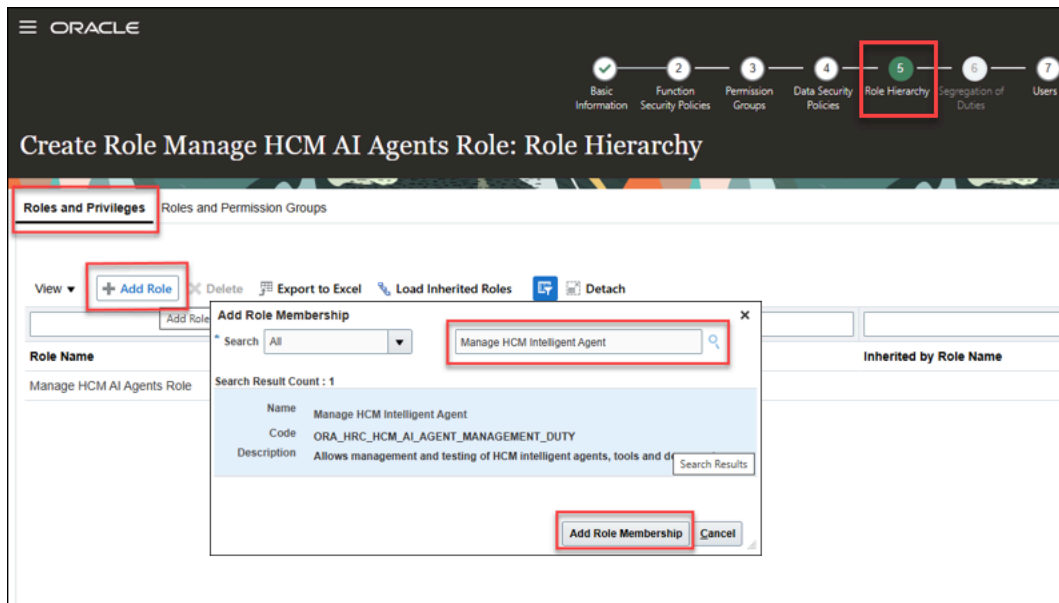
*Role Code

*Role Category

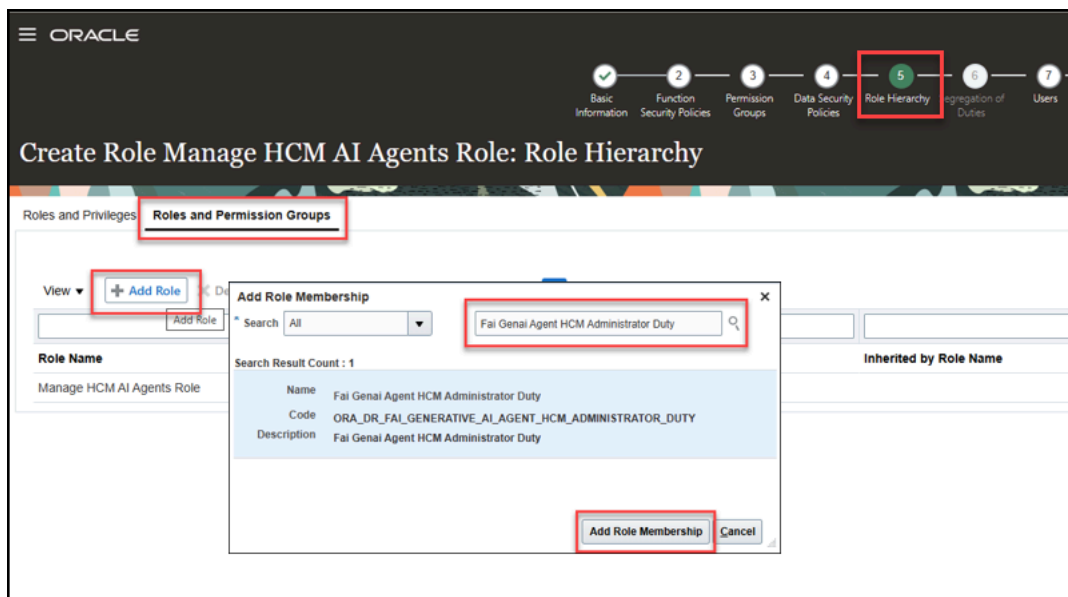
Enable Role for Access ☐
from All IP Addresses

Description

2. Go to the Role Hierarchy page.
 - o Open the Roles and Privileges tab, and add the Manage HCM Intelligent Agent (ORA_HRC_HCM_AI_AGENT_MANAGEMENT_DUTY) duty role.



- o Open the Roles and Permission Groups tab, and add the Fai Genai Agent HCM Administrator Duty (ORA_DR_FAI_GENERATIVE_AI_AGENT_HCM_ADMINISTRATOR_DUTY) role.

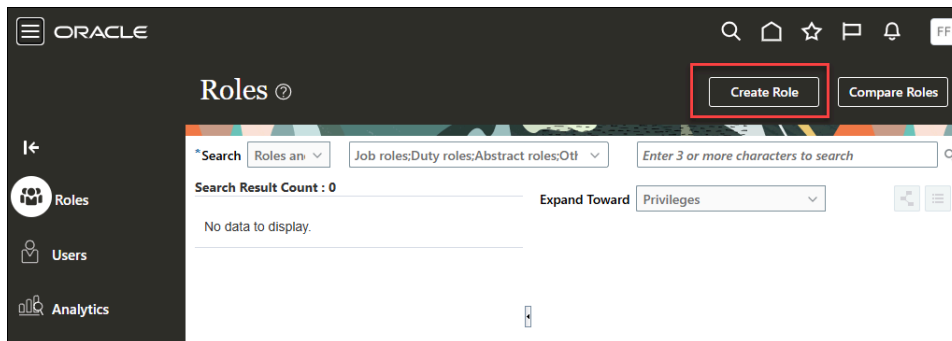


3. Save the custom role and assign to the appropriate job roles.

To give access to users with the Human Capital Management Application Administrator Job Role:

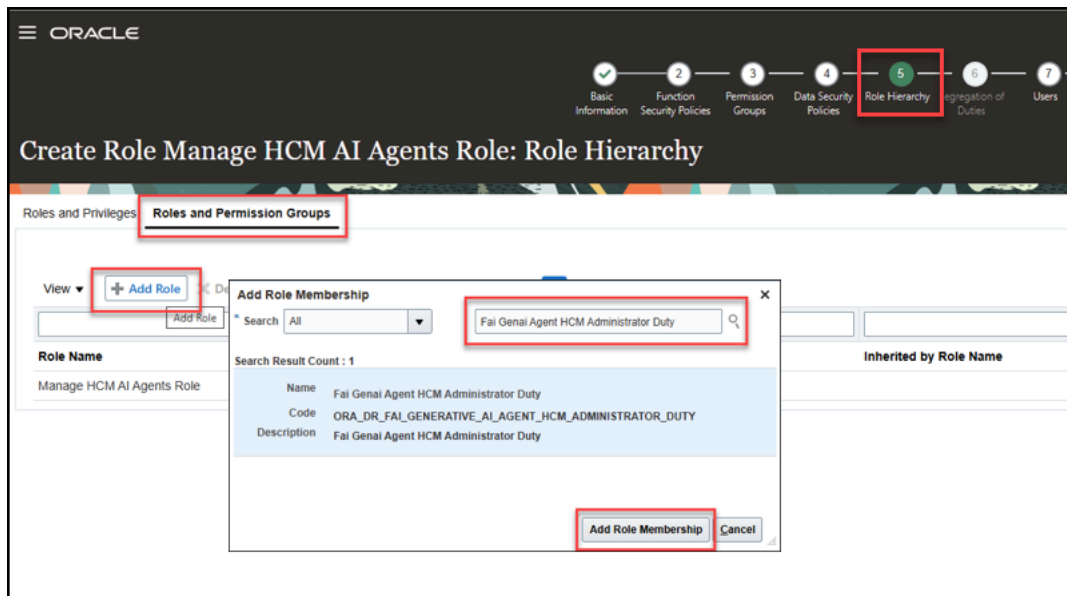
1. Go to the Security Console and create a new custom job role.

Note: Make sure to enable permission groups.



*Role Name	Manage HCM AI Agents Role
*Role Code	MANAGE_HCM_AI_AGENTS_ROLE
*Role Category	Common - Job Roles
	Enable Permission Groups
Enable Role for Access	☐
from All IP Addresses	
Description	Role to manage HCM AI Agents

2. On the Role Hierarchy page, open the Roles and Permission Groups tab, and add the Fai Genai Agent HCM Administrator Duty (ORA_DR_FAI_GENERATIVE_AI_AGENT_HCM_ADMINISTRATOR_DUTY) role.



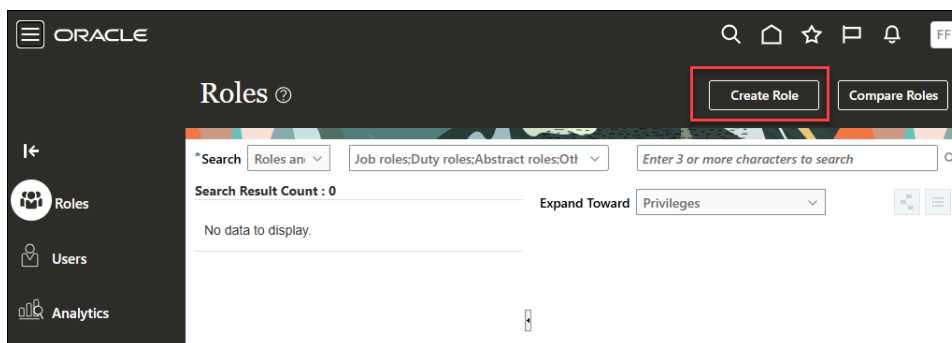
3. Save the custom role and assign to the appropriate job roles.

Provide Access to Configure AI Agents in Oracle Fusion Cloud Supply Chain & Manufacturing

To give access to users without the Supply Chain Application Administrator Job Role:

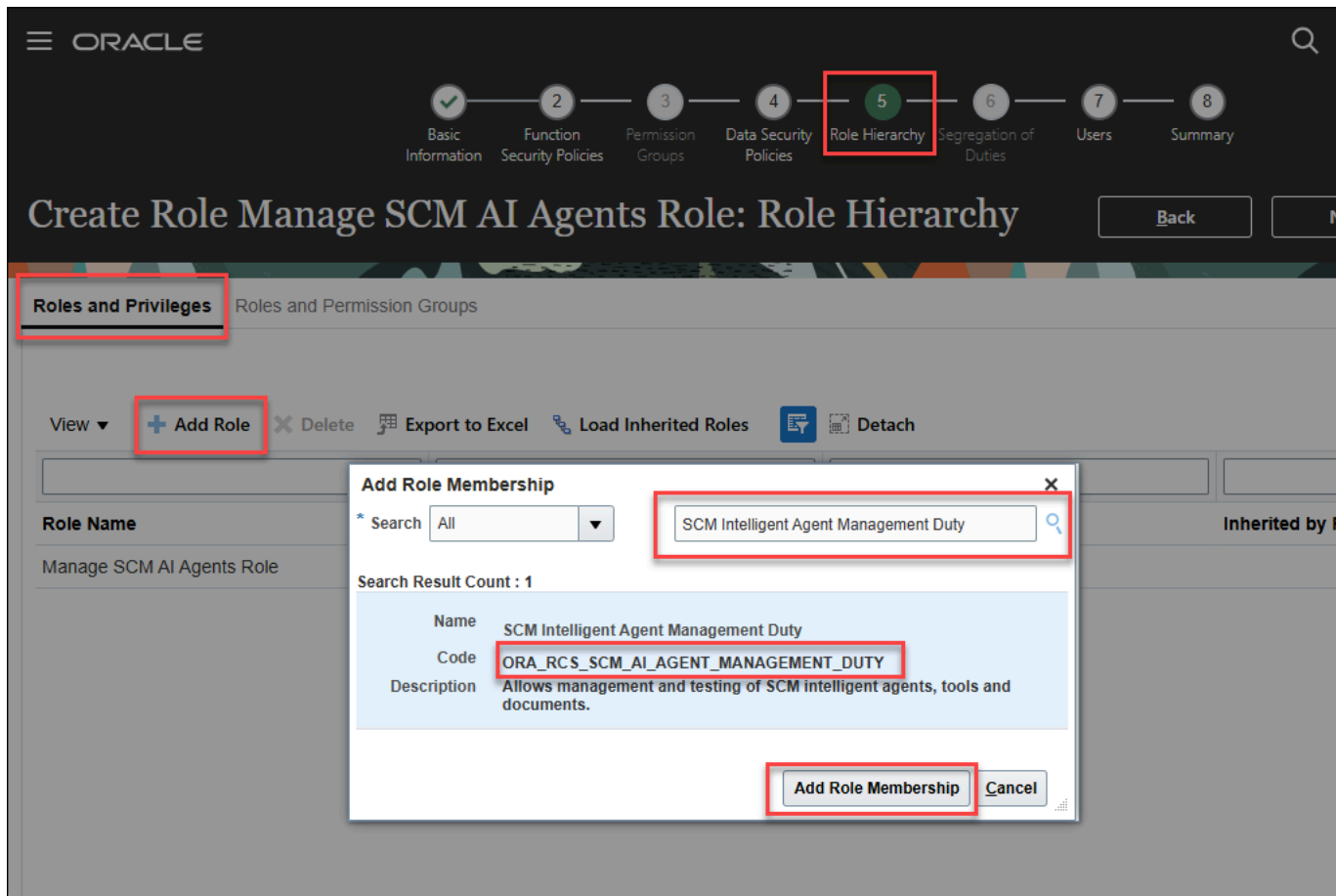
1. Go to the Security Console and create a new custom job role.

Note: Make sure to enable permission groups.

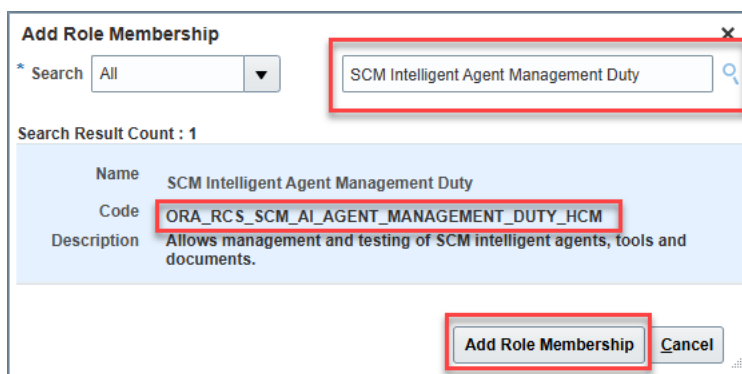


*Role Name	Manage SCM AI Agents Role
*Role Code	MANAGE_SCM_AI_AGENTS_ROLE
*Role Category	Common - Job Roles v
	Enable Permission Groups
Enable Role for Access	☐
from All IP Addresses	
Description	

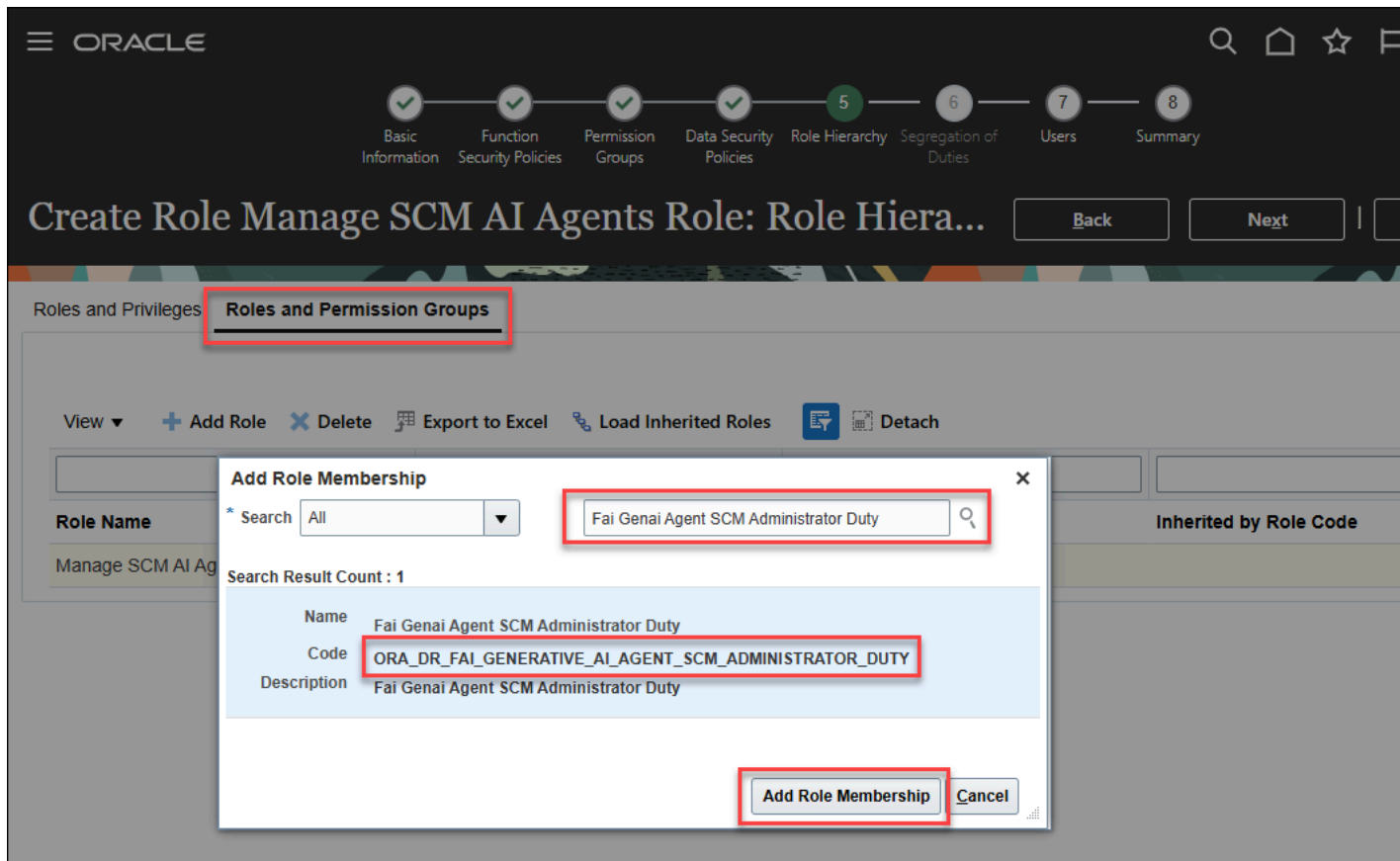
2. On the Role Hierarchy page, open the Roles and Privileges tab and add these roles:
 - SCM Intelligent Agent Management Duty (ORA_RCS_SCM_AI_AGENT_MANAGEMENT_DUTY)



- SCM Intelligent Agent Management Duty (ORA_RCS_SCM_AI_AGENT_MANAGEMENT_DUTY_HCM)



3. Open the Roles and Permission Groups tab and add the Fai Genai Agent SCM Administrator Duty (ORA_DR_FAI_GENERATIVE_AI_AGENT_SCM_ADMINISTRATOR_DUTY) duty role.

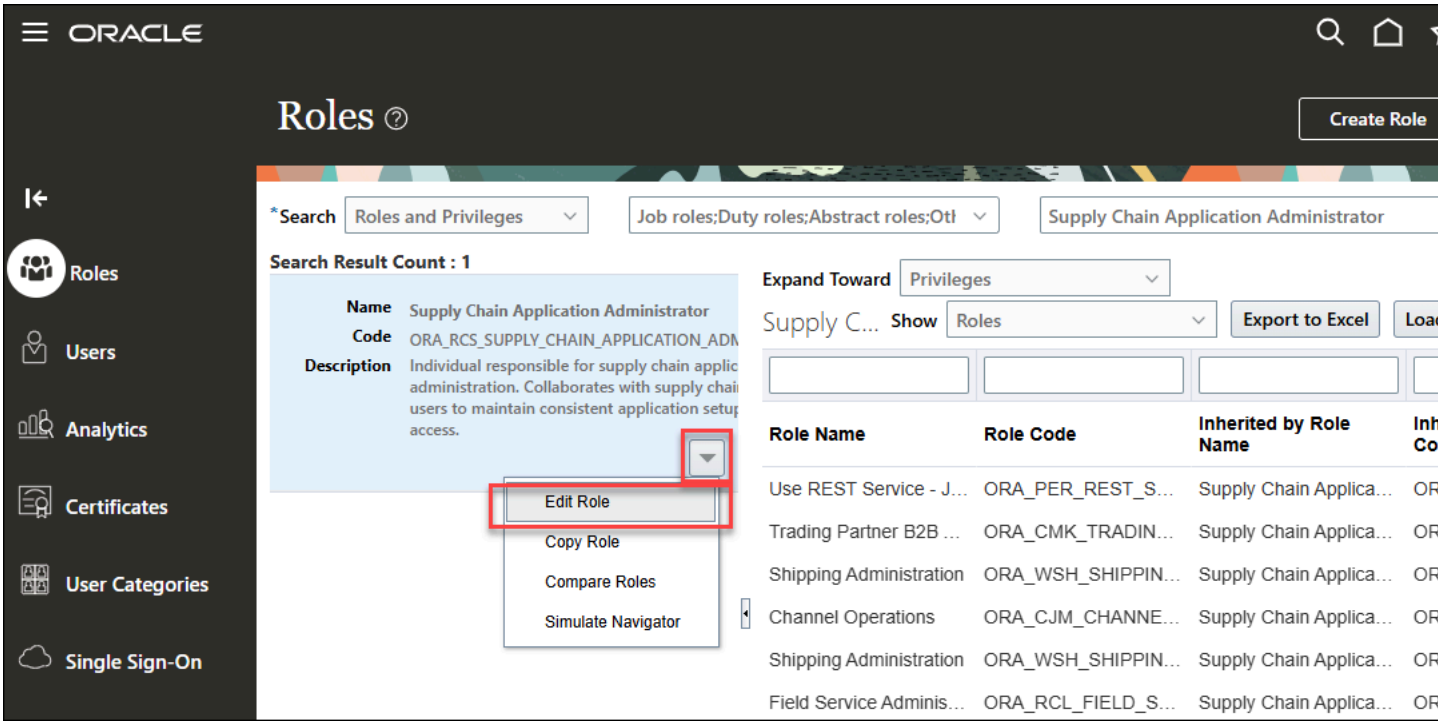
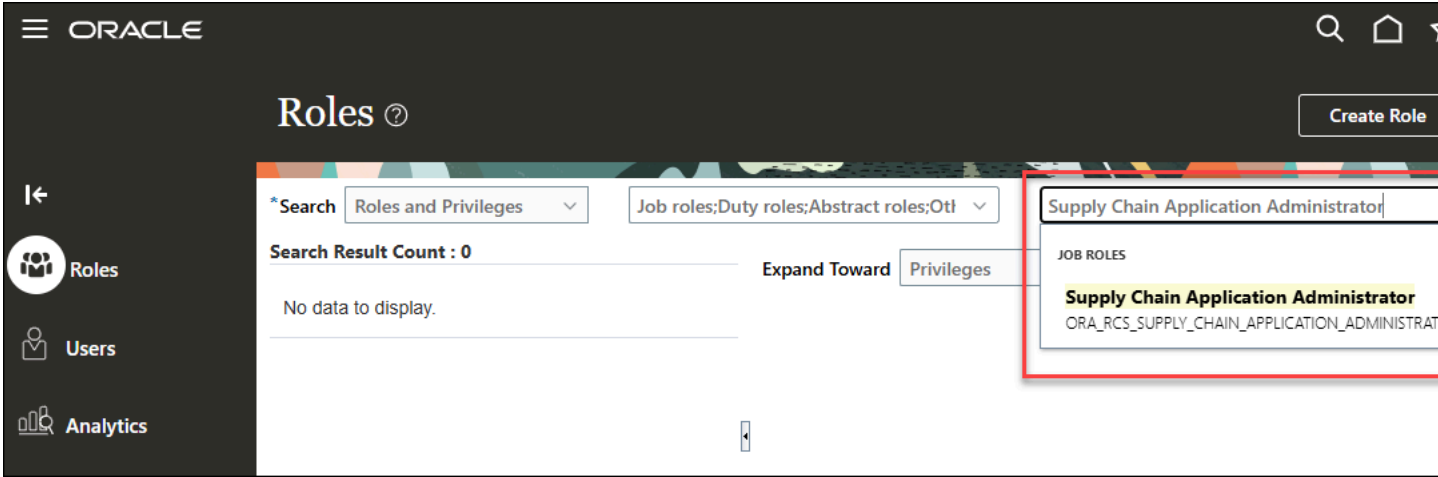


4. Save the custom role and assign this custom role to the appropriate job roles.

To give access to users with the Supply Chain Application Administrator Job Role:

1. Go to the Security Console.

2. Search for the Supply Chain Application Administrator (ORA_RCS_SUPPLY_CHAIN_APPLICATION_ADMINISTRATOR_JOB) job role, and edit it.



3. Enable permission groups and save the job role.

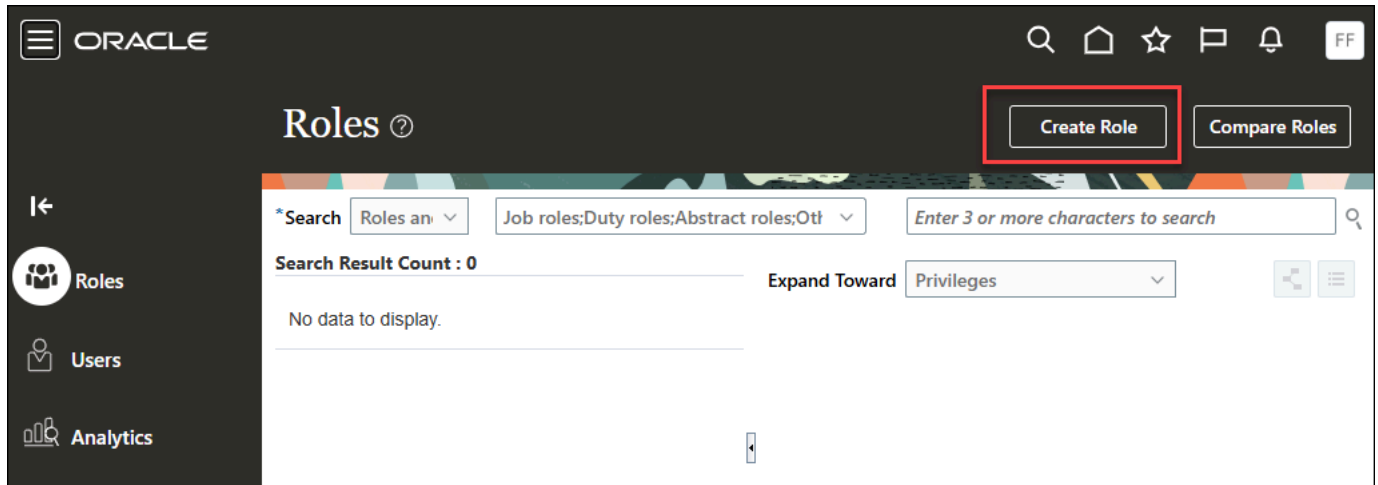
The screenshot shows the Oracle AI Agent Studio interface for editing a role. At the top, there is a navigation bar with the Oracle logo and a progress indicator with five steps: 1. Basic Information, 2. Function Security Policies, 3. Permission Groups, 4. Data Security Policies, and 5. Role Hierarchy. The main heading is 'Edit Role Supply Chain Application Administrator: Basic'. Below this, the role details are displayed: Role Name: Supply Chain Application Administrator, Role Code: ORA_RCS_SUPPLY_CHAIN_APPLICATION_ADMINISTRATOR_JOB, and Role Category: SCM - Job Roles. A red box highlights the 'Enable Permission Groups' button. Below this button, there are checkboxes for 'Predefined Role' (checked) and 'Enable Role for Access from All IP Addresses' (unchecked). At the bottom, there is a 'Description' field with the text: 'Individual responsible for supply chain application administration. Collaborates with supply chain application users to maintain consistent application setup, rules, and access.'

Provide Access to Configure AI Agents in Oracle Fusion Cloud Procurement

To give access to users without the Procurement Application Administrator Job Role:

1. Go to the Security Console and create a new custom job role.

Note: Make sure to enable permission groups.



*Role Name

*Role Code

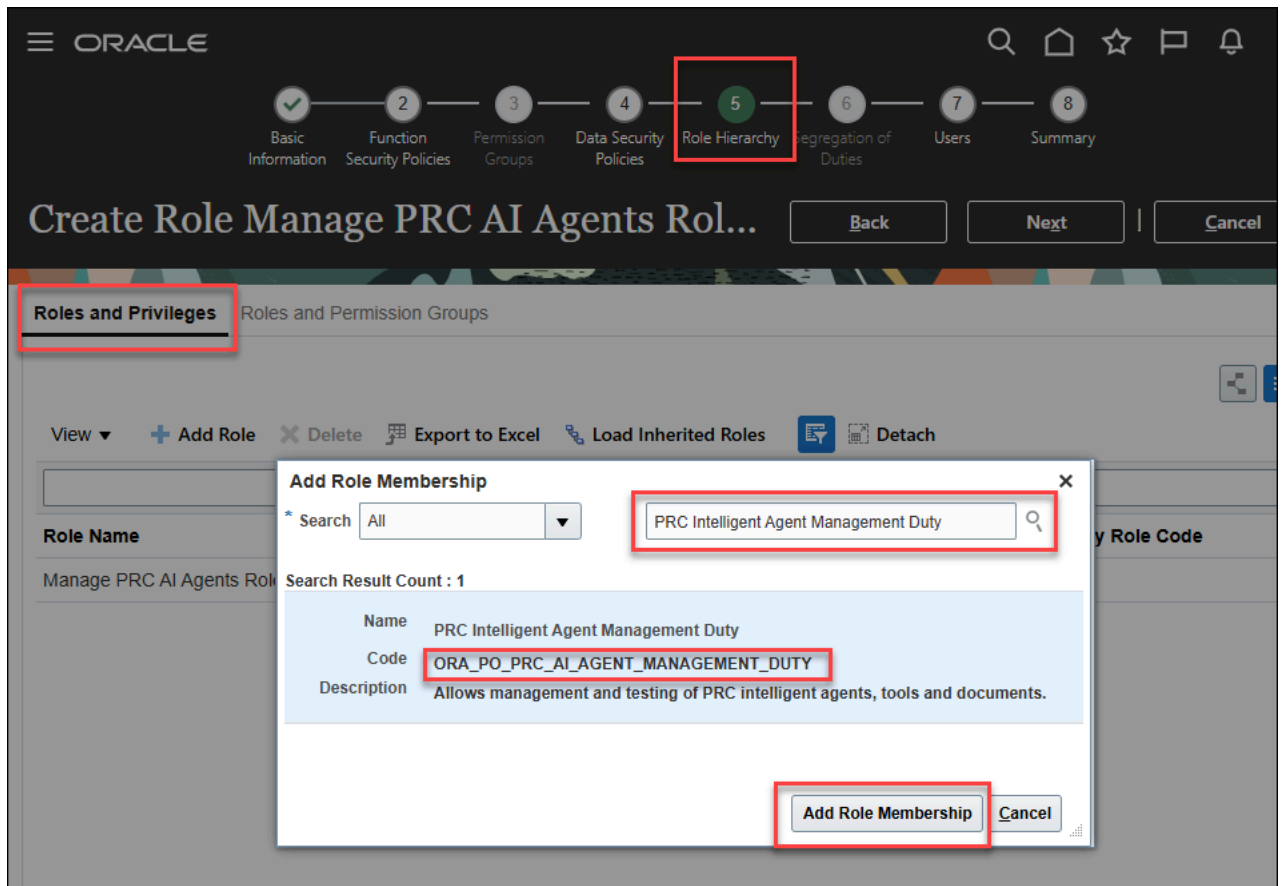
*Role Category

Enable Role for Access ☐

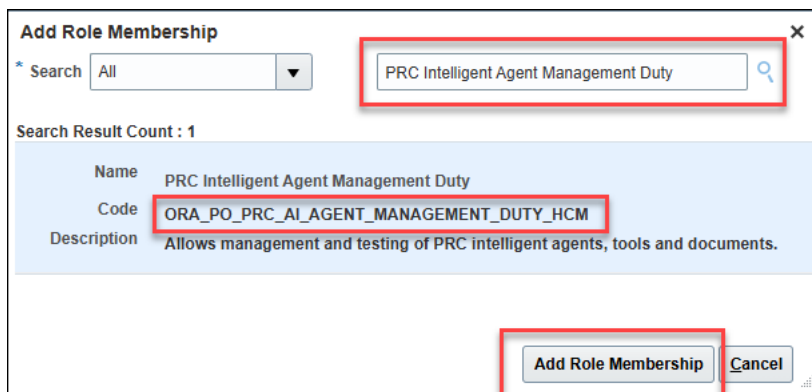
from All IP Addresses

Description

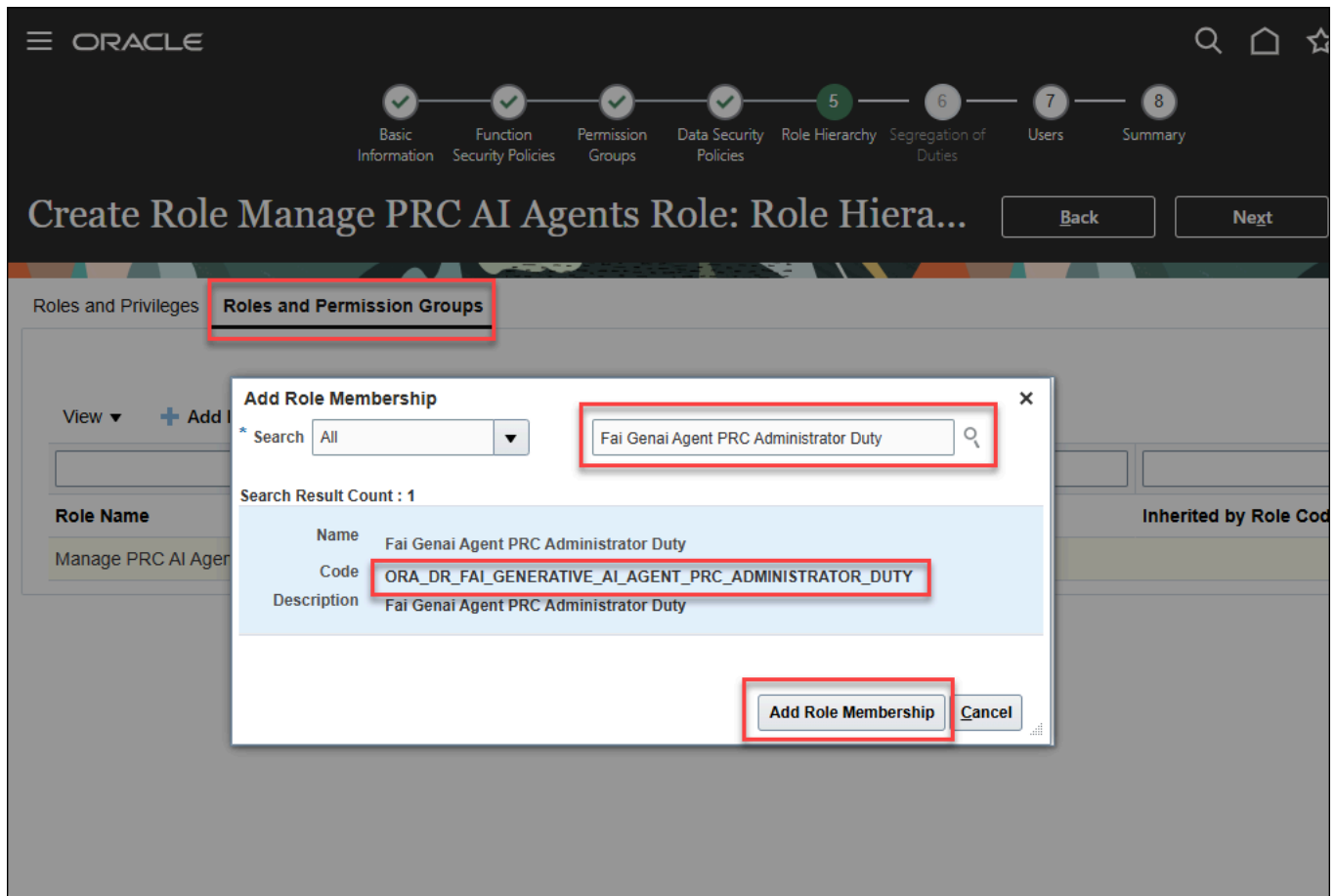
2. On the Role Hierarchy page, open the Roles and Privileges tab and add these roles:
 - o PRC Intelligent Agent Management Duty (ORA_PO_PRC_AI_AGENT_MANAGEMENT_DUTY)



- o PRC Intelligent Agent Management Duty (ORA_PO_PRC_AI_AGENT_MANAGEMENT_DUTY_HCM)



3. Open the Roles and Permission Groups tab and add the Fai Genai Agent PRC Administrator Duty (ORA_DR_FAI_GENERATIVE_AI_AGENT_PRC_ADMINISTRATOR_DUTY) duty role.

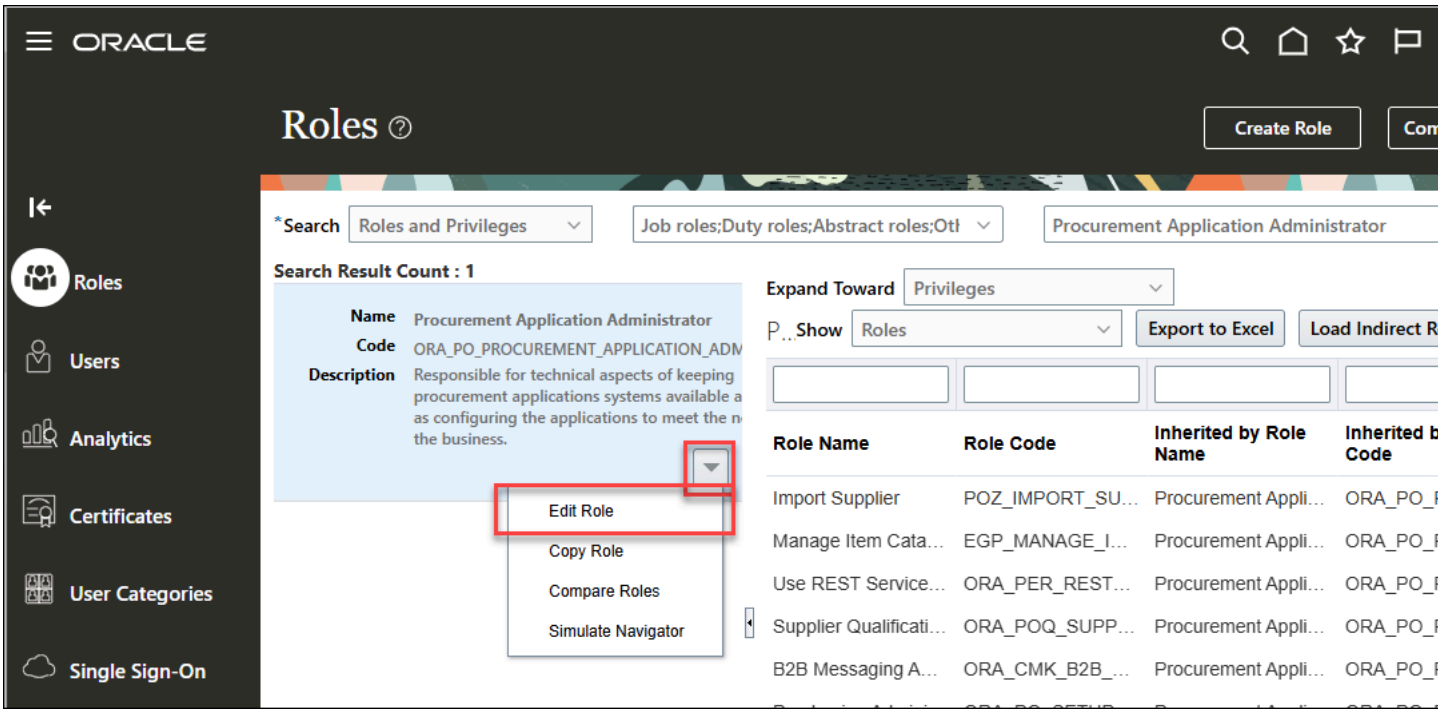
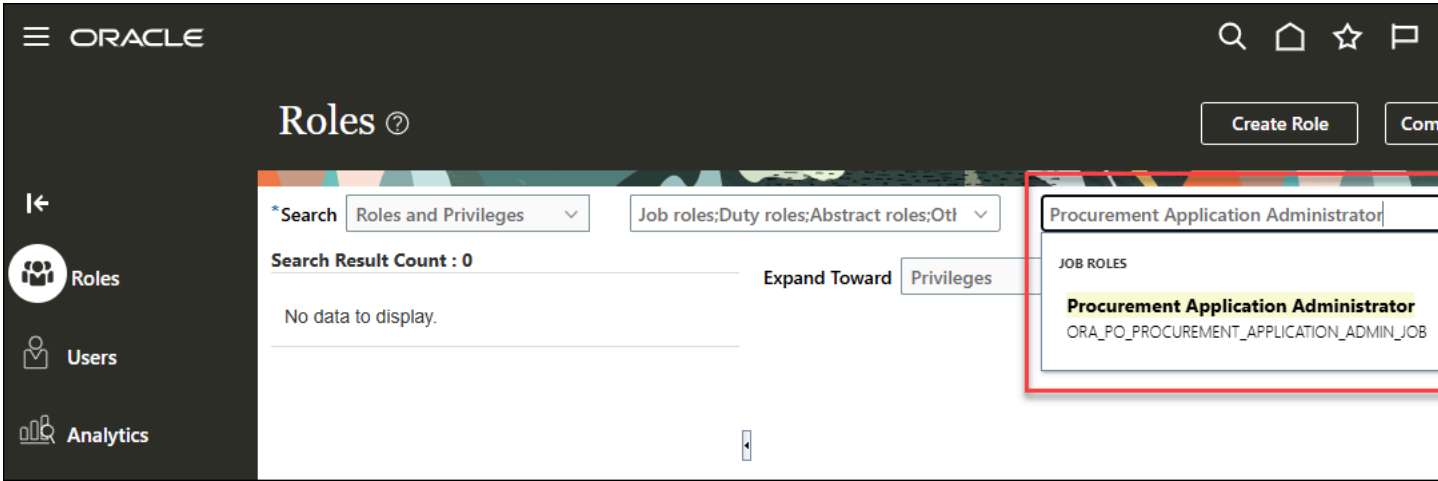


4. Save the custom role and assign this custom role to the appropriate job roles.

To give access to users with the Procurement Application Administrator Job Role:

1. Go to the Security Console.

2. Search for the Procurement Application Administrator (ORA_PO_PROCUREMENT_APPLICATION_ADMIN_JOB) job role, and edit it.



3. Enable permission groups and save the job role.



The screenshot shows the Oracle 'Edit Role' page for the 'Procurement Application Administrator' role. At the top, there is a navigation bar with five steps: 1. Basic Information, 2. Function Security Policies, 3. Permission Groups (highlighted), 4. Data Security Policies, and 5. Role Hierarchy. The main title is 'Edit Role Procurement Application Administrator'. Below this, the role details are listed: Role Name: Procurement Application Administrator, Role Code: ORA_PO_PROCUREMENT_APPLICATION_ADMIN_JOB, and Role Category: Procurement - Job Roles. A red box highlights the 'Enable Permission Groups' button. Below this, there are checkboxes for 'Predefined Role' (checked), 'Enable Role for' (unchecked), and 'Access from All IP Addresses' (unchecked). The description of the role is: 'Responsible for technical aspects of keeping procurement applications systems available as well as configuring the applications to meet the needs of the business.'

Provide Access to Configure AI Agents in Oracle Fusion Cloud Financials

1. Go to the Security Console and create a new custom job role.
Note: Make sure to enable permission groups.
2. Go to the Role Hierarchy page.
 - o Open the **Roles and Privileges** tab, and add the Manage Financials Intelligent Agent (ORA_FUN_MANAGE_FIN_AI_AGENT_HCM) and Manage Financials Intelligent Agent (ORA_FUN_MANAGE_FIN_AI_AGENT) duty roles.
 - o Open the **Roles and Permission Groups** tab, and add the Fai Genai Agent FIN Administrator Duty (ORA_DR_FAI_GENERATIVE_AI_AGENT_FIN_ADMINISTRATOR_DUTY) duty role.
3. Save the custom role and assign it to the appropriate job roles.

Provide Access to Configure AI Agents in Oracle Fusion Cloud CX

To give access to users without the Sales Administrator Job Role:

1. Go to the Security Console and create a new custom job role.
Note: Make sure to enable permission groups.
2. On the Role Hierarchy page, open the Roles and Privileges tab and add these roles:
 - Manage CX AI Agents (ORA_ZCA_MANAGE_CX_AI_AGENTS)
 - Manage CX AI Agents (ORA_ZCA_MANAGE_CX_AI_AGENTS_HCM)
3. Open the Roles and Permission Groups tab and add the Fai Genai Agent CX Administrator Duty (ORA_DR_FAI_GENERATIVE_AI_AGENT_CX_ADMINISTRATOR_DUTY) duty role.
4. Save the custom role and assign this custom role to the appropriate job roles.

To give access to users with the Sales Administrator Job Role:

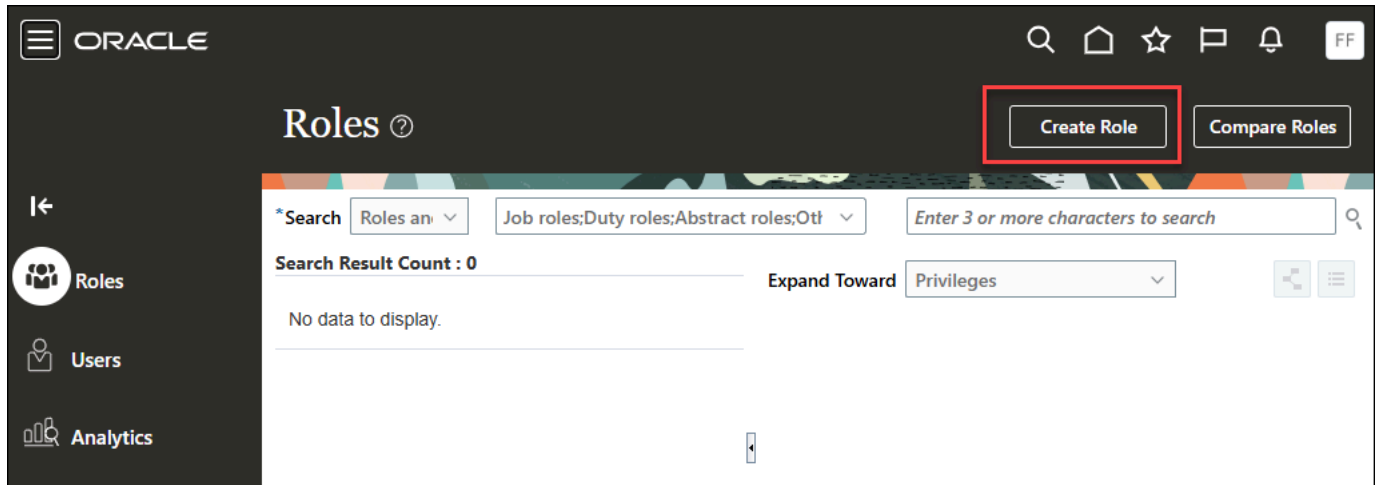
1. Go to the Security Console and search for the Sales Administrator (ORA_ZBS_SALES_ADMINISTRATOR_JOB) job role, and edit it.
2. Enable permission groups and save the job role.

Provide Access to Configure AI Agents in Oracle Permitting and Licensing

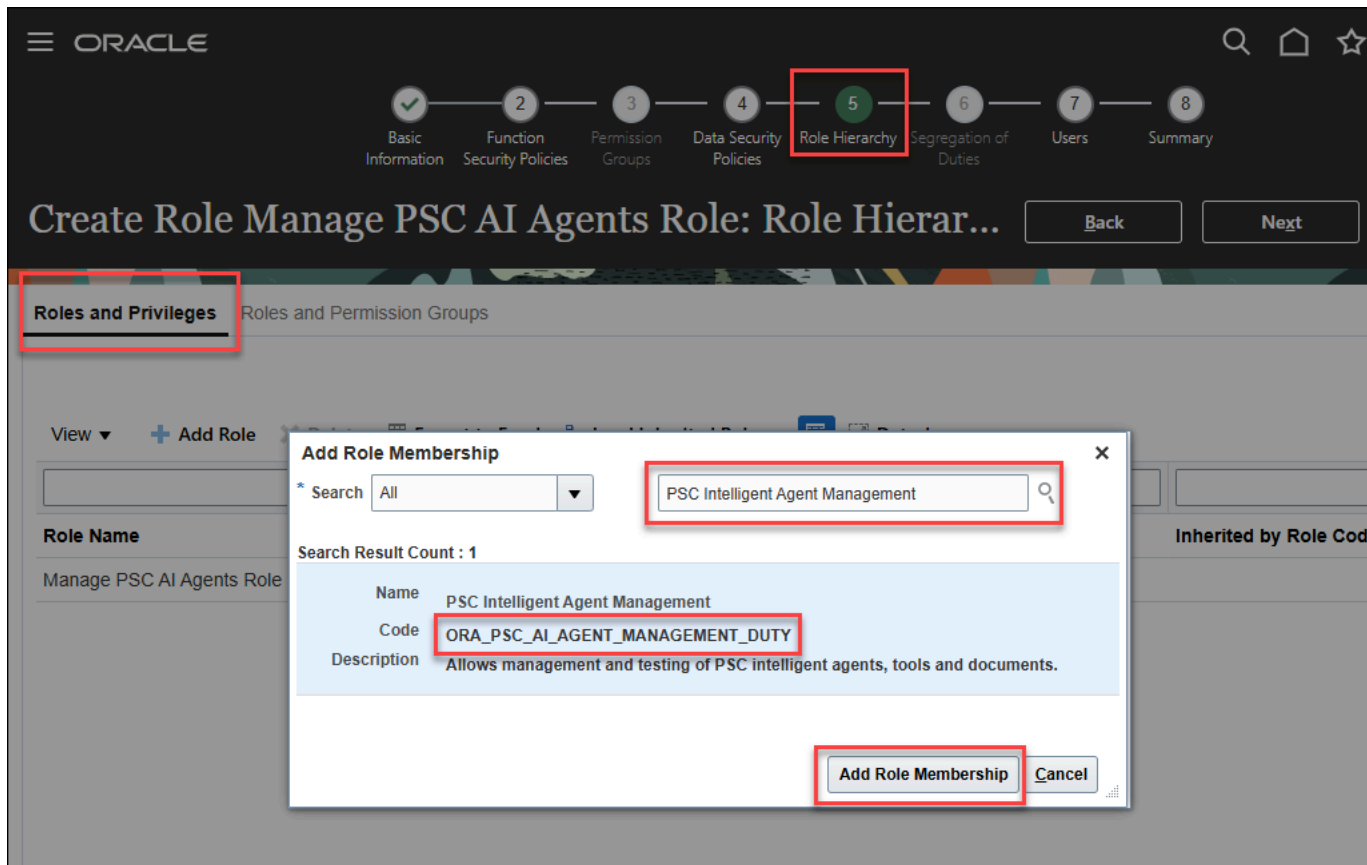
To give access to users without the PSC Application Administrator Job Role:

1. Go to the Security Console and create a new custom job role.

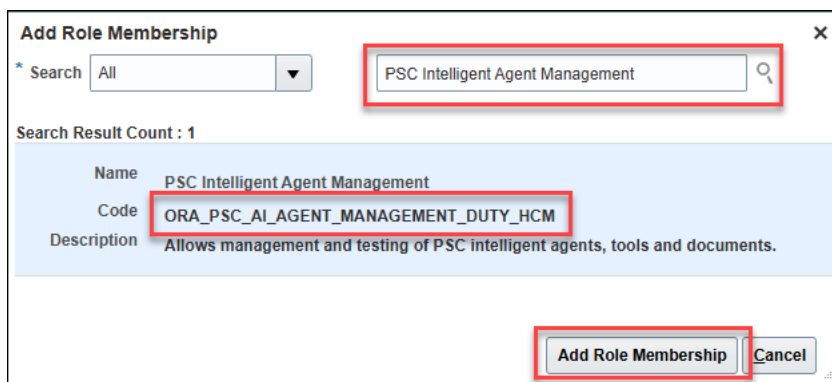
Note: Make sure to enable permission groups.

A screenshot of the 'Create Role' form in the Oracle Security Console. The form has several fields: '*Role Name' with the value 'Manage PSC AI Agents Role', '*Role Code' with the value 'MANAGE_PSC_AI_AGENTS_ROLE', and '*Role Category' with a dropdown menu showing 'Common - Job Roles'. A red box highlights the 'Enable Permission Groups' button. Below this, there is a checkbox for 'Enable Role for Access from All IP Addresses' which is currently unchecked. At the bottom, there is a 'Description' field which is empty.

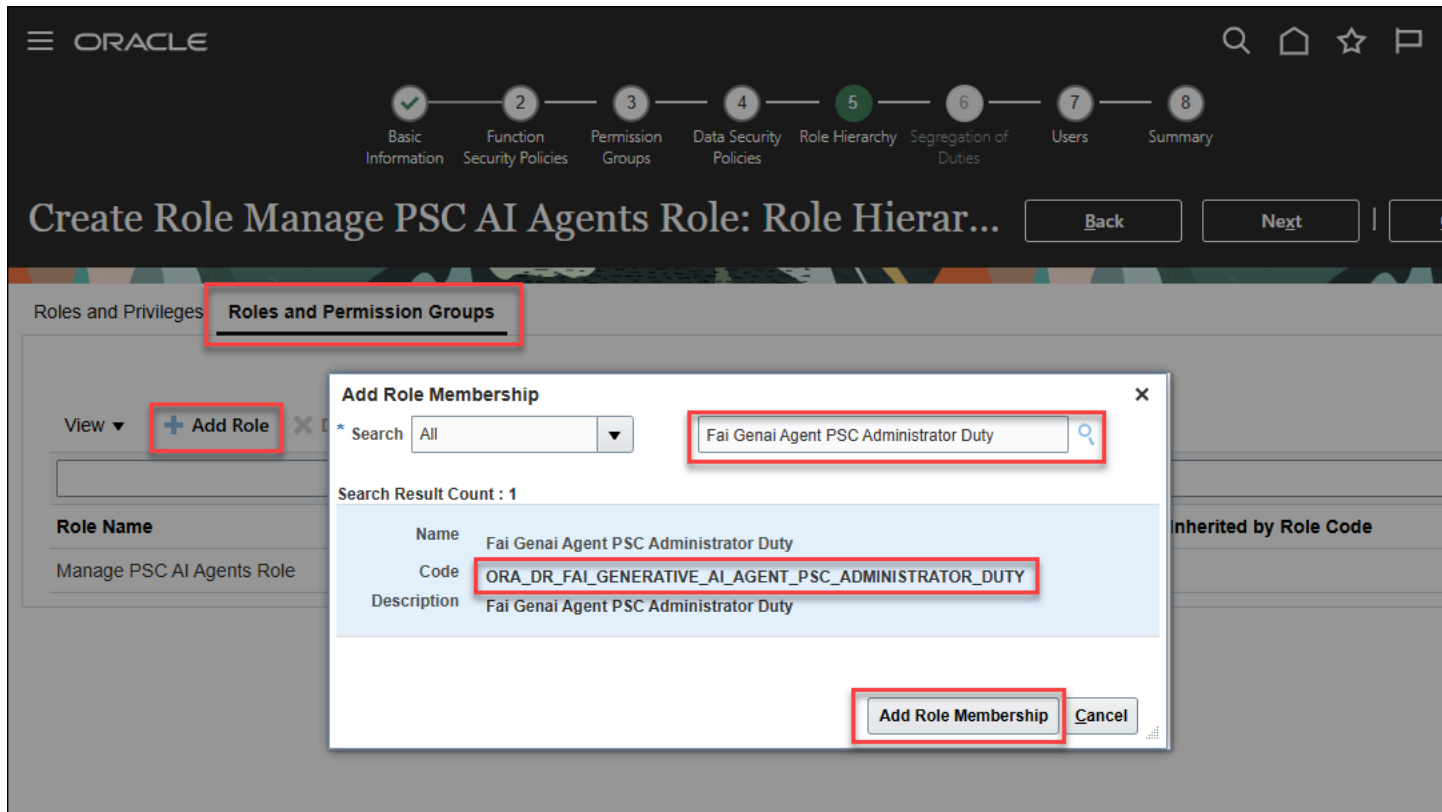
2. On the Role Hierarchy page, open the Roles and Privileges tab and add these roles:
 - PSC Intelligent Agent Management (ORA_PSC_AI_AGENT_MANAGEMENT_DUTY)



- PSC Intelligent Agent Management (ORA_PSC_AI_AGENT_MANAGEMENT_DUTY_HCM)



3. Open the Roles and Permission Groups tab and add the Fai Genai Agent PSC Administrator Duty (ORA_DR_FAI_GENERATIVE_AI_AGENT_PSC_ADMINISTRATOR_DUTY) duty role.

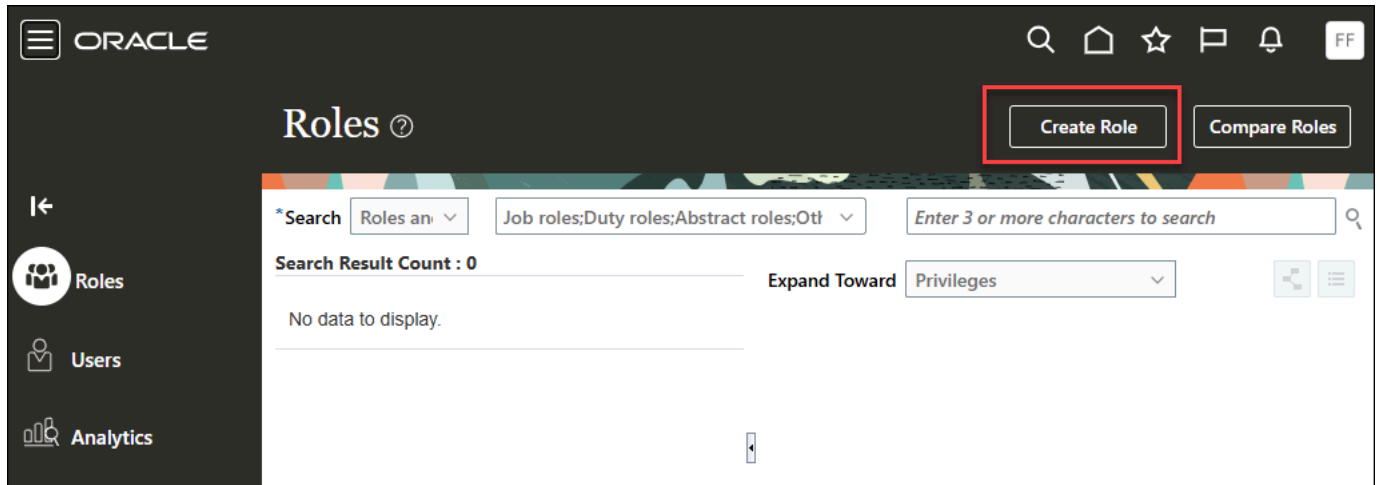


4. Save the custom role and assign this custom role to the appropriate job roles.

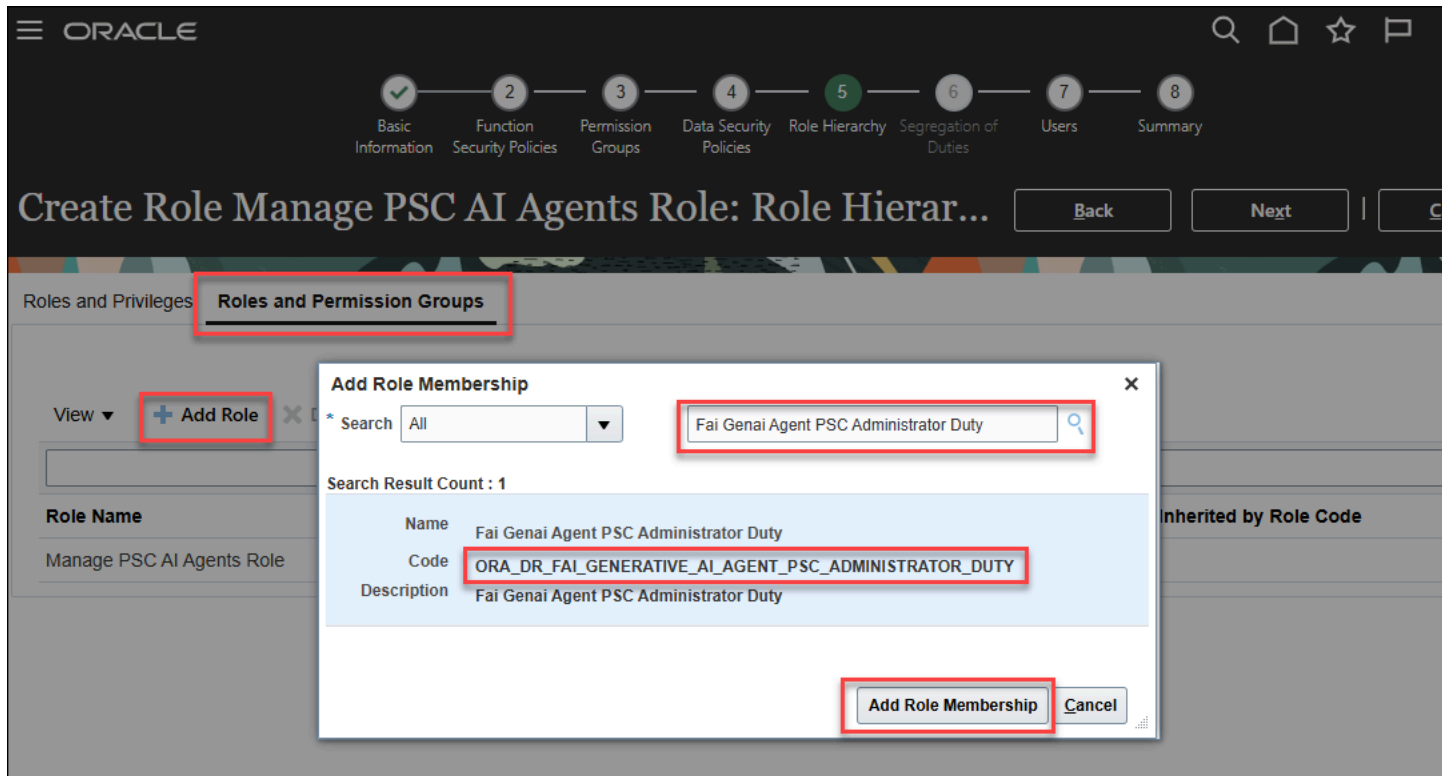
To give access to users with the PSC Application Administrator Job Role:

1. Go to the Security Console and create a new custom job role.

Note: Make sure to enable permission groups.

A screenshot of the 'Create Role' form in the Oracle Security Console. The form contains several fields: '*Role Name' with the value 'Manage PSC AI Agents Role', '*Role Code' with the value 'MANAGE_PSC_AI_AGENTS_ROLE', and '*Role Category' with a dropdown menu set to 'Common - Job Roles'. Below these fields, there is a button labeled 'Enable Permission Groups' which is highlighted with a red rectangle. At the bottom, there is a checkbox for 'Enable Role for Access from All IP Addresses' and a 'Description' field.

- On the Role Hierarchy page, open the Roles and Permission Groups tab and add the Fai Genai Agent PSC Administrator Duty (ORA_DR_FAI_GENERATIVE_AI_AGENT_PSC_ADMINISTRATOR_DUTY) duty role.



- Save the custom role and assign this custom role to the appropriate job roles.

3 Before You Begin

Choose How to Create AI Agents

You can either use one of the preconfigured templates or create custom agents. If you want to create new agents or modify existing agents by adding additional tools and topics, a subscription is required. For more information, see [Fusion Custom AI Agents - Pricing FAQ \(KB864473\)](#).

Here are some examples of how you choose between creating custom agents or using preconfigured templates, based on your use case.

Use Case	How to Create AI Agents
You want to turn on Benefits Agent, which allows employees to ask questions about their benefits, within AI Agent Studio. To ensure that this agent can understand the benefits specific to the organization, the documents specific to that organization must be uploaded.	Create an AI agent using a preconfigured agent team template. You can edit the existing agent, without adding any additional tools. Note: You can either use or copy a template and edit it. When you copy an agent template, you can automatically add a suffix to all the components to easily differentiate them.
You want to create a custom agent to help with onboarding by answering new hire questions and providing deep links to internal new hire resources if the agent is unable to answer. You can tailor the agent's topics and prompts to ensure they effectively address the specific needs of new hires.	Create a custom agent, and add the necessary tools and topics.
You want to create a Contract Assistant Agent to generate a new contract with auto-filled fields. This assistant will be provided with multiple contract templates to generate the new contract. The agent also reviews existing contracts to perform risk analysis, recommend changes and route the contract for final approvals. For these activities that need distinct expertise (template management, legal validation, recommendation, and routing), the administrator will need to build a supervisor agent that collaborates with other worker agents.	You can use either of these methods: • Create a team of custom agents, with one agent added as a supervisor to manage the others. • Use a preconfigured agent team template, and add more agents and a supervisor agent to it.

Use Case	How to Create AI Agents
You want to create a Supplier Quote Assistant agent to parse through quotes from multiple suppliers. The agent processes the quote document and maps the attributes. It then creates a draft requisition and submits for approval. For these activities, administrator will need to create a workflow agent that runs the tasks in the specified order.	Create an agent team of type Workflow, to run the predetermined set of tasks. Each node can perform a defined function, for example, extracting data, calling a business object, running an LLM, or sending an email in the order you define.

Get Started with AI Agent Studio

You can start with a preconfigured agent template or create your own agent team.

When using a preconfigured agent team template, the artifacts such as agents, tools, and topics, aren't directly editable. To change these artifacts, you create a copy of the artifact and add the copy to the agent team.

Note: Agents created directly within an agent team are directly editable within that team.

Here's a broad outline of the tasks involved in creating an agent team.

Task	Details
Define tools	To effectively define the tools required by an agent, you need to first identify the types of questions users might ask and then decide which tools the agent needs to answer those questions accurately. These are the available tools: • Oracle Fusion Cloud Applications business object: Business Object tools allow AI agents to retrieve, update, create, or delete business object records within Fusion Applications. Using this tool, agents can securely access information or call specific functions in the application. You can control what data the agents can access, by selecting the business objects and fields to use or ignore. AI agents adhere to the native security and role-based access controls of Fusion Applications, ensuring protection and privacy for your enterprise data. • Connector: You can use this tool to connect your AI agents to various data sources such as SharePoint, web pages, and other custom content source. For more information, see How do I use connectors in AI Agent Studio? • Document tool: You can upload specific documents to be used by the AI agent, and provide natural language instructions on how the agent should use these documents. The agent can then search for information in the documents to provide a more exact answer to a user's question. • Email: This tool can access the email client to send emails that include summaries of interactions or details pulled from a knowledge store. • Deep link: A deep link will send a user directly to the part of Fusion Applications where they can update underlying information. For example, if the user moved and wants to update their home

Task	Details
	address in the HR system, a deep link can quickly route the user to the page for making that update. - External REST: You can connect to internal and external SaaS applications or public APIs using External REST tool. To connect to internal and third-party services, add the authorization parameters. For example, you can retrieve the real-time weather information for a specific location by creating an External REST tool that connects to the appropriate external API. - MCP: You can use the Model Context Protocol (MCP) tool and securely connect to external MCP servers, without building additional REST wrappers or plugin logic.
Define topics	Topics define the focus of the agent to a specific area of expertise. They are prompts that you can reuse across multiple systems or summarization prompts. For example, within a Benefits Administrator agent, we might define topics such as health policy coverage, vision policy coverage, and benefits enrollment. Use Topics to efficiently streamline your interactions with the agent. - Specify the instructions that help the agent decide which tools to use. - Enable the agent to better understand user intent by letting it identify and select the relevant topic based on the user's question. - Give each topic a clear, specific name, and include natural language instructions to ensure it's used correctly. You can reuse topics across agents.
Define credentials	To enable access to more services, you need to provide the necessary connection credentials. You can add credentials for this artifact: Custom LLM: In addition to the LLMs provided by Oracle, you can use other LLMs you've access to. You can add credentials for your LLM, and select it while creating your agent, node, or agent team.
Build new agents	Define the capabilities and scope of the agent. - Agent name - Product area the agent will work in - Natural language instructions to allow the agentic flow, or other agents, to understand the capabilities of this particular agent - Tools and topics the agent will need to use. In addition to the tools and topics you created, you can also add the predefined ones. In the Tools and Topics tabs, the predefined tools and topics are indicated by the  icon.
Add a user (human) in the loop	If required, add an approval step for some actions that your AI agent will perform. The review step can be added at any point in the process for oversight and control over key actions, such as sending an email or updating a record.
Build the agent team	Create an agent team and add agents and other artifacts to it. Types of agent teams are: - Supervisor: A supervisor agent manages other agents and artifacts in the agentic flow. - Workflow: An agentic flow that does the tasks in a predetermined order. Agents and artifacts are added as nodes in the workflow, and each node performs a defined function, for example,

Task	Details
	extracting data, calling a business object, running an LLM, or sending an email. The node then passes its output to the next step. Note: You can also add predefined agents to your agent team. In the Agents tab, the predefined agents are indicated by the  icon.
Test the agents	Make sure to test the agent before deploying to production. Ask a test query, and determine the accuracy and relevance of the agent's response. You can also see the instructions the agent is following, and the actions the agent has taken to arrive at the response.
Deploy the agent team	After defining and testing your agent team, you can deploy it directly from AI Agent Studio. • Embed the agent conversation chat experience into any website or application. • Trigger the agent from an external resource using Webhooks, or seamlessly embed the chat experience into HTML and React web pages.

Migrate RAG Agents to AI Agent Studio

If you've previously created any RAG agents in Fusion Applications, we recommend replacing your existing agent with a new one created in AI Agent Studio. For more information, see [Migrate Document Tools of RAG Agents](#).

Add External REST Tool

You can enable your agents to securely connect to internal or external SaaS applications and public APIs by adding External REST tools in AI Agent Studio. Make sure you have a role with the Create and Edit Backends for Visual Builder Studio (ORA_FND_TRAP_PRIV) privilege assigned to it.

1. Go to **AI Agent Studio**.
2. From the Tools tab, click **Add**.
3. Add a new tool of type **External REST**.
4. Enter the name, code, and description for the tool, and select the appropriate family and product.
5. Select **Require Human Approval** to ensure that a person reviews and approves any action before the tool runs.
 - a. Message: Enter instructions for the reviewer.
 - b. Actions: Select the required action.
6. In the Authorization subtab, add connection details for the third-party service.
 - a. Instance URL: Enter the base domain of the external API, with no paths, parameters, or trailing slashes. For example, <https://api.weather.gov> is correct.
 - b. Authentication: Define how your agents will authenticate with the external system.
 - c. Description: Provide details about the tool, including what the tool does, and when the agent must call the tool.

7. In the Functions subtab, provide endpoint details.
 - a. Name: Assign a unique name to this endpoint. For example, **getWeatherByCoordinates**.
 - b. Operation Type: Select the HTTP method required to process the REST call against the target endpoint.
 - c. Resource Path: Specify the relative resource path to be appended to the Instance URL. Don't include the protocol or base domain. For example, **/points/{latitude},{longitude}** is correct.
 - d. Description: Provide a clear summary of the endpoint's function and the specific task it performs within the workflow.
8. In the Parameters subtab, define each dynamic variable in the path, including data type and purpose. For example, you can define latitude as the first parameter and longitude as the second.
9. In the Sample Queries subtab, list example user queries that should trigger this tool.
10. In the Headers subtab, specify any required metadata and security credentials for the external server.

The tool now appears in the Tools tab, and you can select it when creating agents, nodes, or agent teams.

Add MCP Tool

Using Model Context Protocol (MCP) tool, you can securely connect to external MCP servers, and use their capabilities inside agents and nodes, without building additional REST wrappers or plugin logic.

1. Go to **AI Agent Studio** and open the Tools tab.
2. Add a new tool of type **MCP**.
3. Enter the name, code, and description for the tool, and select the appropriate family and product.
4. Add MCP server connection details.
 - o Instance URL: Enter the endpoint URL of the MCP server your agent will connect to.
 - o Transport Type: Select the communication method:
 - Server Sent Events (SSE): Enables real-time, one-way streaming of data from a server to a client.
 - StreamableHTTP: Enables servers to independently handle multiple client connections using the HTTP POST and GET requests.
 - o Credential Type: Select the authentication method required by the MCP server.
5. Update the details and create the tool.

Your tool now appears in the Tools tab, and you can select it when creating agents, nodes, or agent teams.

Add Your LLM

In addition to the LLMs provided by Oracle that are available for selection when creating an agent, node, or agent team, you can also use other LLMs by adding the credentials for it. These are the providers you can choose to add.

- OpenAI (through Microsoft Azure)
- Google Gemini (through Google Vertex AI)
- Google Gemini (through direct API)
- Anthropic (through Google Vertex AI)

To add the LLM, do these steps:

1. Go to **AI Agent Studio** and open the Credentials tab.
2. In the LLM subtab, add the API key.
3. Add the details for the provider and save it.
 - Model: Choose the model you need from the available options.
 - API Key: Add the API key for the selected model.
 - Basic URL: Enter the URL to send LLM requests to.

Your LLM now appears in the LLM tab, and you can select it when creating agents, nodes, or agent teams.

Add Email Accounts

To use an email trigger or to use a human approver in your workflow, you must register an email account with AI Agent Studio.

1. Go to **AI Agent Studio** and open the Credentials tab.
2. In the **Email Accounts** subtab, select **Add Account**.
3. Add either a **Microsoft** or **Google** email account.
4. Select **Inbound** or **Approver** to designate the purpose of the email account.
 - Inbound: Receives emails used to initiate a workflow
 - Approver: Receives emails to either approve or reject requests

These email account designations are configured in the same way and they differ only in where they appear in the UI. You can't use the same email account for both the inbound and approver accounts.

5. Enter the account information.

Field	Description
Email Folder	Folder name in the email account that's used for monitoring requests
Polling Interval	Interval at which AI Agent Studio checks the folder in the email account

Microsoft and Google accounts require different authorization details to manage the account. You can obtain the required information from the vendor's cloud console.

Account	Authorization Fields
Microsoft	◦ Tenant ID ◦ Client ID ◦ Client Secret
Google	◦ Service Account Email

Account	Authorization Fields
	Private Key

Create Documents to Add to Document Tool

You can upload files to your document tools for agents to query from. Each document is a logical collection of attachments (files). You can add up to 500 files (attachments) to a single document, and each attachment can consist of up to 300 pages. Make sure that the total size of all attachments doesn't exceed 2GB. You can also upload multimodal files.

These file types are supported for attachments:

- PDF-TAGGED / SCANNED
- HTML
- JSON
- MARKDOWN
- XML
- TXT
- DOCX
- DOCX-MULTIMODAL
- XLSX
- CSV
- PPTX
- JPG
- PNG
- ZIP

The content and structure of a document are key aspects to fetch the correct answer for your query. Here are some best practices for attachments:

- Make sure that a single document doesn't contain more than five images or scanned pages. This applies to the following types of files:
 - Fully scanned PDFs
 - Image-only documents
 - Multimodal files combining text and images.

If your content includes more than five images, split it into multiple smaller documents before uploading.

- Avoid heavy files. To ensure reliable processing and avoid latency or timeouts, make sure that your attachments follow these standards:
 - Avoid documents larger than 25 MB
 - Use image compression where possible
 - Prefer optimized PDFs over high-resolution raw scans

- Remove unnecessary pages, blank pages, or duplicated content before uploading

Document Structure

It's highly recommended to structure your document using headings, subheadings, and so on.

Headings must clearly describe the meaning of the content in the subsequent section. Each section (under a heading) must contain semantically related concepts. If you find that a section contains a mix of concepts, it's advisable to create sections. Also, the various heading levels must be clearly distinguishable. For example, use font size 12 for Heading 1, font size 10 for Heading 2, font size 8 for Heading 3, and so on.

A well-constructed document structure enables the agent to split the document into semantically meaningful chunks. This improves the quality of the generated answers.

Here's a comparison of two approaches to structuring a document:

Approach 1	Approach 2
Scope of Plan Coverage This plan covers Employees of ACME. An "Employee" shall mean a common law Employee of ACME, Inc. (ACME), ACME International Corporation and ACME Software Technology GmbH. Independent contractors and/or "leased workers" engaged by a staff leasing company aren't Employees for purposes of the Plan and therefore aren't eligible to participate in the Plan. Full and part time employees are covered as follows – You might participate in one of the medical, dental, vision, life/accidental death and dismemberment (AD&D), long term disability (LTD), and health care and dependent care flexible spending accounts (FSA) benefits offered under the Plan when you meet one of the eligibility requirements listed below. - Regular Full-Time Employee on the ACME U.S. payroll scheduled to work 30 or more hours per week. - Regular Part-Time Employee on the ACME U.S. payroll scheduled to work 20-29 hours per week. - Interns and temporary Employees aren't eligible to participate in the	Scope of Plan Coverage This plan covers Employees of ACME and their dependents. Eligible Employees An "Employee" shall mean a common law Employee of ACME, Inc. (ACME), ACME International Corporation and ACME Software Technology GmbH. Independent contractors and/or "leased workers" engaged by a staff leasing company aren't Employees for purposes of the Plan and therefore aren't eligible to participate in the Plan. FULL & PART TIME EMPLOYEES You might participate in one of the medical, dental, vision, life/accidental death and dismemberment (AD&D), long term disability (LTD), and health care and dependent care flexible spending accounts (FSA) benefits offered under the Plan when you meet one of the eligibility requirements listed below. - Regular Full-Time Employee on the ACME U.S. payroll scheduled to work 30 or more hours per week. - Regular Part-Time Employee on the ACME U.S. payroll scheduled to work 20-29 hours per week. - Interns and temporary Employees aren't eligible to participate in the Plan as they're not regular Employees of ACME. INPATRIATES An Inpatriate (and eligible Dependents) residing in the United States for 90+ days and scheduled to work 30 -- 40 hours/week for an ACME Entity will be enrolled, upon notifying , in the following benefits: - ACME Premium PPO Medical - Dental Plan II - Vision Service Plan (VSP) Option 1

Approach 1	Approach 2
Plan as they're not regular Employees of ACME. INPATRIATES are covered as follows – An Inpatriate (and eligible Dependents) residing in the United States for 90+ days and scheduled to work 30 -- 40 hours/ week for an ACME Entity will be enrolled, upon notifying , in the following benefits: • ACME Premium PPO Medical • Dental Plan II • Vision Service Plan (VSP) Option 1 • An Inpatriate (and eligible Dependents) are not eligible for Life/Accidental Death and Dismemberment (AD&D) and Long-Term Disability (LTD) coverage. Your eligible dependents may be covered by the medical, dental, vision, and life insurance options under the Plan when Your Dependent meets one of the eligibility requirements listed in this section. NOTE: Your Dependents may not enroll in a particular benefit option offered under the Plan unless You are also enrolled	• An Inpatriate (and eligible Dependents) are not eligible for Life/Accidental Death and Dismemberment (AD&D) and Long-Term Disability (LTD) coverage. ELIGIBLE DEPENDENTS Your eligible Dependents may be covered by the medical, dental, vision, and life insurance options under the Plan when Your Dependent meets one of the eligibility requirements listed in this section. NOTE: Your Dependents may not enroll in a particular benefit option offered under the Plan unless You are also enrolled.

Approach 2 is highly recommended. Breaking the document into semantically related content, organized by the various coverage categories, enables the agent to provide more relevant answers to questions about benefits coverage. Also, note that the various heading levels are visually distinguishable.

Add Connector Tool

You can use the Connector tool to connect your AI agents to various data sources such as SharePoint, web pages, and other custom content sources. For more information, see *How do I use connectors in AI Agent Studio?*

4 Use Cases

Use Cases for Agents of Type Supervisor

Single Agent with Multiple Workers

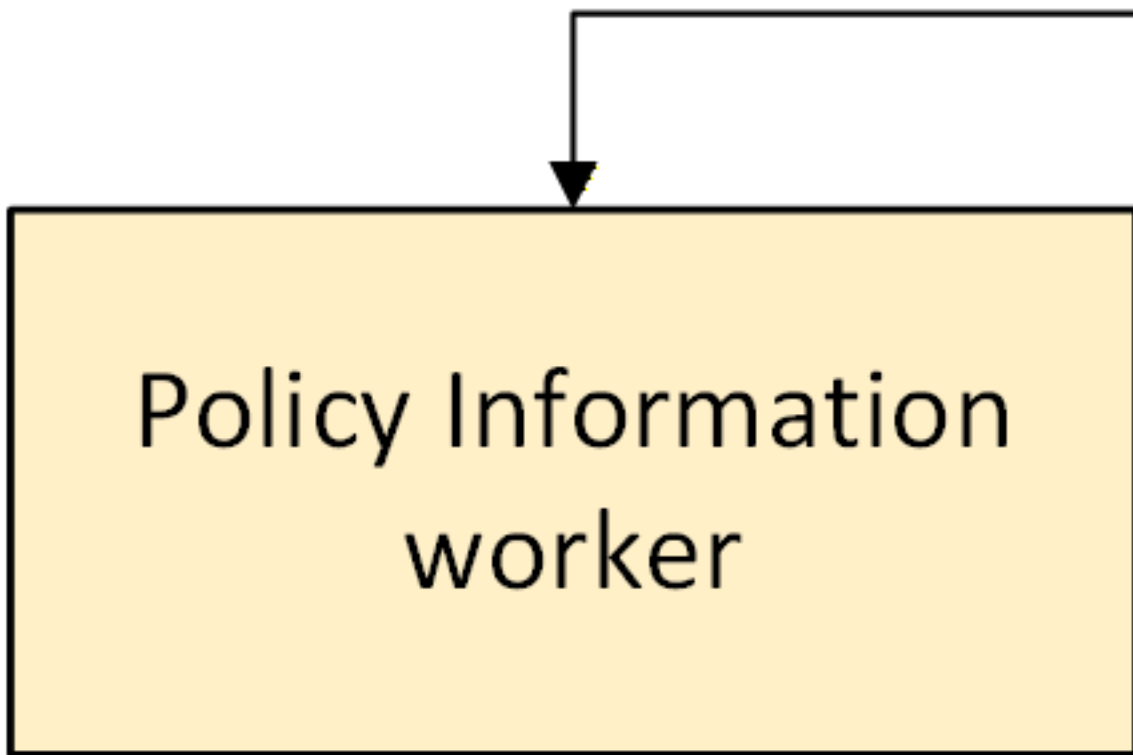
A worker is a specialized agent designed to perform a narrowly defined task or skill, and act as a modular building block for supervisor and workflow agents. Each worker is configured with specific instructions and has access to the tools required to perform that role effectively. Workers don't make high-level decisions or independently orchestrate tasks; instead, they perform well-scoped operations with precision.

Workers are designed to operate as part of a single or multi-agent system, and not as standalone entities. Workers are typically embedded within supervisor or workflow agents that coordinate their usage. This includes deciding when and how to initiate workers, pass appropriate context, and integrate the output from the worker into the overall task operation.

For example, a supervisor agent, based on the problem context, might dynamically select and initiate workers in a non-deterministic manner. A workflow agent might call workers in a predefined, deterministic sequence. In both cases, workers remain task-focused and don't control the broader flow of operations.

In this pattern, a Benefits Advisor acts as a central agent, helping employees understand and make the most of their benefits packages. The agent coordinates several specialized workers to provide a personalized support for medical, dental, and vision coverage.

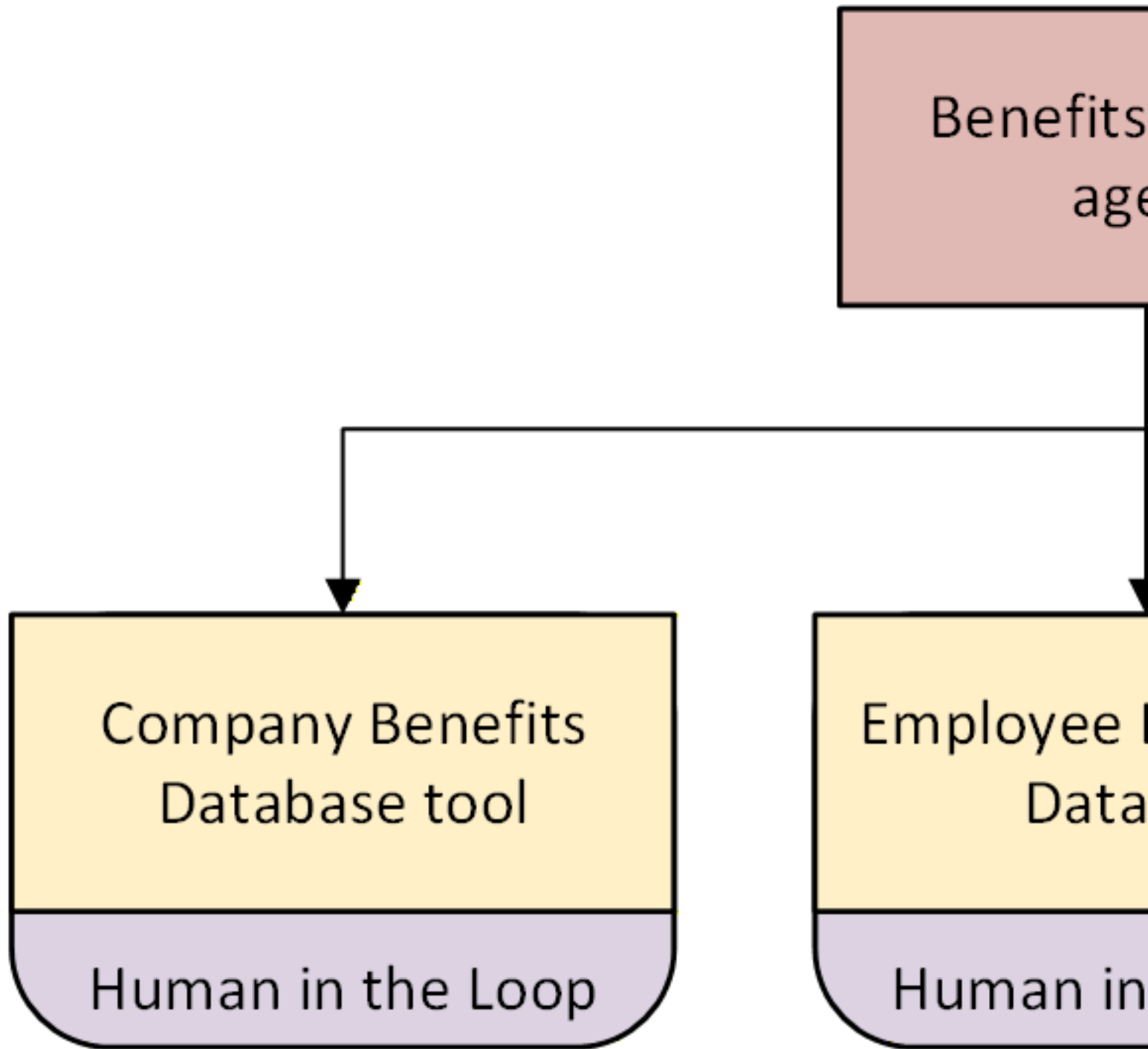
- The **Benefits Advisor** agent serves as the initial contact, helping the employee understand their benefits, and answers any preliminary questions.
- The **Policy Information** worker provides the employee with detailed information on available medical, dental, and vision benefits.
- The **Personal Enrollment** worker uses the employee's enrollment details to offer tailored advice on how they can optimize their benefits package.
- The **Eligibility and Assistance** worker answers queries regarding eligibility, coverage, helping to resolve issues related to benefits.



Single Agent with Multiple Tools and a Human in the Loop

This pattern shows the Benefits Advisor as a central agent that helps employees understand and optimize their benefits. This agent coordinates specialized workers to provide personalized insights on medical, dental, and vision coverage.

- The **Benefits Advisor** agent starts the interaction with a friendly introduction and asks for the employee's needs regarding benefits.
- The **Company Benefits Database** tool retrieves information on the company's benefits policies, eligibility, and plan details.
- The **Employee Enrollment Data** tool uses the employee's personal enrollment information to provide tailored advice, helping them understand and optimize their coverage.
- The **Benefits Optimizer** tool analyzes the employee's selections and offers personalized suggestions for improvement.
- The *Human-in-the-Loop* flag determines whether a user must provide explicit approval before the agent retrieves sensitive data, such as salary or personal information.



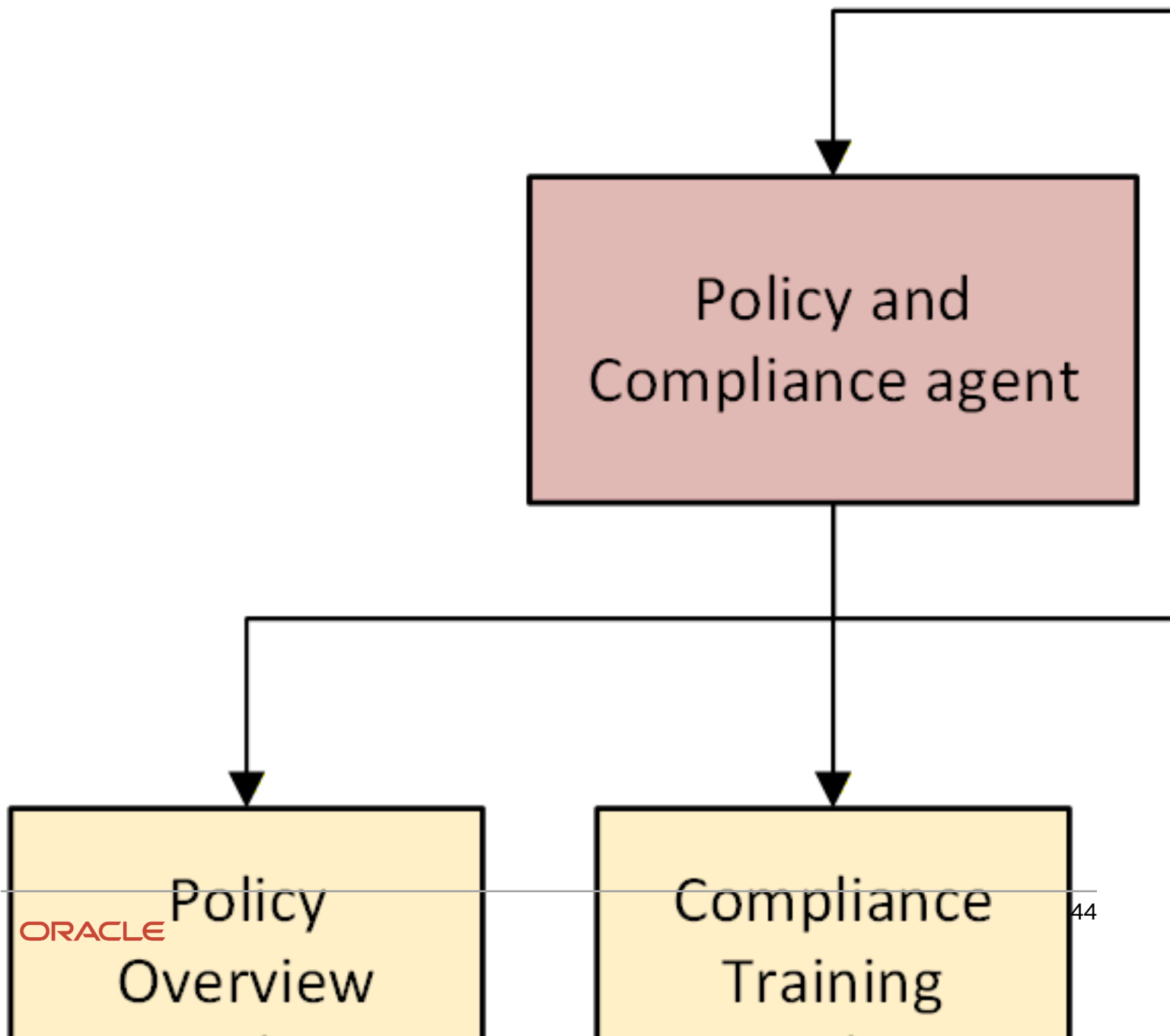
Multiple Agents With Multiple Tools and Delegations

In this pattern, the Central Onboarding Assistant agent acts as a supervisor, guiding new employees through their onboarding process. The supervisor agent delegates tasks to specialized agents, each focused on a specific aspect of onboarding, supported by dedicated workers.

Agent	Description
Policy and Compliance Agent	This agent helps employees understand and comply with company policies, and is supported by these workers. • Policy Overview worker: Shares key policies like leave, dress code, and workplace behavior. • Compliance Training worker: Provides access to mandatory training. For example, safety and social media policies. • Document Management worker: Ensures the employee completes and submits all required forms, such as tax forms and nondisclosure agreements.
Cultural Integration Agent	This agent helps employees adapt to the company's culture and connect with teams, and is supported by these workers. • Company Values worker: Explains the vision, mission, and goals of the organization. • Community Introduction worker: Connects employees with internal groups, events, and networking opportunities. • Workplace Norms worker: Provides guidance about unwritten rules, such as communication etiquette and meeting culture.
Tools and Resources Agent	This agent ensures that employees have access to tools, systems, and knowledge required for their job, and is supported by these workers. • Account Setup worker: Helps employees set up email, communication tools, and HR platforms. • Resource Guide worker: Provides access to handbooks, FAQs, and internal documentation. • Technical Support worker: Troubleshoots issues with tools or platforms.

Process

- The Central Onboarding Assistant agent acts as the entry point that gathers basic information about the employee's role and specific needs, and delegates tasks to specialized agents based on the query.
- The Policy and Compliance agent handles queries, or tasks, related to company rules and legal documentation, such as, "What's the process for requesting leave?"
- The Cultural Integration agent handles queries that focus on helping employees feel part of the team and aligned with the company values, such as, "How do I join company events?"
- The Tools and Resources agent handles queries about gaining access to systems and essential knowledge, such as, "How do I log into the time-tracking tool?"



Product-Specific Use Case Examples

Oracle Fusion Cloud CX

Agent Type	Key Characteristics	When to Use	Example
Customer service agent	- Interactive - Rule-based - Escalation-friendly - Customer-focused	- Repetitive, high-volume queries - Customer support - Query resolution	Chatbot for customer support agent answers FAQs and resolves issues.
Learning agent	- Adaptive - Feedback-driven - Continuously improves - Explainable	- Adaptive systems - Chatbot improvement - Demand forecasting refinement	Chatbot improvement agent learns from customer feedback to improve responses.

Oracle Fusion Cloud HCM

Agent Type	Key Characteristics	When to Use	Example
Copilots	- Collaborative - Proactive - Decision-making - Domain-specific	- Complex tasks requiring expertise - Draft documents - Input validation	Hiring copilot agent helps to build job descriptions and suggests interview questions.
Document referencing agent	- Knowledge-intensive - Accurate - Up-to-date - Query-driven	- Knowledge-intensive tasks - Answers to policy or FAQ questions - Compliance checks	Benefits agent answers employee questions about benefits.
Evaluating agent	- Assessment-focused - Criteria-based - Actionable insights - Feedback-driven	- Performance evaluation - Candidate ranking - Feedback analysis	Candidate evaluation agent ranks candidates based on interview performance.
Scheduling agent	- Task-oriented - Reactive - Limited decision-making - User-centric	- Routine task scheduling - Calendar management - Appointment booking	Interview scheduling agent coordinates availability between candidates and hiring managers.
Task automation agent	- Repetitive task handler - Efficient - Error-resistant	- Repetitive, high-volume tasks - Payroll processing	Payroll processing agent automates time sheet collection and payslip generation.

Agent Type	Key Characteristics	When to Use	Example
	Trigger-based	Form generation	

Oracle Fusion Cloud SCM

Agent Type	Key Characteristics	When to Use	Example
Collaborative workflow agent	- Multi-stakeholder - Real-time coordination - Role-based task allocation - Communication-driven	- Complex processes requiring collaboration - Supplier collaboration - Crisis management	Supplier collaboration agent coordinates with procurement, supplier, and warehouse bots to work together.
Decision-making agent	- Data-driven - Recommendation-focused - Transparent - Adaptive	- Data-driven tasks - Supplier evaluation - Candidate shortlisting	Supplier evaluation agent evaluates suppliers based on cost and quality.
Goal-seeking agent	- Optimization-focused - Real-time adjustments - Objective-driven - Explainability	- Resource allocation - Route planning - Workforce scheduling	Route optimization agent plans optimal delivery routes.
Monitoring agent	- Proactive - Alert-driven - Data-intensive - Response-oriented	- Proactive management - Inventory monitoring - Social media tracking	Inventory monitoring agent tracks in-stock levels and alerts for replenishment.

5 Create AI Agents

Create AI Agents Using Preconfigured Templates

You can create and deploy AI agents using preconfigured templates available in these two subtabs:

- **AI Agents:** Access the preconfigured templates built and provided by Oracle. When using a preconfigured agent team template, the artifacts such as agents, tools, and topics aren't directly editable. To change these artifacts, you create a copy of the artifact and add the copy to the agent team.
- **Marketplace:** Access the preconfigured templates built and provided by Oracle partners. View template details using the ⓘ icon. To create a copy of any template, click **Create**.

To use preconfigured agent team templates from the AI Agents subtab:

1. Go to **AI Agent Studio**.
2. Select **Copy Template** from the required agent team, enter a suffix for the new agent team and then select **Continue**.

Tip: You can choose the **Use Template** option to provide all details for the new agent team before creating it.

3. Your copy of the template is created. If needed, select **Agent Team Settings** (⚙️) to edit the agent team details, and select **Update**.

Note: You can add expressions to fields using **Insert Expression** ⓘ. For more information, see *Expressions in AI Agent Studio*.

Details Tab

Field	Description
Family	Select the family to which this agent team belongs.
Product	Select the product within the family to which this agent team belongs.
Maximum Interactions	Indicate the number of times an agent within this agent team can interact with the topics and tools assigned to it.

LLM Tab

Field	Description
Provider	Choose to use the default model or select a model. When using a custom model, specify the model properties.

Security Tab

Field	Description
Add	Select the roles which will have access to this agent team.

Questions Tab

Field	Description
Starter Questions	Enter initial questions for the agent team.
Follow-up Questions	Enable this to indicate that the agent team can ask follow-up questions based on the user's conversation history.
Prompt	Enter the prompt to be used for the follow-up questions. For example: using {chatHistory} generate 3 follow up questions in json format. JSON Schema format mentioned below. Remove the ``json markdown from the output. Here is the JSON Schema format the output should adhere to: [{"question": "<put first question generated here>"}, {"question": "<put second question generated here>"}, {"question": "<put third question generated here>"}]
Insert Expression	Add additional variables to the prompt. For example, to add the current system date to your prompt, select the Current Date Time option.

Chat Experience Tab

You can enable users to upload attachments while interacting with agents. Users can upload up to five files, with a total combined size limit of 50 MB. These are the supported file types users can upload to the chat: PDF-Tagged / Scanned, HTML, JSON, MARKDOWN, XML, TXT, DOCX, DOCX-Multimodal, XLSX, CSV, PPTX, JPG, PNG, and ZIP.

Field	Description
Enable file upload	Select this option to allow users to upload files from local storage.

Field	Description
Enable third-party upload	Select this option to allow users to upload files from connected cloud storage accounts. To enable this functionality, add credentials for your provider by navigating to the Credentials tab in AI Agent Studio and completing the Chat Experience setup.

Note: To make sure your agent can process attachments, add the common tool of **Runtime File Processor** type, **MultiFileProcessor**.

For more information, see [How can I attach files while chatting with an agent?](#)

Output Tab

Define the overall structure of the agent's output using JSON schema, to specify the exact output.

Field	Description
Specification Mode	Select this mode to directly modify the JSON schema for the output.
Simple Mode	Select this mode to define the output values and types. The corresponding JSON schema will be generated automatically and displayed in the specification mode for any further changes.

- Continue to edit and add details for the agent team. If you want to edit the artifacts in the agent team including agents, tools, and topics, you can create a copy of the artifact and edit it. After editing, remove the original artifact from the agent team, and then add the copy you created. See [Can I edit a preconfigured agent team?](#)

Note: The agents and other artifacts included within an agent template are optimized to provide the best usage of the agent team. We recommend not to change the basic functionality of the artifacts because that might impact your agent team's performance.

- If needed, use  to test the agent team. For any required fine-tuning, you can edit the agent team using .
- Publish your agent team.

Users can view the published agents from the AI Agents page. To access this page, go to Me → Quick Actions → Show More → AI Agent Studio → AI Agents. If you're using **Home with Ask Oracle** as your home page layout, search for the **AI Agents** page.

Note: Make sure that your users have access to interact with the AI agents. For information, see [How can I give users access to AI agents?](#).

Create Custom AI Agents of Type Supervisor

The supervisor agent orchestrates and plans tasks for your agent team in Supervisor flows. It directs other agents on how to interact and generate responses. You can configure supervisor agents as a single agent or within a multiple agent pattern. To see some of the common supervisor agent patterns, go to [Use Cases for Agents of Type Supervisor](#).

Tip: For recommendations about improving your agent's efficiency, see *How do I make agents respond faster?*.

1. Go to **Navigator > Tools > AI Agent Studio**.
2. From the Tools tab, add the required tools.

For example, to create an HR benefits administrator agent that can answer questions related to medical, vision, dental, retirement, and stock plans, these are some of the tools needed.

- Document tools, with the organization's health and financial benefits documents.

Note: Before you add any document tools to an agent, make sure you've set the status of the document tool as **Ready to Publish**.

- Business object tools, to fetch the employee enrollments data.
- Calculator tool, to check balance amounts and percentages.

3. From the Topics tab, add the required topics. Make sure you include instructions about these key areas.

- What the topic is about
- Tools to use with the topic
- Examples of possible questions
- Any guidelines and guardrails

4. From the Agents tab, add an agent.

- In **Maximum Interactions** field, specify the number of times the agent can interact with the topics and tools assigned to it.
- Describe the persona and role of the agent, including the tone to use.
- Add the prompt for the agent, and select any variables from **Insert Expression**.

5. Assign the required tools and topics to the agent, and create it.

6. From the **Agent Teams** tab, add an agent team and provide details for the new agent team.

Note: You can add expressions to fields using **Insert Expression** . For more information, see *Expressions in AI Agent Studio*.

- **Details Tab**

Field	Description
Family	Select the family to which this agent team belongs.
Product	Select the product within the family to which this agent team belongs.
Type	Select Supervisor to create an agent team where one agent acts as supervisor and manages other agents and artifacts.

Field	Description
Maximum Interactions	Indicate the number of times an agent within this agent team can interact with the topics and tools assigned to it.

- **LLM Tab**

Field	Description
Provider	Choose to use the default model or select a model. When using a custom model, specify the model properties.

- **Security Tab**

Field	Description
Add	Select the roles which will have access to this agent team.

- **Questions Tab**

Field	Description
Starter Questions	Enter initial questions for the agent team.
Follow-up Questions	Enable this to indicate that the agent team can ask follow-up questions based on the user's conversation history.
Prompt	Enter the prompt to be used for the follow-up questions. For example: using {chatHistory} generate 3 follow up questions in json format. JSON Schema format mentioned below. Remove the ``json markdown from the output. Here's the JSON Schema format the output should adhere to: [{"question": "<put first question generated here>"}, {"question": "<put second question generated here>"}, {"question": "<put third question generated here>"}]
Insert Expression	Add additional variables to the prompt. For example, to add the current system date to your prompt, select the Current Date Time option.

- **Chat Experience Tab**

You can enable users to upload attachments while interacting with agents. Users can upload up to five files, with a total combined size limit of 50 MB. These are the supported file types users can upload to the

chat: PDF-Tagged / Scanned, HTML, JSON, MARKDOWN, XML, TXT, DOCX, DOCX-Multimodal, XLSX, CSV, PPTX, JPG, PNG, and ZIP.

Field	Description
Enable file upload	Select this option to allow users to upload files from local storage.
Enable third-party upload	Select this option to allow users to upload files from connected cloud storage accounts. To enable this functionality, add credentials for your provider by navigating to the Credentials tab in AI Agent Studio and completing the Chat Experience setup.

Note: To make sure your agent can process attachments, add the common tool of **Runtime File Processor** type, **MultiFileProcessor**.

For more information, see [How can I attach files while chatting with an agent?](#)

- **Output Tab**

Define the overall structure of the agent's output using JSON schema, to specify the exact output.

Field	Description
Specification Mode	Select this mode to directly modify the JSON schema for the output.
Simple Mode	Select this mode to define the output values and types. The corresponding JSON schema will be generated automatically and displayed in the specification mode for any further changes.

7. Select **Create**.
8. Select  and click **New Supervisor Agent**.
9. Enter details for the supervisor agent and create it. This supervisor agent is specific only to the agent team in which it's created, and can't be reused.
10. Select  and add any existing agents as worker agents, or create worker agents.

When you create worker agents from within the agent team, those agents are available only for this agent team.

11. Add any needed artifacts such as tools and topics.
12. If needed, use  to test the agent team. For any required fine-tuning, you can edit the agent team using .
13. Publish your agent team.

Users can view the published agents from the AI Agents page. To access this page, go to Me → Quick Actions → Show More → AI Agent Studio → AI Agents. If you're using **Home with Ask Oracle** as your home page layout, search for the **AI Agents** page.

Note: Make sure that your users have access to interact with the AI agents. For information, see [How can I give users access to AI agents?](#).

Create Custom AI Agents of Type Workflow

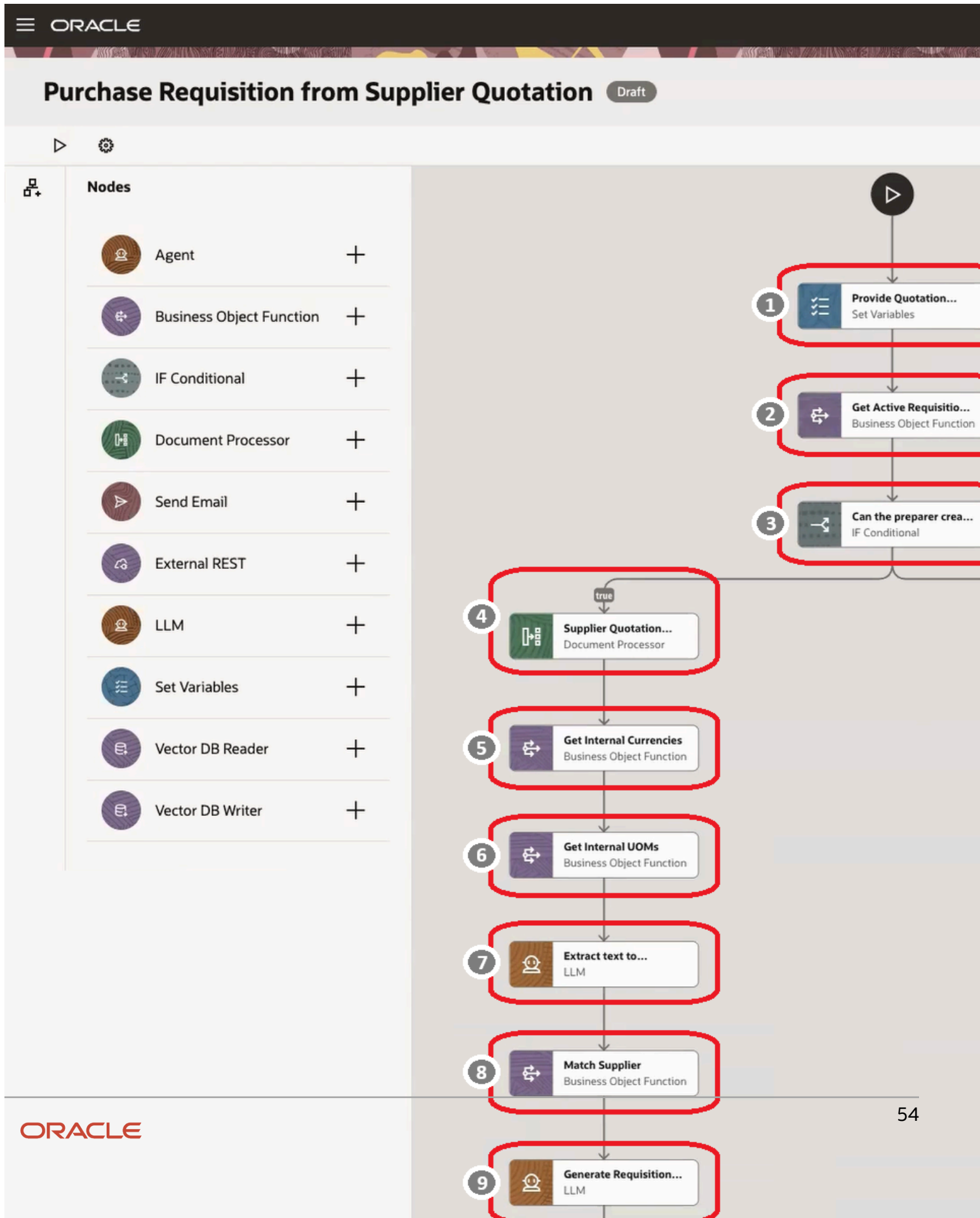
Agent teams of type Workflow consist of a sequence of nodes used for deterministic, rule-based orchestration of tasks in which every step is preconfigured. Workflows are ideal for scenarios where compliance, repeatability, and governance are essential. The workflow tasks are represented as a connected sequence of nodes. Each node within the workflow performs a defined function, such as extracting data, calling a business object function, running an LLM, or sending an email. The output from each node is seamlessly passed to the following step, ensuring a controlled and efficient process flow.

When designing your workflow, keep these process breakdown steps in mind:

- Model the manual workflow. Start from a high-level process and break it down into smaller, automatable steps.
- Identify task dependencies and group nodes by type. For example, distinguish between back-end logic tasks and those driven by LLMs. Doing so helps to streamline operation and minimize unnecessary LLM usage.
- For each task, define the node type, required input values, expected outputs, and any transformation logic needed.
- For recommendations about improving your agent's efficiency, see [How do I make agents respond faster?](#)

Let's look at an example to illustrate how a workflow operates in practice. Suppose an employee needs to buy a product or service and contacts a supplier to request a quotation. The supplier responds with a quotation, which might vary in format and content, but typically includes key information such as the seller's details, a list of items, and the total quoted price. The quotation is processed internally, and a requisition order is created.

Here's an example workflow for this scenario, created in AI Agent Studio.



Callout Number	Node Name	Node Type	Description
1	Provide Quotation	Set variables	Sets the values for the variables.
2	Get Active Requisition	Business Object Function	Retrieves the default preferences of the requester.
3	Can the preparer create a requisition?	IF conditional	Validates whether the user has permission to create a requisition. - If true, the workflow continues to the following nodes. - If false, the workflow indicates that a requisition wasn't created, sends a failure email.
4	Supplier Quotation	Document processor	Downloads the document from the WebCenter Content server and parses the content.
5	Get Internal Currencies	Business Object Function	Pulls the valid internal currency value.
6	Get Internal UOM	Business Object Function	Pulls the valid internal units of measure value.
7	Extract text to LLM	LLM	Uses LLM to convert the unstructured text from upstream nodes to a structured JSON payload.
8	Match supplier	Business Object Function	Checks and validates the supplier information extracted earlier.
9	Generate requisition payload	LLM	Maps all gathered data from the document and database to generate a valid post requisition payload.
10	Create purchase requisition	Business Object Function	Calls the API to create the draft requisition. After this call, the requisition is saved to the database.
11	Is requisition created?	IF conditional	Checks if the requisition is successfully created. - If true, the workflow proceeds to the success confirmation node. - If false, the workflow proceeds to the send failure email node.
12	Success confirmation	LLM	Displays a message indicating the successful creation of the requisition.
13	Send failure email	Send email	Sends an email to the user with failure details.

Here are the steps to create your own workflow.

1. Go to **AI Agent Studio**.

2. From the **Agent Teams** tab, add an agent team and provide details for the new agent team.

Note: You can add expressions to fields using **Insert Expression**  . For more information, see *Expressions in AI Agent Studio*.

Details Tab

Field	Description
Family	Select the family to which this agent team belongs.
Product	Select the product within the family to which this agent team belongs.

Field	Description
Type	Select Workflow .

LLM Tab

Field	Description
Provider	Choose to use the default model or select a model. When using a custom model, specify the model properties.

Security Tab

Field	Description
Add	Select the roles which will have access to this agent team.

Triggers Tab

Create triggers to begin the workflow. Triggers define the conditions or events that start the workflow and identifies when and why it begins.

Field	Description
Type	Select the data type of the trigger.

For more information about setting up triggers, see *Triggers*.

Variables Tab

Define workflow-level variables, making them accessible to all nodes within the workflow. Variables are ideal for storing IDs, constants, or any values you need to share and use across different parts of the workflow.

Field	Description
Type	Select the data type of the variable.

Chat Experience Tab

You can enable users to upload attachments while interacting with agents. Users can upload up to five files, with a total combined size limit of 50 MB.

Note: To make sure your agent can process attachments, add a worker agent with the tool of **Runtime File Processor** type. For more information, see *Create Custom AI Agents of Type Supervisor*.

These are the supported file types users can upload to the chat: PDF-Tagged / Scanned, HTML, JSON, MARKDOWN, XML, TXT, DOCX, DOCX-Multimodal, XLSX, CSV, PPTX, JPG, PNG, and ZIP.

Field	Description
Enable file upload	Select this option to allow users to upload files from local storage.
Enable third-party upload	Select this option to allow users to upload files from connected cloud storage accounts. To enable this functionality, add credentials for your provider by navigating to the Credentials tab in AI Agent Studio and completing the Chat Experience setup.

For more information, see [How can I attach files while chatting with an agent?](#)

Output Tab

Define the overall structure of the workflow's output using JSON schema, to specify exactly what you'll receive when the workflow is completed. If no schema is specified, the output type defaults to that of the individual node. When a schema is provided, the system validates the outputs against the schema and enables auto-complete for those fields.

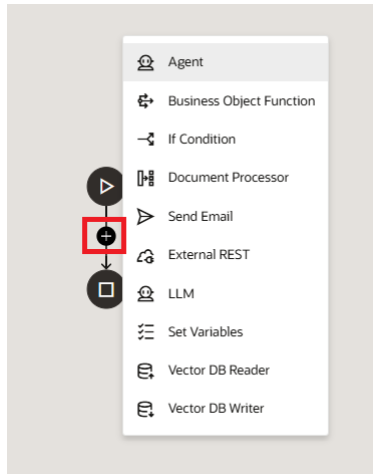
Field	Description
Specification Mode	Select this mode to directly change the JSON schema for the output.
Simple Mode	Use this mode to define output values and types. The corresponding JSON schema will be generated automatically and displayed in the specification mode for you to edit.

Error Handling Tab

Specify where to send an email when a workflow instance runs into a permanent error. You can also use context expressions in these fields for more clarity.

3. Click **Create**.

4. Add nodes to the workflow by hovering over the line between the start and end of the workflow. Click  to choose the type of node you want to add.



Tips

- o Connect nodes in sequence to control how input values are received, processed, and forwarded throughout the workflow.
 - o You can insert new nodes before, after, or between existing nodes as needed.
 - o For more information about Vector nodes, see [Vector Write and Read Nodes](#).
 - o Once created, you can't rearrange the nodes. To change a node's position, delete it and create it again in the desired spot.
5. If needed, use  to test the agent team. For any required fine-tuning, you can edit the agent team using .
 6. Publish the agent team.

Users can view the published agents from the AI Agents page. To access this page, go to Me → Quick Actions → Show More → AI Agent Studio → AI Agents. If you're using **Home with Ask Oracle** as your home page layout, search for the **AI Agents** page.

Note: Make sure that your users have access to interact with the AI agents. For information, see [How can I give users access to AI agents?](#).

Vector Write and Read Nodes

To automate business processes effectively, your AI workflows require access to high-quality, well-structured, and reusable knowledge. You can use Vector Write and Read nodes in AI Agent Studio to store knowledge as embeddings and retrieve it intelligently with semantic search and metadata filters. By leveraging these nodes, you can design workflows that capture, manage, and reuse critical business knowledge across processes, thereby improving accuracy, scalability, and reliability in an enterprise environment.

Here's what you need to know to effectively use Vector Write and Read nodes in your workflow agents, along with some best practices for organizing and managing enterprise knowledge for the best possible retrievals.

Operational Limits and Defaults

To ensure efficient vector operations, keep node entries within specified sizing and retrieval limits.

Parameter	Recommendation
Document size per entry	50KB
Number of retrievals	15 retrievals by default, consisting of 10 semantic and five text
Maximum results per read	Up to five

Here are some sizing tips:

- Keep the number of retrievals to a minimal amount of relevant items.
- Don't store more than 500 objects per workflow and use strong metadata filters to improve precision.
- Apply a semantic match score threshold before using results. If the scores are low or have no value, implement a fallback path to a deterministic look-up or a human-in-the-loop intervention.
- Avoid noisy, raw, or sensitive data.
- Keep entries fresh with UPSERT or OVERWRITE for updates and DELETE for stale records.

Vector Write Node

Vector write nodes store high-value knowledge as embeddings for future semantic retrieval. These nodes are also referred to as vector store nodes.

When you create a node, you can configure all values as expressions that resolve dynamically at runtime. Here's an example of the node creation window. Each numbered field is described in detail in the following table.

New node

Vector DB Writer

Name

1

Required

Operation Type

Overwrite

4

▼

Index Name

5

i

Type {{ to start seeing context-sensitive expressions.

Content

6

i

Type {{ to start seeing context-sensitive expressions.

Content Type

7

i

Type {{ to start seeing context-sensitive expressions.

Document ID

8

i

Type {{ to start seeing context-sensitive expressions.

Callout Number	Field	Description	Example
1	Name	Design-time name for the node	Use descriptive names, like <code>WriteResolutionSummaryToVectorDB</code> , so the workflow is easy to understand.
2	Code	Programmatic identifier in the workflow schema	This field is auto-generated but can be changed to a user-defined one, such as <code>write_resolution_vector</code> . Use lowercase and underscores.
3	Error Handler	Fallback path on failure	Select an error branch or a dedicated handler node to manage failures cleanly.
4	Operation Type	Method of how the document is written to the index	Use <code>INSERT</code> for new entries, <code>OVERWRITE</code> to replace content, <code>UPSERT</code> to safely update or create, or <code>DELETE</code> to remove entries.
5	Index Name	Name of the vector index to write to	Choose an existing index or specify a new one, such as <code>support_ticket_summaries</code> or <code>product_docs_index</code> .
6	Content	Textual data to embed	Summarize this in a clean and structured manner. Avoid raw logs. Instead, use LLM-generated summaries, extracted facts, or curated knowledge.
7	Content Type	Type of content being indexed or embedded	Typical values include <code>json</code> or <code>text</code> .
8	Document ID	Unique identifier for this record	Use stable identifiers, such as <code>ticket_1123</code> or <code>customer_450_profile</code> .
9	Properties	Other optional metadata key-values	<code>{objectId:"a12345", region:"NA", severity:"High"}</code>

Best Practices for Building Vector Write Nodes

A vector store works best when you write clear, meaningful, metadata-rich knowledge while avoiding noise, duplication, and topic drift.

Best Practice	Description
Store only public, reusable knowledge	Exclude sensitive, permission-controlled, or private information, for example, personally identifiable information (PII), credentials, confidential documents, or any information requiring permission checks.
Focus on durable, high-value content	Store information that will have long-term value for future workflows, such as case resolutions, structured summaries, or validated insights. Avoid writing noisy, one-off, or ephemeral content, like raw chat logs or temporary instructions.
Clean and normalize before writing	Always standardize content, summarize documents, deduplicate overlaps, remove irrelevant details, and attach key metadata.

Best Practice	Description
Keep data up to date	Vector writes must reflect the latest truth. Stale entries decrease accuracy and lead to incorrect answers. • Set auto-refresh triggers to update vectors whenever business objects change (for example, case closures, policy updates, PO resolutions). • Use OVERWRITE or UPSERT to keep summaries up to date. • Delete outdated or unused entries to prevent stale data from accumulating. • Maintain freshness so workflows rely on correct, up-to-date knowledge.
Follow smart update practices	Update vectors in a controlled and intentional manner. Don't create more entries for each update, or store conflicting or outdated data. • Avoid creating new entries for every update. Refine existing data instead of duplicating. • Use UPSERT to update and improve existing summaries. • Use OVERWRITE for significant changes or new, higher-quality content. • Prune redundant or conflicting entries to keep the vector store clean. • Establish clear versioning strategies for policies, product versions, and configuration changes to prevent fragmentation.
Always include rich metadata	Tag with business object IDs, product numbers, version information, and other searchable attributes. Use a consistent metadata schema across agents and indexes.
Prevent data pollution	Check for existing knowledge to avoid duplicates and inconsistent tags. Regularly remove low-value content.
Exclude business object data	Duplicating this data in vector memory creates noise, redundancy, and version drift.

Vector Read Node

Vector read nodes retrieve the most relevant knowledge using semantic similarity and metadata filters.

Here's an overview of what the node creation window looks like. When creating a node, all values can be configured as expressions that will be resolved at runtime.

New node

Vector DB Reader

Name

1

Enter a unique name

Index Name

4

i Type {{ to start seeing context-sensitive expressions.

Query

5

i Type {{ to start seeing context-sensitive expressions.

Document ID

6

i Type {{ to start seeing context-sensitive expressions.

Data Fields

7

i Type {{ to start seeing context-sensitive expressions.

Filter Criteria

8

Callout Number	Field	Description	Example
1	Name	Name of the node	Use clear names like <code>RetrieveTicketContextFromVectorDB</code> to keep the workflow readable.
2	Code	Internal programmatic identifier	This field is auto-generated but can be changed to a user-defined one, like <code>read_ticket_context_vector</code> .
3	Error Handler	Defines what occurs on failure	Route to an error handler or fallback logic to avoid empty or invalid agent responses.
4	Index Name	Vector index to search	Choose the same index used by the writer, such as <code>support_ticket_summaries</code> or <code>employee_profile_index</code> .
5	Query	Natural language search query	Avoid vague queries and ask for intent-specific info, such as, "What troubleshooting steps were taken?"
6	Document ID	Unique identifier for this record.	Specify this when you want details tied to a known record, such as <code>ticket_12345_summary</code> .
7	Data Fields	Metadata fields to return	Specify an array of strings that will be used in the filter.
8	Filter Criteria	Logical filters applied before ranking	Specify this field to enable high precision and to constrain results from a retrieval, such as <code>product = Payroll, region = US, Or severity >= High</code> .
9	Maximum Results	Maximum number of ranked results returned	Specify an integer value for the max number of results to be displayed.

Best Practices for Building Vector Read Nodes

A vector read node works best when you write clear, specific queries that use metadata filters to ensure semantically correct responses, while preventing topic drift and hallucinations.

Best Practice	Description
Write clear, intent-driven queries	Ensure queries are specific and aligned with the exact workflow or agent task. For example, use precise prompts like "What resolved similar issues?" instead of vague ones like "Help me with this."
Use metadata filters for precision	Apply filters (for example, entity type, business object ID, and product number) to target only relevant content.
Validate retrieved results before using them	Check metadata and context before trusting or using results in workflows. This validation prevents incorrect answers, silent workflow failures, and LLM hallucinations grounded in bad evidence.

Best Practice	Description
Set an appropriate <code>maxResults</code> value	Set the <code>maxResults</code> field to return three to five, or more results. Don't depend on a single query result.
Add graceful fallback logic	Implement backup steps, such as business objects or API lookups, to handle cases where queries return no results.
Avoid broad or irrelevant retrieval	Scope queries to just what's needed. Skip irrelevant or noisy indexes to boost performance and accuracy of retrieved results.

Triggers

Workflow triggers are automated, predefined events or conditions that initiate a workflow when specific criteria are met.

Note: Triggers activate after the workflow is published.

Webhook

A webhook is an automated, event-driven mechanism for sending real-time data from an application over HTTP. You can configure a webhook trigger to call a workflow.

1. Open the workflow, select **Agent Team Settings** , and go to the **Triggers** tab.
2. Select **Add** for webhook trigger, and enter the input name and type.
3. Save and publish the workflow.

Email

You can configure an email trigger to call a workflow when an email is received in a monitored account. Each workflow can monitor only one unique inbound email account. You're not allowed to use the same inbound email account for another workflow.

To allow the workflow to be triggered by incoming emails:

1. Register the inbound email account in **AI Agent Studio**. See [Add Email Accounts](#).
2. Open the workflow, select **Agent Team Settings** , and go to the **Triggers** tab.
3. Select **Add** for email trigger and select your registered inbound email account.
4. Save and publish the workflow.

To verify that emails received in the selected inbound email account will now trigger this workflow:

1. Go to the **Monitoring and Evaluation** tab and open the **Activities** subtab.
2. Go to **Inbound Emails** to view the contents of the trigger email messages and verify the messages are initiating the workflow correctly.

Schedule

You can configure a scheduled trigger based on a defined time interval, to call a published workflow without user interaction or API calls. Scheduled triggers are useful when workflows need to run regularly or at fixed intervals.

Note: Make sure you have the FAI Batch Job Manager (*ORA_DR_FAI_BATCH_JOB_MANAGER_DUTY*) permission assigned in the Security Console.

To allow the workflow to be triggered at a scheduled date and time:

1. Open the workflow, select **Agent Team Settings** , and go to the **Triggers** tab.
2. Select **Add** for schedule triggers and choose the type of schedule to create.

Type	Description
Interval	Run the workflow between fixed time periods, for example every 100 minutes, every 10 hours, or every 5 days.
Recurrence	Run once at a specific time or repeatedly on calendar patterns, such as daily, weekly, or monthly. Enter a start time and an optional date and time for the recurrence to end.

3. Save and publish the workflow.

To verify that your scheduled trigger is starting the workflow:

1. Go to the **Monitoring and Evaluation** tab and open the **Activities** subtab.
2. Go to **Schedules** to view the entry and verify the scheduled trigger is initiating the workflow correctly.

Expressions in AI Agent Studio

You can use expressions to capture, store, and reuse values instead of deriving them each time, across various steps in your agent team, from AI Agent Studio. For example, you can use expressions to reuse extracted fields such as `personId`, intermediate node outputs, and so on. Expressions are different from static values. Static values can be anything you specify, whereas expressions are represented in the `{{any expression}}` format. You can directly enter expressions in the

required format, or you can add it from the **Insert Expression**  menu that's displayed with the fields. Here are the types of values allowed in expressions:

- String
- Date
- Integer
- Number
- Boolean
- Object
- Array

Variables Specific to Workflow Agent Teams

Expressions contain different types of variables that provide the necessary context. They've a defined set of scopes in workflow agent teams.

Scope	Description
Global	Values that can be referenced anywhere in the workflow. For example, Current Date , Trigger Type , and variables defined at the workflow level.
Node	Values such as the node input variable's value and the output of each node in the workflow.
Trigger	Every workflow can have multiple triggers and each trigger will have its own set of inputs.

Here are the types of variables available in expressions that are specific to agent teams of type Workflow:

- System Variables
- Workflow Variables
- User Variables
- Trigger Variables
- User-defined Variables
- Node Variables

System Variables

System Variables are predefined runtime read-only values provided by the application. These variables can be referenced in inputs, agent mappings, and prompts using the `{{context.$system. . . }}` syntax. For example, `{{context.$system.$currentDate}}`, `{{context.$system.$inputMessage}}`.

Scope	Expression	Description
Current Date	<code>{{context.\$system.\$currentDate}}</code>	Current system date (ISO format)
Current Date and Time	<code>{{context.\$system.\$currentDateTime}}</code>	Current system time stamp
Trigger Type	<code>{{context.\$system.\$triggerType}}</code>	Trigger that started the workflow
Input Message	<code>{{context.\$system.\$inputMessage}}</code>	Raw user chat input
User Name	<code>{{context.\$system.\$chatHistory}}</code>	Full conversation history for the current session (prior user and assistant turns)

Workflow Variables

Workflow variables are workflow-scoped runtime values that are created and managed within a workflow process.

Scope	Expression	Description
Workflow Name	<code>{{ \$context.\$workflow.\$name }}</code>	The ID of the workflow being run. The variable is always available at runtime, and remains stable across runs unless the workflow is renamed.
Trace ID	<code>{{ \$context.\$workflow.\$traceId }}</code>	Unique identifier for a specific workflow run instance that's generated per run. This variable is critical for debugging and support.
Conversation ID	<code>{{ \$context.\$workflow.\$conversationId }}</code>	Identifier for the conversation or session associated with the workflow run.

User Variables

User variables are predefined runtime values that describe the user currently interacting with the application or chat. The user's details are available using the `{{ $context.$user. }}` syntax. For example, user name can be retrieved using the expression `{{ $context.$user.$name }}`.

Scope	Expression	Description
User Name	<code>{{ \$context.\$user.\$name }}</code>	ID of the authenticated user who initiated the workflow execution

Trigger Variables

When a workflow agent team is called by an email trigger or REST trigger, trigger variables are available for use.

- REST Trigger

Scope	Expression	Description
Triggers > REST > Input > field	<code>{{ \$context.\$triggers.REST.\$input.<field> }}</code>	Specific request field

- Email Trigger

Scope	Expression	Description
Triggers > Input > subject	<code>{{ \$context.\$triggers.EMAIL.\$input.subject }}</code>	Subject line of the incoming email that triggered the workflow
Triggers > Input > fromAddress	<code>{{ \$context.\$triggers.EMAIL.\$input.fromAddress }}</code>	Email address of the sender
Triggers > Input > content	<code>{{ \$context.\$triggers.EMAIL.\$input.content }}</code>	Body of the email
Triggers > Input > headers	<code>{{ \$context.\$triggers.EMAIL.\$input.headers }}</code>	Full set of email headers

Scope	Expression	Description
Triggers > Input > attachments	<code>{{context.\$triggers.EMAIL.\$input.attachments}}</code>	Attachments included in the email

User-defined Variables

In addition to the system provided variables, you can also add variables that can be shared across nodes in an agent team of type Workflow. You can define these variables when creating the workflow agent team, and they're available throughout your agent team's flow. You can use the defined user variables in user questions or conversations.

To add user defined variables, do these steps:

1. When creating or editing a workflow agent team, select the Variables tab.
2. Select **Add** and enter the details:
 - o Name: Enter the name for the variable
 - o Type: Choose the type, for example, string.
 - o Scope: Choose the scope as **User Question** or **Conversation**.
3. Apply the changes and save.

You can now reference the variable using `{{context.$variables.<variableName>}}`.

To assign a value for the variable you created, do these steps:

1. Edit the workflow agent team in which you want to add the user-defined variable.
2. Add a node of type **Logic > Set Variables** and enter the details to create the variable node.
 - o From the Variables section, select **Add**.
 - o From the **Name** field, choose the user defined variable you had created.
 - o Enter the value for the variable. You can enter a static value, or a dynamic value that's derived from the runtime context.
3. Save your changes.

Scope	Expression	Description
Variables > <name>	<code>{{context.\$variables.<name>}}</code>	Workflow-level variable

Node Variables

• Node Inputs

Node inputs are variables used to pass input to the node in the workflow agent team.

Scope	Expression	Description
Entire input	<code>{{context.\$nodes.<nodeCode>.\$input}}</code>	Returns the complete input object passed to the specified node.

Scope	Expression	Description
Specific input field	<code>{{context.\$nodes.<nodeCode>.\$input.<field>}}</code>	Returns a single field from a node's input object.

• Node Outputs

Node outputs are variables that contain the output of the node in the workflow agent team.

Scope	Expression	Description
Entire output	<code>{{context.\$nodes.<nodeCode>.\$output}}</code>	Returns the complete output object produced by the specified node (schema-based if defined).
Specific output field	<code>{{context.\$nodes.<nodeCode>.\$output.<field>}}</code>	Returns a single field from a node's output object.

- - Use entire input or output objects when flexibility is needed or when passing data to an LLM for reasoning.
- Use specific fields when doing these actions:
 - Writing conditions
 - Assigning workflow variables
 - Populating email subjects or concise prompts
 - Referencing a field that doesn't exist at runtime resolves to empty or null

• Node Error Details

Node error details are variables that are available when a node in the workflow agent team fails and the Error Handler is called. They're scoped to the node that encountered the error.

Scope	Expression	Description
Error information	<code>{{context.\$nodes.<nodeCode>.\$error.\$info}}</code>	Detailed error information returned by the failed node
Error code	<code>{{context.\$nodes.<nodeCode>.\$error.\$code}}</code>	Machine-readable error code associated with the node failure

- **Node Execution Status**

Node execution status variable contains the status of the node in the workflow agent team.

Scope	Expression	Description
Node status	<code>{{context.\$nodes.<nodeCode>.\$status}}</code>	Processing status of the node

Variables Specific to Supervisor Agent Teams

Here are the types of variables available in expressions that are specific to agent teams of type Supervisor:

- System variables
- Input variables
- Self or agent scoped input variables

System Variables

These variables are runtime, read-only variables.

- `{{context.$system.$currentDate}}`: The current date at runtime.
- `{{context.$system.$currentDateTime}}`: The current date and time at runtime.
- `{{context.$system.$inputMessage}}`: The user's latest message that triggered the agent run.
- `{{context.$system.$chatHistory}}`: The conversation history available at runtime.
- `{{context.$system.$availableAgents}}`: The list of agents available for the supervisor to route work to.

Input Variables

These variables are the inputs configured for the supervisor agent team. To pass values to the supervisor agent team as configured inputs, use the `{{context.$input.<name>}}` syntax.

To configure an input variable, do these steps:

1. When creating or editing a supervisor agent team, select the Input Variables tab.
2. Select **Add** and enter the details:
 - Name: Enter a unique name for the variable. The name is the reference key. For example, name is used in the `{{context.$input.<name>}}` format.
 - Choose the type, for example, string.
3. Enter the value for the variable. You can enter a static value, or select **Insert Expression**  to add a context expression.
4. Add a description and save your changes.

Self or Agent Scoped Input Variables

Self variables (`$agents.$self`) provide the current agent's own context, that's, its configured inputs and added topics. To reference these variables, use the appropriate format:

- For agent input : `{{context.$agents.$self.$input.<name>}}`
- For specific topic: `{{context.$agents.$self.$topics.<name>}}`

6 Edit and Test AI Agents

Edit an Agent Team

To edit an agent team, do these steps:

1. From the Agent Teams tab of AI Agent Studio, search for and select the agent team you want to edit.
2. Select **Edit**  to modify the artifacts such as tools, topics, and external tools in the agent team.

Note: If you see **View**  instead of **Edit** , it indicates that the agent team is a preconfigured one, which can't be edited or directly modified. For more information, see [Can I edit a preconfigured agent team?](#)

3. To edit the agent team details such as the large language model (LLM), chat experience, questions and agent team security, select **Agent Team Settings** . For descriptions of the fields on these tabs, see [Create Custom AI Agents of Type Supervisor](#) and [Create Custom AI Agents of Type Workflow](#).
4. If needed, use **Debug**  to test the agent team.
5. Publish your agent team.

Edit and Test AI Agents Iteratively in Playground

You can use **Edit in Playground** to edit and test your AI agents and large language model (LLM) nodes. This option enables iterative, real-time testing and refinement of agent instructions and parameters, without deploying changes to the production environment. It bridges the gap between high-level intent and step-by-step execution, so you can safely test, adjust, and fine-tune your agent's behavior before deploying it to production.

To edit and test AI agents:

1. Go to **AI Agent Studio**.
2. Choose the appropriate tab based on your project:
 - Agent Teams: For collaborative or multi-agent flows
 - Agents: For individual agent configurations
3. Search for the agent or agent team and then select **Edit**  to modify their settings.
4. From the editing area, select **Edit in Playground** to edit and test your AI agents.

5. Edit the agent details to adjust the prompt logic and model parameters. To edit and view real-time results in a dual-pane layout, you can select **View Results**  .

Prompts Tab

You can design templates and instructions that guide the agent's behavior.

Note: The system prompt and summarization prompts are specific to supervisor type of agents.

Field	Description
System Prompt	Edit the system prompt to define your agent's identity, job description, and rules.
Summarization Prompt	Select the summarization mode, and edit the summarization prompts to include only essential instructions relevant to your specific use case.

Note: You can add expressions to fields using **Insert Expression**  . For more information, see *Expressions in AI Agent Studio*.

LLM Tab

Choose the type of model and change the custom model properties if needed. You can either use the default model or choose a model.

Field	Description
Provider	Choose to use the default model or select a model. When using a custom model, specify the model properties.

Input Tab

Enter the specific test data, variables, or context required for an agent or LLM node to run. You can simulate real-world triggers and verify that your agent processes incoming information correctly before saving your changes.

Field	Description
User Input	Enter your queries or actionable commands that initiates the agent's logic.
Evaluation	Use evaluation sets to assess your agent's performance. An evaluation set contains one or more test questions, the expected agent responses, and the metrics to be measured. For more information, see <i>Evaluate Agents</i> .
Variables	Configure variables, making them accessible to all nodes within the workflow type of agent.

Field	Description
Additional Variables	Add additional variables to the prompt. For example, to add the current system date to your prompt, select the Current Date and Time option using Insert Expression  .

Output Tab

Configure the overall structure of the agent's output using JSON schema, to specify the exact output.

Field	Description
Specification Mode	Select this mode to directly modify the JSON schema for the output.
Simple Mode	Select this mode to define the output values and types. The corresponding JSON schema will be generated automatically and displayed in the specification mode for any further changes.

6. Select **Save and Test** to verify the changes you made and test the agent or the LLM node. While the test is running, you can monitor the results and performance in these sections:
 - Input: Review the messages you entered in the Input Message field on the Input tab. To view the JSON schema, select **JSON**. To view the final response, select **Human**.
 - Output: Select a session to open the detailed trace view. The trace shows a step-by-step timeline of the run, including which tools were called, how long each step took, and metrics for each step. Select the color-coded trace lines to view details, such as latency and input or output token usage. To view the final response, select **Human**.
7. Select **Run History**  to view, compare, and reuse previous test runs. You can view a detailed record of every run and track your agent's performance.
 - Select **Edit**  to add a comment in the change log.
 - Select **Apply**  to load a proven configuration into the panel and resume iterative testing from a high-quality baseline. This action only affects the testing environment and won't update your production agent until deployed.
 - Use **Expand**  to view change log details of the test.
 - You can compare any two runs by selecting them. Select **Compare** to view a pane displaying the details of each run and highlighting their differences.
 - Select **Save and Test** to save your changes to the agent, and run the agent with the test configuration.
8. Select **Done**.
9. After testing your changes, publish your agent team.

7 Monitor and Evaluate AI Agents

Monitor and Evaluate AI Agents

Monitor and gain insights into how your AI agents are performing, and also evaluate the agents for accuracy. You can also track the interactions with your agents, understand real-world usage patterns, identify common errors, and measure overall performance.

- **Monitoring:** Monitoring tracks performance and provides insight on how your agents behave in production. Monitor agents to ensure that your quality bars for response time and token counts are maintained over time. You can also see any errors logged here.
- **Evaluation:** Evaluate agents before you deploy them, to ensure that they're ready for production. Test your agents for response correctness, response time, and token usage to meet your quality standards. You can also check the quality of answers generated through the document tool to assess how effectively agents utilize the retrieved context from the retrieval-augmented generation (RAG) metrics. After making any changes to your agent, or after a model update, rerun evaluations to confirm that your agent continues to perform as expected. This proactive approach helps you maintain high-quality experiences for your users.



This table summarizes some key metrics, their descriptions, and their availability for monitoring or evaluation.

Metric	Description	Available to Evaluate	Available to Monitor
Error Rate	Percentage of user sessions that ended in an error.	Yes	Yes
Error Count	Total number of errors recorded.	Yes	Yes
Session Count	Total number of conversations initiated with agents.	Yes	Yes
P99 Latency	The maximum wait time in milliseconds for 99% of users, revealing any areas where you should review and optimize the prompts or structure of the agent.	Yes	Yes
P50 Latency	The maximum wait time in milliseconds for 50% of users, helping identify performance issues.	Yes	Yes

Metric	Description	Available to Evaluate	Available to Monitor
	You can view this metric in the details of the monitoring or evaluation results.		
Total Tokens	Cumulative number of tokens used by all agents.	Yes	Yes
Input Token Count	Total tokens sent to the LLM for requests. This includes system prompts, user messages, retrieved or context data, chat history, and tool or function definitions.	No	Yes
Output Token Count	Total tokens generated by the LLM for requests sent to it.	Yes	Yes
Median Correctness	The 50th percentile of correctness scores across evaluation runs. Each score (0–1) is computed by comparing the agent’s answer to the reference answer provided in the evaluation set.	Yes	No
Session Count	The number of unique conversational sessions between a user and an AI agent. One session can include multiple messages or evaluation runs.	Yes	Yes
Groundedness	Alignment of the generated answer with the retrieved source content, indicating how faithfully the response reflects the supporting information.	Yes	No
Answer Relevance	Degree to which the answer directly addresses the user’s question, measuring how fully and precisely the content covers the required subject matter.	Yes	No
Context Relevance	Quality and appropriateness of the retrieved information, assessing whether it meets the necessary standards to be considered reliable evidence.	Yes	No

Prerequisites

Aggregate the metrics that are displayed on the Monitoring and Evaluation tab in AI Agent Studio.

1. Go to **Navigator > Tools > Scheduled Processes**.
2. In Scheduled Processes, click **Schedule New Process**.
3. Leave the type as **Job**.
4. Search for and select **Aggregate AI Agent Usage and Metrics**.
5. Run the **Aggregate AI Agent Usage and Metrics** scheduled process.

You can schedule this process to run on a recurring basis, for example, once a day.

The process aggregates the metrics that are displayed in the Monitoring and Evaluation tab of AI Agent Studio.

Monitor Agents

1. Go to **Navigator > Tools > AI Agent Studio**.
2. From the Monitoring and Evaluation tab, go to the Monitoring subtab.
 - The Monitoring subtab displays aggregated metrics of all the agent runs over the selected time frame.
 - All supervisor agent runs are monitored, including agents in draft status.
3. Select the agent to view additional details.

Each row represents a single session and displays the number of turns (back-and-forth messages), the session's completion status (successful or error), and the number of tokens used.

4. Select any session in the list to open the detailed trace view. This view displays a step-by-step timeline of the entire conversation, showing exactly which tools were called, the duration of each step, and the metrics for each step.

Evaluate Agents

Use evaluation sets to assess your agents' performance. An evaluation set contains one or more test questions, the expected agent responses, and the metrics to be measured. Evaluation sets are specific to each agent, and an agent can have multiple evaluation sets. For more information, see [Best Practices for Creating Evaluation Sets](#).

1. Create an evaluation set for an agent.
 - a. Go to **AI Agent Studio**.
 - b. Select the Monitoring and Evaluation tab.

All evaluations run on the agents are displayed in the Evaluation subtab.

- c. To create an evaluation set, click **Manage Evaluations** and select .
- d. Enter a name, code, and description for the evaluation set, and select the agent team to be evaluated.
- e. Choose the run mode.

Sequential: Runs the questions in the exact order you define them. Use this if one question depends on the context of the previous one.

Random: Runs the questions in a random order.

- f. Select **Enable document tool evaluation** if you want to view RAG metrics in the evaluation report.
- g. From the Questions subtab, add common questions that users are likely to ask the agent and the answers you would like the agent to deliver. Ensure both the questions and answers are concise, user-friendly, and reflect best practices.

You can either add your questions and the expected answers individually, or upload a CSV file with the questions in the first column and the expected answers in the second column.

- h. From the **Metrics** subtab, edit each metric to set the pass and fail criteria. For example, to indicate that the test fails if the correctness score is less than 0.7, choose **<** as the threshold condition and enter 0.7 as the threshold value.
 - i. Select **Create** to save the evaluation set.
2. Run the evaluation set.
 - a. On the Manage Evaluations page, select the **Initiate Evaluation Run** action for your evaluation set.
 - b. Choose the version of the agent team to evaluate and run the evaluation.
 3. Analyze the results.
 - a. Click the evaluation set to view the Evaluation Runs page.
 - b. Select the evaluation run and select the **View Run Results** action.

Subtab	Information Displayed
Response performance	- Comparison of the expected response versus the actual response from the agent for each question, along with the metrics for each question in the evaluation. - Trace provides information about the detailed timeline for each question in the evaluation.
Correctness	Detailed breakdown of the correctness score. The LLM provides an initial score and feedback, and you can add your own feedback for record keeping in the Correctness Score by Human column.

Compare Evaluation Runs

You can see a side-by-side comparison of two different runs of the same evaluation, and easily spot regressions or improvements in latency, correctness, and token usage. Doing so, you can understand how an agent's performance changes over time, especially after you've made modifications.

1. From the Evaluation subtab, select the evaluation.
2. Select any two runs and click **Compare**.
 - o The Summary subtab displays a high-level overview of the performance differences between the runs.
 - o The Details subtab provides a granular, question-by-question breakdown of the runs. For each question in the evaluation set, you can directly compare the actual response from Run 1 against the actual

response from Run 2. You can also compare the specific latency, tokens used, and trace links for each question, making it easy to pinpoint exactly where and why performance or accuracy has changed.

View the Measured Value of Agents

Use the **Value** dashboard to understand the business impact of your AI agent teams. This visual dashboard provides a consolidated view of your AI fleet, performance by Oracle Fusion Cloud Applications products, and metrics for individual agent teams. These metrics help you estimate and visualize the tangible ROI of your AI initiatives. You can track deployment, usage, and estimated time and cost savings for your published agent teams in one view.

Configure Performance Metrics for an Agent Team

To configure performance metrics for a specific agent team:

1. Go to **AI Agent Studio**.
2. From the Agent Teams tab, search for the agent team you want to edit, and select **Edit** .
3. Select **Agent Team Settings**  to edit the agent team details.
4. On the Details tab, scroll to the **Capture estimated Cost and Time savings** section.
5. Configure the estimated values for ROI tracking metrics by setting **Time saved each agent run** and **Cost saved each agent run**. If you don't enter any values, the default values apply.
6. Select **Update** to save your changes.
7. Publish your agent team.

You can now view the measured value of the agent team using the Value subtab of the Monitoring and Evaluation tab.

Monitor AI Agent ROI and Key Efficiency Indicators

1. Go to **AI Agent Studio**.
2. From the Monitoring and Evaluation tab, select the Value subtab. You'll see a dashboard that displays real-time ROI per agent and team, allowing you to filter by Family or Product to monitor key metrics such as **Total Time Savings**, **Total Cost Savings**, **Total Usage**, and **Total Agents**.
3. Select any item on the dashboard to see the total time saved (hours) and total cost saved (USD) across agent teams.
4. You can also configure the estimated ROI metric values for an agent team. Select the **Agent Teams** link, then search for the agent team and select **Edit**  to update the settings. Select **Apply**, and then save your changes.

Best Practices for Creating Evaluation Sets

Creating a good evaluation set helps in improving the efficiency of your evaluation. Here are some best practices for creating evaluation data sets and testing protocols for agent teams.

The foundation of a successful evaluation is a high-quality data set. This data set consists of paired questions or inputs, and expected responses. The expected response must be factually correct and grounded in the source context.

Example Data Set

Question	Expected Response
Is aromatherapy covered?	No, aromatherapy isn't covered. According to the provided context, aromatherapy is listed under "Alternative Treatments" which aren't covered by United Healthcare Medical Plans.
Do you pay for thermometers?	Based on the provided context, thermometers aren't covered. The document "MEDICAL SUPPLIES AND APPLIANCES" lists thermometers as excluded supplies.
Is laser surgery for eyes covered?	Based on the provided context, laser surgery for eyes isn't covered. The "VISION" section states that surgery to correct nearsightedness, including laser surgery, is listed under plan exclusions.

General Evaluation Guidelines

These principles apply to all agent teams, to ensure comprehensive testing.

- **Safety and grounding:** Validate that the agent adheres to safety policies. It must avoid hallucinations, that is, if no relevant information is found, it should state that, rather than guessing.
- **Ambiguity and multipart queries:** Evaluate the agent's ability to handle vague queries, implicit reasoning, or single turns containing multiple distinct questions.
- **Negative testing:** Introduce out-of-scope or unanswerable questions, for example, topics not covered by the available tools, to ensure the agent responds with an appropriate "I don't know" or redirect.
- **Multiturn dialogue:** If the agent is conversational, design tests that require maintaining context and history across several turns, for example, answering a follow-up question based on the previous answer.
- **Latency and performance:** Measure response time and efficiency, particularly in scenarios involving calls to multiple tools or complex reasoning.

Evaluation Guidelines for Supervisor Agent Teams

Agent teams of type Supervisor are evaluated based on the tools they use.

Tool	Guidelines
Document Tool (RAG)	Questions designed for Retrieval-Augmented Generation (RAG) must test the agent's ability to handle complexity, not just simple keyword retrieval. • Long Range Context: Test if the agent can resolve dependencies scattered across distant sections or several pages of a document. • Distributed Context: Ensure the agent can aggregate information from multiple noncontiguous parts of the document to answer comprehensively. • Concealed Context: Test the ability to find and extract specific, obscure details deep within the text. • Reasoning: Check if the agent can apply reasoning or logic to the retrieved information to provide a correct answer. • Table-Sourced: Test the ability to interpret and pull accurate data from tables within the document.
Business Object	• Function Coverage: Ensure evaluations test all business functions available in the business objects. • Parameter Variation: Test the same business function from different angles and with different parameters. For example, if a BO creates an object, test it with various input types.

Tool	Guidelines
REST	- Endpoint Coverage: Ensure evaluations test all functions available in the REST tool. - Scenario Variation: Test the same REST tool for different scenarios and parameters, for example, handling different payload structures or update types.
Deep Link	Validation: Confirm that deep links are generated correctly and for the appropriate scenarios.
Model Context Protocol (MCP)	- Tool selection: Validate that the correct MCP tool is called. - Function accuracy: Validate that the correct functions are called within the MCP tool.

Evaluation Guidelines for Workflow Agent Teams

Evaluations for Workflow agents must test the overall logic flow and the robustness of individual nodes.

Workflow Structure and Logic

- Path coverage: If the workflow has multiple paths or branches, ensure the evaluation tests all paths.
- Scenario depth: Test multiple distinct scenarios for the same path to ensure consistency.
- Unsupported scenarios: Include tests that identify scenarios the workflow doesn't support, to verify graceful failure or error handling.

Workflow Nodes

A workflow includes multiple node types, and the evaluation must cover the key scenarios applicable to each type.

Node	Scenario
LLM	Test the LLM prompt to identify if it can handle all types of questions and formatting instructions.
Code	- Robustness: Include questions that test the stability of the code. - Edge Cases: Test for scenarios that cover different edge cases to ensure the code doesn't break the flow.
- Business Object - REST	Apply the same guidelines as specified for agent teams of type Supervisor. Test functions from different angles and with varied parameters.
RAG Document Tool	Apply the same guidelines as specified for agent teams of type Supervisor.
Document Processor	Format Handling: Test the node for various attachment types (PDF, txt, and so on) to ensure consistent text extraction.
Vector DB Reader	Retrieval Accuracy: Test if the node retrieves the most semantically relevant chunks based on the input query.

For the other supported nodes, make sure evaluation tests cover other nodes including Tools, Vector DB Reader, and Vector DB Writer.

8 Migrate RAG Agents to AI Agent Studio

Migrate Document Tools of RAG Agents

If you've created any RAG agents in Oracle Fusion Cloud Applications previously, we recommend that you replace your existing agent with an agent you create in AI Agent Studio.

You can migrate the Document tool you created for your existing agent to AI Agent Studio.

1. Go to the **Configure RAG Agents** work area.

Tip: You can use global search to get to the work area.

2. From the Tools tab, click  to migrate your tool.
3. Enter a unique tool name and code. That will help you easily find your migrated tool in AI Agent Studio.

After migrating your tool, create a new agent in AI Agent Studio using an appropriate template, add your migrated tool, and publish your agent. After testing and verifying your new agent, you can delete the original agent in the **Configure RAG Agents** work area.

9 Promote Published Agents

Migrate AI Agents from One Instance to Another

You can migrate an individual agent team from one environment to another using the import and export options available in AI Agent Studio. For example, you can export from the test or staging environment and import into the production environment.

After importing the agent team, you must publish it in the new environment to make it available for use. Also, if any of your agents use the following tools, make sure you complete the corresponding post-import tasks for each tool:

- Document tool: Upload and republish the document to ensure its content is correctly processed in the new environment.
- Email tool: Reconfigure the email tool to make sure necessary alerts are generated properly in the new environment.
- External REST tool: Reconfigure authentication settings for external REST API tools.

To migrate an agent team:

1. From the source environment, go to **AI Agent Studio**.
2. From the Agent Teams tab, locate the agent team you want to export and click the  icon.
3. Enter the export details and click **Export** to download the JSON file to your local system.
4. From the target environment, go to **AI Agent Studio** and select the Agent Teams tab.
5. Click **Import** and upload the previously downloaded JSON file.
6. Use the **Draft** button to edit the imported agent team.
7. Publish the agent team.

10 Create Channels in AI Agent Studio

Create Microsoft Teams Channel for AI Agent Studio

You can integrate Microsoft Teams with AI Agent Studio, and enable users to access Oracle AI agents from Microsoft Teams. The integration preserves user identity and data security across Oracle Fusion Cloud Applications and Microsoft accounts. After integration, users can select **Ask Oracle** app in Microsoft Teams and interact with the AI agents available to them.

Integration with Microsoft Teams involves configuring several Microsoft 365 services, including Microsoft Entra (Azure Active Directory), Microsoft Bot Framework, and the Teams App Manifest. These components are closely interdependent and must be configured correctly across environments and tenants. Proper coordination and version management are essential for identity flows, manifest packaging, channel registrations, permission scoping, and endpoint exposure.

Here are the steps to set up Ask Oracle in Microsoft Teams:

1. Register Azure bot in Microsoft Entra (Azure Active Directory). Ensure to save the bot ID, password and tenant ID after registration.
2. Create a manifest file for Microsoft Teams in AI Agent Studio, and download it.
3. Upload the manifest file downloaded from AI Agent Studio in Microsoft Teams.

Register Azure Bot in Microsoft Entra

1. As an Azure/Exchange administrator, sign in to the *Microsoft Azure portal*.
2. Click **Create a resource** and search for **Azure Bot**.
3. Create the bot by adding these details:
 - Enter a unique bot handle.
 - Choose your active subscription.
 - Select an existing resource group or create one. For **Type of App**, select **Single Tenant**.
 - Select **Create new Microsoft App ID** as **Creation Type**.
4. Click next and create the bot.
5. After deployment is complete, click **Go to resource**.
6. Enter the URL of the bot service in **Messaging endpoint**. The URL will be of this pattern:

```
https://<pod name>/api/fusion-ai/orchestrator/collabTools/v1/teams-message
```

Replace <pod name> with the base URL of your Oracle application environment.

Note: Make a note of the App ID and Tenant ID because you'll need them when creating the manifest file in AI Agent Studio.

7. Generate a client secret:
 - a. Click **Manage Password** next to **Microsoft App ID** and create a new client secret.
 - b. Provide a description, for example, Oracle Agents Secret, and select a desired expiry date.

Note: Make a note of the client secret because you'll need it when creating the manifest file in AI Agent Studio.

8. Configure Application Permissions:
 - a. From the left panel, open **API Permissions**.
 - b. Go to **Add Permission > Microsoft Graph > Application Permissions**.
 - c. In the **User** folder, search for **User.ReadBasic.All** and add it.
 - d. Click **Grant admin consent** for the added permission.
9. Configure Delegate Permissions:
 - a. From the left panel, open **API Permissions**.
 - b. Go to **Add Permission > Microsoft Graph > Delegate Permissions**.
 - c. In the **User** folder, search for **User.ReadBasic.All** and add it.
 - d. Click **Grant admin consent** for the added permission.
10. Enable Microsoft Teams Channel:
 - a. Go back to your bot resource, and open **Channels** from the left panel.
 - b. Select the **Microsoft Teams** channel, agree to the terms and conditions, and click **Apply**.

Create Manifest File in AI Agent Studio

Prerequisite: Make sure that your role has the necessary permission groups. For more information, see Assign Permission Groups to Use Channels in Credentials in [Access Requirements for AI Agent Studio](#).

1. In **AI Agent Studio**, go to the Credentials tab, and then open the Channels subtab.
2. Configure **Microsoft Teams**.
3. Enter details for the channel manifest file.
 - o Short Manifest Name: Enter the short name for the app as displayed in Microsoft Teams. The default value is **Ask Oracle**.
 - o Full Manifest Name: Enter the full name for the app as displayed in Microsoft Teams. For example, Ask Oracle Fusion Agent.
 - o Manifest Code: Enter a unique ID to be used in AI Agent Studio (not part of the Microsoft Teams schema). For example, b3e4a289-7cbb-43d3-9a4c-d52bc874e830
 - o Version: Select the manifest schema version.
 - o Manifest Schema Version: Select the Microsoft schema version that's to be used for the manifest file.
 - o Oracle Instance URL: Enter the URL for the Oracle tenant for Fusion Applications.
Format is
`https://<fully qualified pod name>:<optional port>`
Create separate manifests for each tenant.
 - o Bot ID or App ID: Enter the App ID that was generated during the Azure bot setup.
 - o Tenant: Enter the Tenant ID that was generated during Azure bot setup.
 - o Secret: Enter the client secret that was generated during the Azure bot setup.
 - o Advanced Settings: List of valid domains or websites the bot can access or redirect to. Include all API or identity-related endpoints.

Note: You must add token.botframework.com with the fully qualified pod name.

4. Create the manifest file.

5. Select **Edit** and download the manifest file, to be used in Microsoft Teams.

Upload Bot Artifacts in Microsoft Teams Console

1. As an administrator, go to *Microsoft Teams Admin Center*.
2. Open **Manage Apps** and select **Upload new app** from the **Actions** menu.
3. Upload the manifest file downloaded from AI Agent Studio.
4. Publish the app. Your app now appears in the list of available apps.
5. In **Manage Apps**, go to **Setup Policies > Org Wide Policy** and add the app you published.

Your app is now available to all users included in Org Wide Policy.

Create Slack Channel for AI Agent Studio

You can integrate Slack with AI Agent Studio, and enable users to access the published Oracle AI agents from Slack. After integration, users in the workspace can select the **Ask Oracle** app in Slack and interact with the AI agents available to them.

Note: Make sure that the Slack user has the necessary access permissions to interact with AI agents. For more information, see *Fusion AI | How can I give users access to AI agents?*

The setup process involves these high-level steps:

1. Create the Slack channel manifest in AI Agent Studio.
2. Create the Ask Oracle app in *api.slack.com/apps* and add the manifest file.
3. Update the manifest file created in AI Agent Studio and add the credentials to the Bot configuration section.
4. Add the Ask Oracle app in Slack.

Create Manifest File in AI Agent Studio

Before you start, make sure your role has the necessary permission groups. For more information, see **Assign Permission Groups to Use Channels** in *Access Requirements for AI Agent Studio*.

1. In AI Agent Studio, go to the Credentials tab, and then open the Channels subtab.
2. Configure Slack.
3. Enter details for the channel manifest file.
 - Manifest Short Name: Enter the short name for the app. The default value is Ask Oracle.
 - Manifest Full Name: Enter the full name for the app. For example, Ask Oracle Fusion Agent.
 - Version: Enter the manifest file version.
 - Manifest Schema Version: Select the schema version that's to be used for the manifest file.
4. Create the manifest file.
5. Copy the manifest details from the Manifest.json section. You'll need this information when creating the app on *api.slack.com/apps*.

Create Ask Oracle App in Slack

1. Go to api.slack.com/apps and select **Create New App**.
2. Select **From a manifest**, choose the workspace, and click **Next**.
3. In the JSON tab, paste the manifest details copied from the manifest file created in AI Agent Studio.
4. Click **Next** and create the app.
5. Select the app you created and go to **Settings > Basic Information**.
6. Note down these credentials for later use:
 - App ID
 - Client ID
 - Client Secret
 - Signing Secret
7. Go to **Settings > Install App** and install the app.
8. Make a note of the **Bot User OAuth Token** value.
9. Go to **Features > Event Subscriptions** and check if the **Request URL** is successfully verified. If it isn't verified, select **Retry**.

Update Manifest File in AI Agent Studio

1. In AI Agent Studio, go to the Credentials tab, and then open the Channels subtab.
2. Edit the Slack manifest file.
3. In the Bot configuration section, enter the credentials you saved from the Slack application setup.
 - Bot ID or App ID
 - Client ID
 - Secret
 - Token
 - Signing Secret
4. Update the manifest file.

Add the Ask Oracle App in Slack

1. Open the Slack application and select **Add app**.
2. Add Ask Oracle.

Users in the workspace can now interact with the Oracle AI agents they've access to.

11 Allow External Access to Fusion Applications Agents

Enable Applications to Access Fusion Applications Agents

You can access the agents of Oracle Fusion Cloud Applications from other Oracle applications and from external applications, using the `/invokeAsync` API. The access is based on the role assigned to you, and whether that role can access agent teams. The roles that can access an agent team are specified in the Security tab when creating or editing the agent team in AI Agent Studio.

Access from Other Oracle Applications

To use `/invokeAsync` API, you must authenticate through Oracle Cloud Infrastructure Identity and Access Management (OCI IAM) using an OAuth 2.0 bearer token. For agent operations to function, integration must be bidirectional. Both systems must securely access each other's services.

As an administrator, you create the confidential application in Fusion Applications IAM, representing the other Oracle application service. Then create a matching confidential application in the other application service, representing the AI Agent Studio's service. These two applications form the trusted link for the two-way communication.

Set Up Two-Way Trust Between Applications

1. Create a confidential application in Fusion Applications IAM:

This application acts as a representation of the external Oracle application in Fusion Applications IAM. After creating the confidential application, enable access to the AI Agent Studio service so the application can call the `/invokeAsync` API exposed by this service.

- Identify the identity domain (authorization server) where you'll create your confidential application.
- Specify OAuth grant types. For example, a 2-legged OAuth.
- Define the access scope.
- Generate the access token and use it to make REST API calls.

For more information, see [Configure OAuth Using the Fusion Applications Identity Domain](#).

2. Create a confidential application in the external Oracle application's IAM:

Similar to the previous confidential application you created, create another confidential application that acts as a representation of the Fusion Application service in the external Oracle application's IAM. After creating the confidential application, configure it to allow AI Agent Studio to call the appropriate service endpoints.

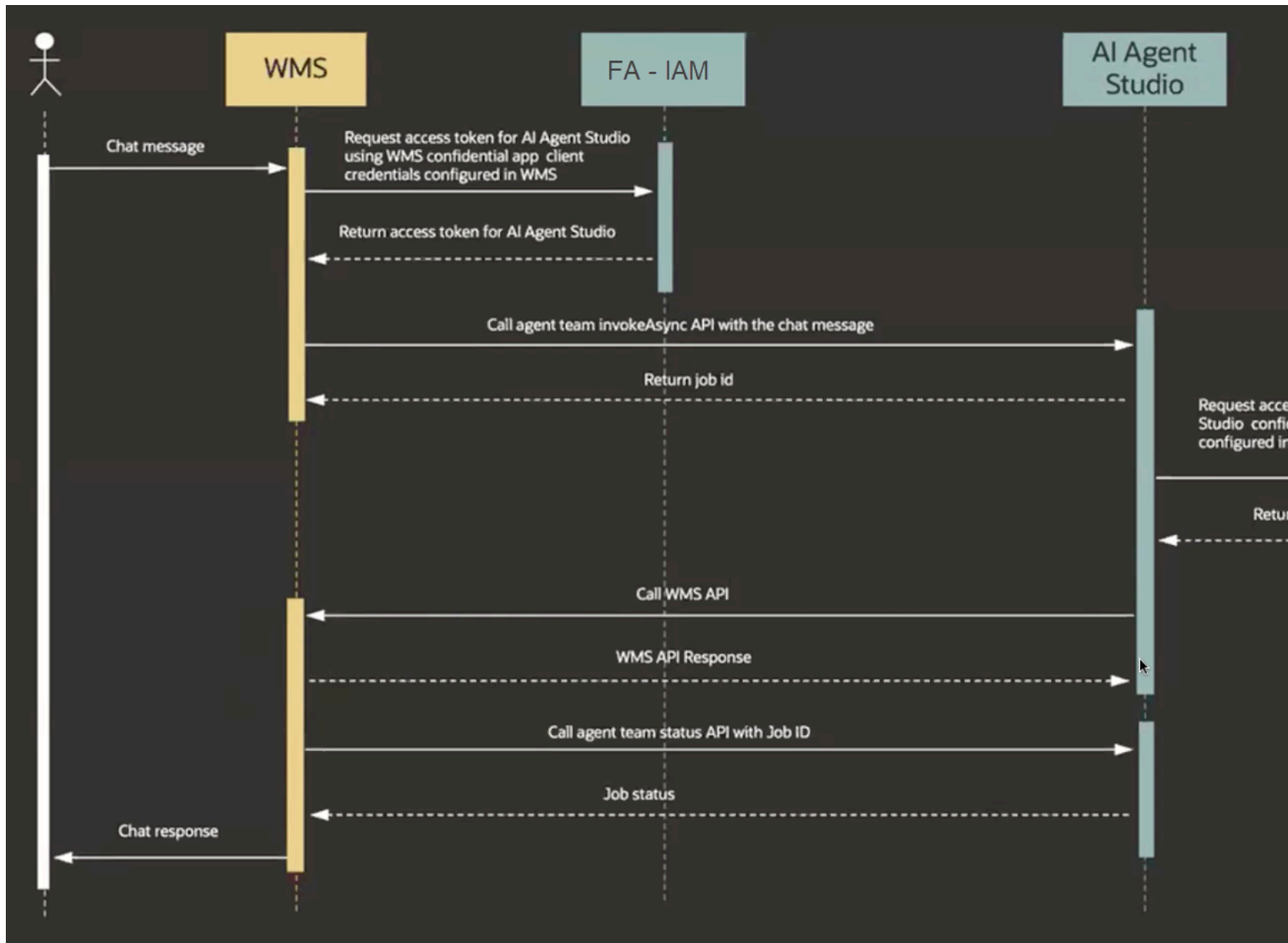
For more information, see [Configure OAuth Using the Fusion Applications Identity Domain](#).

3. Register the confidential application created in the external Oracle application's IAM (from step 2) in AI Agent Studio:
 - a. Go to **AI Agent Studio** and open the Credentials tab.
 - b. Go to the Data Source Applications subtab and add the details of the external confidential application.
 - Base URL: Enter the base URL of the external API
 - IDCS URL: Enter the URL of the Fusion Applications IAM, to enable AI Agent Studio to get the OAuth token
 - Scope: Specify what access needs to be given whenever that OAuth token is retrieved.
 - Public and Private keys: Enter the keys for signing authentication requests. Used by IDCS to verify that the request is from a trusted and authorized source.
4. Create a business object to access the external Oracle application:
 - a. In **AI Agent Studio**, open the Business Object tab.
 - b. Add a new business object with the **Resource Type** as **Other Data Source Application**.
 - c. From the **Oracle Data Source Application** list, select the data source application that you've registered in step 3.

Access Flow During Runtime

At runtime, the external Oracle application obtains an access token from Fusion Applications using the configured credentials. The application then uses this token to call the `/invokeAsync` API in AI Agent Studio, enabling access to the agent team.

This illustration shows the interactions between Warehouse Management System (WMS), which is an Oracle application outside Fusion Applications, Fusion Applications IAM, AI Agent Studio, and WMS IAM.



- The user interacts with the WMS application and sends a chat message.
- WMS requests an access token from Fusion Applications IAM using its configured credentials.
- WMS calls the `/invokeAsync` API in AI Agent Studio, passing the user's chat message and access token.
- AI Agent Studio returns a Job ID to WMS, which WMS uses to track progress and get the response.
- To call WMS API, AI Agent Studio requests an access token from WMS IAM.
- AI Agent Studio calls the WMS API using the access token.
- WMS returns the API response and requests for the job status.
- AI Agent Studio returns the job status.
- Final chat response is processed and delivered to the user.

Access from External Applications

Use `/invokeAsync` API to access the agents of Fusion Applications. For more information, see [Agent Team REST Endpoints](#).

12 Build Agentic Apps Using Agents from AI Agent Studio

Overview of Agentic Apps

Agentic apps help in combining your agents to form active, intelligent apps, enabling their use as your decision-making partners. You can consider the agents as workers and the agentic app is the productized solution that bundles those workers with the integrations, knowledge, and controls that are needed. The apps can include multiple agents, each of which is purpose-built for a specific category or domain.

Instead of asking one agent about everything from sales performance to inventory management, you can build specialized agents, for example, a Sales agent, an Inventory agent, a Finance agent; each with deep expertise in their domain. These agents can be used to create your agentic app. Doing so provides better insights and actions because each agent can maintain focused context, domain-specific reasoning, and tailored prompting for its area of responsibility. You can also scale it up by adding additional agents as needed.

The best agentic apps don't simply observe and report; they analyze, prioritize, and recommend actions that drive business forward.

- **Proactive alerting over passive monitoring:** Agents identify anomalies, opportunities, and risks in real-time, surfacing them with context and urgency. Instead of making users hunt for problems, the agents bring critical issues directly to their attention when they matter.
- **Decision support, not data dumps:** Apps filter signals from noise by analyzing situations and presenting only actionable decisions. Instead of providing users with every data point, provides synthesized insights with clear recommendations and trade offs.
- **Action-oriented intelligence:** Every agent interaction leads toward concrete actions, such as, approve a proposal, adjust a strategy, resolve a blocker. If it's purely informational, we can use an API or dashboard instead of agentic apps.
- **Context-aware prioritization:** Apps understand business goals and user workflows to determine what's urgent versus what can wait. Not all alerts are equal; agents should respect user's time and attention.

Here are the main pillars of agentic apps:

- **Information Displays**

Shows you data. These are agent-generated visualizations, such as lists, charts, tables, that show relevant data in real-time using predefined patterns. Agents built for agentic apps answer questions by creating information displays, except when intent clarification, confirmation, or a brief response is more appropriate. Each information display is generated with a topic key that captures the context of the user request. As conversation topics shift, the page template may hide or remove displays that are no longer relevant.

Agents must be configured to support specific information display types, granting them the capability to generate the displays. The implementation details are provided using prompting. At application startup, each agent may generate an initial information display that's shown to the user immediately. The agent determines whether to show an initial display and which display to show, and the host application can override either decision through prompting.

- **Advisors**

Answers your questions. Advisors are agents that answer user questions either independently or in coordination with other agents.

- Single agent mode: Users focus on a specific agent as their advisor
- Multi agent mode: Questions are routed to two or more capable agents, and outputs are consolidated into a unified response

Agents typically respond with both answers and supporting information displays. By default, information displays prioritize showing detailed data within the visualization itself, with accompanying text summarizing key insights. When a single agent acts as advisor, only that single agent produces an information display. In multi-agent scenarios, only selected agents generate displays, with the final answer summarizing their collective output.

- **Actions**

Helps you make decisions. Actions are agent-generated decision points that require human approval. Here are the artifacts in actions:

- Title describing the topic
- Description of what the action performs
- Instructions for the agent to execute if the action is called (including commands and parameters)

This configuration is agent-generated and not modifiable by users. Actions are generally associated with a single agent, and when the action is called, that same agent processes the action. The agent either performs the action immediately or engages the user if further clarification is needed.

Agents must be explicitly configured to generate actions. The generation logic and invocation handling are defined in the agent's prompting. At application startup, each agent can produce an initial list of priority actions that represent immediate decision points an app user may make. When answering questions, agents can include one or more actions in their response, which are then presented to the user.

Information displays may also expose actions directly within their pattern. They work identically and their presentation is defined by the associated pattern.

- **Communications**

Helps you reach out. Communications are agent-suggested outbound messages, for example, emails or Slack messages that follow consistent predefined templates. Here are the fields in Communications:

- Title and description explaining what the communication does and who it targets
- Template parameters used to compose the message

Communications always initiate outbound messages; they never intercept or receive incoming messages. The communication suggestions are routed to a specific agent for delivery. This agent doesn't need to be an advisory agent. It can be a generic, reusable agent shared across applications. The designated agent then processes the communication request, performs any additional work, and delivers the message using the appropriate tool, such as email, Slack, and Microsoft Teams.

All communications adhere to the templated patterns for consistency and uniformity. Messages aren't free-form; they conform to specific templates with defined parameter sets. Agents can produce communication suggestions when responding to inquiries and at app initialization. They can prefill template parameters based

on their knowledge and prompting. Communication suggestions can also be app-configured and associated with specific agents or all available agents.

When a communication item is selected, the associated agents attempt to populate any unfilled template parameters. The app user can then modify these values before the message is sent to the template engine for final processing.

Anatomy of an Agentic App

Here's an example of an Executive Advisor agentic app that consists of four agents.

Executive Advisor App

View all your information in one place!

Ask Oracle

Zenith Enterprises Faces Renewal Risk Amid Operational Challenges

Zenith Enterprises' 2025 renewal is at high risk due to a low health score and multiple Sev-2 service requests. The company is over budget on operating expenses, mainly from Sales & Marketing travel and contractor services, while the Supply Chain Digitization project is delayed by two weeks. Despite being under budget on headcount, high attrition in APAC Engineering remains a concern.

Workforce

Headcount Growth Under Budget but Increasing Over Time

Your headcount has been steadily increasing over the past year but remains under the budgeted levels, indicating controlled growth aligned with organizational plans.

950

900

850

800

750

Dec 2024

Mar 2025

Jun 2025

Sep 2025

Current Headcount

Budgeted Headcount

Key Insights

▪ Headcount has steadily increased but remains under budget

▪ Budgeted headcount has also grown, reflecting planned expansion

Urgent: Fill Director, Customer Success Role

Approve Pending Hire: Principal Engineer

Address High Attrition in APAC Engineering

Projects

Critical and High Priority Projects Facing Delays and Risks

The top three critical or high priority projects currently face

Deals

Top Deals: \$1.2M Delta Implementation Stalled, \$900K Polygon Delayed by Legal, \$750K Axios Fully Live

Below are your top 3 deals by total contract value (TCV), ordered highest to lowest.

Status: On Hold — Technical blocker (JIRA-22322)

Delta Innovations — \$1,200,000 TCV: Implementation On Hold

Deal signed; implementation stalled at 12% progress due to install script failures on customer's Kubernetes (JIRA-22322). High timeline risk; VP Customer Success notified.

Progress 12% — Blocker opened 2025-09-02

HIGH RISK

Status: Active — Awaiting customer legal review

Polygon Corp — \$900,000 TCV: Active but Slowing

POC exceeded success criteria (110%). Main delay is customer-side legal MSA review; CIO is champion but legal is backlogged, risking deal velocity.

Progress 45% — Projected close 2025-10-30

AT RISK

Status: Implementation Complete — Healthy account

Axios Solutions — \$750,000 TCV: Closed-Won and Live

Go-live on 2025-06-15. Customer fully ramped with strong health (95/100); upsell opportunity and roadmap session scheduled for 2025-10-15.

100% progress — Live since 2025-06-15

HEALTHY

Financials

Q3 2025 OpEx Over Budget Driven by Sales & Marketing and Operations

The company's operating expenses for Q3 2025 are \$450K over budget, primarily due to higher spending in Sales & Marketing and Operations & G&A. Weekly trends show significant increases in travel and contractor services, driven by end-of-quarter sales activities and extended contractor support, while engineering's cloud infrastructure costs are decreasing due to optimization efforts.

	PREVIOUS	CURRENT	CHANGE	% CHANGE
Sales & Marketing Over Budget by \$300K	\$5.800M	\$6.100M	+\$300,000.00	(+5.2%)
Operations & G&A Over Budget by \$150K	\$2.000M	\$2.150M	+\$150,000.00	(+7.5%)
Engineering On Budget at \$5.5M	\$5.500M	\$5.500M	+\$0.00	(+0.0%)
Travel & Entertainment Weekly Spend Up 12% (...)	\$0.00	\$88,000.00	+\$88,000.00	(n/a)
Contractor & Professional Services Weekly Spe...	\$0.00	\$55,000.00	+\$55,000.00	(n/a)

Callout Number	Name	Description
1	Ask Oracle Advisor	Lets you ask natural language questions against the set of agents included in the app.
2	Summary Section	Displays the most critical information across all the domains in the app's agents. The summary

ORACLE

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Callout Number	Name	Description
		is typically the first content users read when opening the app.
3	Agents	The example agentic app includes four agents, each represented by a tile on the dashboard. An agent can provide its own headline to convey the most important information for the app user. Users can zoom in on any agent to keep focus on that agent. Any subsequent questions, actions, and communications are then scoped to the selected agent.
4	Widget	A widget generated by an agent from the widget library. Each widget has space for the agent to provide its own insights, and capabilities that the user can choose from. Widgets can also have agent-specific actions that the user can act upon.
5	Priority Actions	App-specific critical information that require the user's immediate attention and provide options for the user to take action.
6	Communications	Outbound message suggestions that may be predefined by the app or generated by the LLM. It can start a flow allowing the user to construct the content and send the communication.

Create and Configure Workflow Agent Teams for Use in Agentic Apps

To build agentic apps with agent teams, first configure Workflow type agent teams for optimal results. Only published workflows can be used in agentic apps.

Best Practices

- Use an agent team of type Workflow. Workflow agent teams enable structured inputs and predictable flow.
- Make sure that the description of your agent team clearly defines the agent team's capabilities and specialization. When multiple agent teams are engaged, this description is used to help identify which team is best suited to handle a given query.
- Design smaller, topic-focused agent teams. You can combine multiple agent teams as needed within the agentic app.

- Keep workflows lean to reduce latency. Limit the number of LLM and agent nodes where possible to reduce perceived delay when resources are requested from the agent team.
- Pass user input and history explicitly. In workflows, conversation history and user input aren't automatically passed to internal LLM, agent, or workflow nodes. Use a code node or variable assignment to pass what you need (for example, using `$context.$system.$inputMessage`), and then reference those variables in subsequent LLM or agent calls.
- When possible, end your workflow with an agent or LLM node to help prevent output streaming.

Create and Enable Workflows for Using in Apps

An agentic app is built using agent teams of type workflows. These agent teams provide the response when the agentic app requests for displays, actions, communications, summary text, subtitle text, or user-query answers. The recommended method is for the agent team to receive the request from the agentic app using `OraMessageHint` webhook variable. Create your agent team structure to receive the webhook message, branch the workflow, and then provide the needed output to the agentic app.

To use workflows in agentic apps, you must first enable the workflow agent teams from AI Agent Studio.

1. Go to AI Agent Studio and open the Agent Teams tab.
2. Edit the workflow agent team you want to use in agentic apps.
3. From the Details tab, select **Expose to agentic apps**.
4. Save your changes.

`OraMessageHint` Variable

`OraMessageHint` is the primary app-aware workflow variable, and its value can be retrieved using the expression `$context.$app.$OraMessageHint`. It tells your workflow why the app is calling it. In prompt templates, use expressions such as `{{ $context.$app.$OraMessageHint }}`. In Code nodes, use the value as `$context.$app.$OraMessageHint`.

Here are some common values for `OraMessageHint`:

Value	Description
<code>Summary</code>	Indicates that the app is asking for summary text. Return short summary text for that agent's contribution.
<code>InitSubtitle</code>	Indicates that the app is asking for the dynamic subtitle. Return the subtitle text itself.
<code>InitDisplay</code>	Indicates that the app is asking for startup displays. Return one or more <code>oraInfoDisplay</code> blocks, and it may also include actions and communication suggestions when that improves the startup experience.
<code>InitActions</code>	Indicates that the app is asking for startup actions. Return zero or more action suggestions.
<code>InitCommunications</code>	Indicates that the app is asking for startup communication suggestions. Return zero or more <code>oraComms</code> blocks.
<code>Query</code>	Indicates that the user asked something through the app. Use the input message and answer the user. This response can also include displays, actions, and communications when appropriate.

Value	Description
InvokeAction	Indicates that a UI action sent work back to the workflow. Use this for follow-up handling after a widget command triggers an action path that sends a command to an agent.
AdditionalContent	Indicates that the app is loading an extra panel under a focused agent. Use OraPanelName to tell which panel is being requested.
FillParameters	Indicates that the app is asking one or more agents to fill communication parameters before generation or send.
SendCommunication	Indicates that the app is asking a target workflow to send or complete a communication flow.

Expressions Useful for Agentic Apps

Here are some expressions that you can use in your workflow agent teams:

Expression	Description
<code>{{ \$context.\$app.\$OraMessageHint }}</code>	App stage that's calling the workflow
<code>{{ \$context.\$system.\$inputMessage }}</code>	Current user message or app-supplied input for the current stage
<code>{{ \$context.\$app.\$OraAppContext }}</code>	Current app context value
<code>{{ \$context.\$app.\$OraUserContext.fullName }}</code>	Current user's complete name
<code>{{ \$context.\$app.\$OraUserContext.userId }}</code>	Current user's ID
<code>{{ \$context.\$app.\$OraAttachments }}</code>	Attachments available to the current app query flow
<code>{{ \$context.\$app.\$OraAction }}</code>	Action payload or command supplied to the workflow action path
<code>{{ \$context.\$app.\$OraCommParamsToFill }}</code>	Communication parameters that must be filled
<code>{{ \$context.\$app.\$OraCommNonTemplateParamList }}</code>	Non template communication parameters for sender-style flows
<code>{{ \$context.\$app.\$OraPanelName }}</code>	Additional panel that's requesting content

Here's an example user context object:

```
{
  "fullName": "John Doe",
  "userId": "jdoe1212",
  "userName": "John.Doe@example.com",
```

```
"guId": "sf sdf332323f dsf"
}
```

Variables and Their Values

There are multiple variables that can be used in workflows created for using in agentic apps.

Variable Name	How the Value is Populated
OraMessageHint	Set by the app runtime based on what the app is doing right now: startup display load, startup actions, communications, query, subtitle, and so on.
inputMessage	The current message for that request. For a user query, it's the user prompt. For some action and communication flows, it's the command or content being sent into the workflow.
OraAppContext	Passed from the current app context that may come from the app URL, from app-to-app navigation, or from an action that switches context.
OraUserContext	Passed from the signed-in user and is available to app-aware workflows.
OraAttachments	Populated for query flows when the conversation includes attachments.
OraPanelName	Populated only when the app is loading additional panel content for a focused agent.
OraCommParamsToFill	Populated when the app is asking agents to fill in communication parameters.
OraCommNonTemplateParamList	Available for communication sender flows that need non template parameter information.

Enable Agent Teams to Generate Widgets

In your workflow, the LLM and agent nodes can generate widgets. Initial graphics and insight displays are produced by agent output blocks that carry a `patternId` and a widget-specific `config`. In App Builder, the display prompt tells the runtime agent what widget output to generate.

Here are the types of widgets an LLM or agent node can generate:

Type of Widget	Details	Example
Card	• Use it for: a single alert, update, or status card • Required: <code>subject</code> • Common fields: <code>summary</code>, <code>timestamp</code>, <code>badgeText</code>, <code>variant</code>	<pre><oraInfoDisplay key="maintenance-card"> { "patternId": "cardWidget", "config": { "subject": "Maintenance Complete", "summary": "Scheduled database maintenance finished successfully.", "timestamp": "1 hour ago",</pre>

Type of Widget	Details	Example
	Actions: link and optional <code>additionalLink</code>, each with <code>text</code> and <code>action</code> Variant values: error, warning, info, or omitted for neutral Use commands such as <code>ora.Invoke(...)</code>.	<pre>"badgeText": "Info", "variant": "info", "link": { "text": "View Details", "action": "ora.Invoke(\"viewMaintenanceSummary \")" } } } </oraInfoDisplay></pre>
Messages List	Use it for: alerts, activity feeds, and small collections of highlight items Required: <code>items[]</code> with at least <code>title</code> per item Item fields: subtitle, summary, status, badgeText, priority, timestamp, image Interactivity: item action and item <code>additionalAction</code> with <code>text</code> + <code>command</code> Runtime behavior is considered. <code>additionalAction</code> is only shown when the widget is in a focused panel. Also, the runtime supports success as a priority in addition to alert, warning, and medium.	<pre><oraInfoDisplay key="system-alerts"> { "patternId": "messageListWidget", "config": { "subtitle": "Real-time metrics", "items": [{ "title": "Memory Usage Spike Detected", "subtitle": "82% confidence", "summary": "Memory consumption increased by 40% in the last 5 minutes", "status": "Investigating root cause", "badgeText": "Medium", "priority": "medium", "timestamp": "3 min ago", "action": "ora.Invoke(\"openMemoryIncident\", { \"metric\": \"memory\" })", "additionalAction": { "text": "View Metrics", "command": "ora.Invoke(\"viewMetrics\", { \"metric\": \"memory\" })" } }, { "title": "Network Latency Increase", "summary": "Average response time increased from 120ms to 180ms.", "status": "Under investigation", "badgeText": "Warning", "priority": "warning", "timestamp": "16 min ago" }] } } </oraInfoDisplay></pre>
Change List	Use it for: current-versus-previous metric comparisons Required: <code>displayType</code>, <code>items</code> displayType: percentage, raw, or currency	<pre><oraInfoDisplay key="campaign-change-list"> { "patternId": "changeListWidget", "config": { "subtitle": "Campaign performance comparison", "displayType": "percentage",</pre>

Type of Widget	Details	Example
	items[]: title, currentValue, previousValue Optional: subtitle, messages Values should stay numeric. Formatting comes from displayType, not from embedding percent signs or currency symbols inside the numbers.	<pre> "items": [{ "title": "Email Open Rate", "currentValue": 28.2, "previousValue": 24.5 }, { "title": "Click-through Rate", "currentValue": 4.6, "previousValue": 3.8 }, { "title": "Conversion Rate", "currentValue": 1.9, "previousValue": 2.1 }], "messages": ["Conversion rate drop may indicate landing page issues"] } </oraInfoDisplay> </pre>
Chart	Use it for: trends, comparisons, or proportions Required: type, data type: line, bar, or pie data.labels[]: x-axis labels or categories data.datasets[]: each dataset needs label and numeric data[] Optional: insights[]	<pre> <oraInfoDisplay key="sales-trend-chart"> { "patternId": "chartWidget", "config": { "type": "line", "data": { "labels": ["January", "February", "March", "April", "May"], "datasets": [{ "label": "Sales Revenue", "data": [10000, 25000, 15000, 40000, 30000] }] }, "insights": ["Revenue peaked in April", "Steady growth trend overall"] } } </oraInfoDisplay> </pre>
Record	Use it for: forms and read-only detail views Required: id, fields[] Optional: readOnly Field types: text, textarea, number, date, select, system Field fields: id, type, value, plus label and options when applicable In editable mode, the runtime automatically submits the form through an <code>ora.Agent(<oraFormSubmit ...>)</code> command. You don't configure a separate submit command in the widget itself.	<pre> <oraInfoDisplay key="employee-form"> { "patternId": "recordWidget", "config": { "id": "employee_form", "readOnly": false, "fields": [{ "id": "full_name", "type": "text", "label": "Full Name", "value": "John Doe" }, { "id": "employee_id", "type": "number", "label": "Employee ID", </pre>

Type of Widget	Details	Example
		<pre> "value": 12345 }, { "id": "department", "type": "select", "label": "Department", "value": "engineering", "options": [{ "value": "engineering", "label": "Engineering" }, { "value": "sales", "label": "Sales" }] }, { "id": "session_context", "type": "system", "value": { "conversation_id": "abc123" } }] } } </oraInfoDisplay> </pre>
Multi Record	• Use it for: tabular record lists • Required: <code>cols[]</code> and <code>rows[]</code> • Column forms: plain string, or object with <code>label</code> and optional <code>showOnExpand</code> • Row cells: plain string, or badge object with <code>type</code>, <code>text</code>, and <code>priority</code> • Optional row actions: <code>action</code> button and <code>drillDownAction</code> on the first column • Optional: <code>subtitle</code> for table context and accessibility <code>showOnExpand</code> columns only appear when the widget is focused. This is useful for secondary columns that should stay hidden in compact mode.	<pre> <oraInfoDisplay key="pending-approvals-table"> { "patternId": "multiRecordWidget", "config": { "subtitle": "Pending approvals", "cols": ["Request ID", "Type", "Status"], "rows": [{ "cells": ["REQ-101", "Budget", { "type": "badge", "text": "PENDING", "priority": "warning" }], "action": { "text": "Review", "command": "ora.Invoke(\"reviewRequest\", { \"id\": \"REQ-101\" })" }, "drillDownAction": "ora.Invoke(\"openRequest\", { \"id\": \"REQ-101\" })" }, { "cells": ["REQ-102", "Travel", { "type": "badge", "text": "PENDING", "priority": "warning" }], "action": { "text": "Review", </pre>

Type of Widget	Details	Example
		<pre>"command": "ora.Invoke(\"reviewRequest\", { \"id\": \"REQ-102\" })" }] } } </oraInfoDisplay></pre>
Sankey	• Use it for: flows between stages or states • Required: <code>nodes []</code>, <code>edges []</code> • nodes[]: <code>id</code>, <code>name</code> • edges[]: <code>source</code>, <code>target</code>, <code>value</code> Node IDs are numeric and edges point to those IDs. Keep values positive and keep the flow easy to follow.	<pre><oraInfoDisplay key="support-flow- sankey"> { "patternId": "sankeyWidget", "config": { "nodes": [{ "id": 0, "name": "Web Visitor" }, { "id": 1, "name": "Web Chat" }, { "id": 2, "name": "AI Chat Bot" }, { "id": 3, "name": "Resolved by Bot" }, { "id": 4, "name": "Transfer to Human" }, { "id": 5, "name": "Resolved by Human" }], "edges": [{ "source": 0, "target": 1, "value": 1000 }, { "source": 1, "target": 2, "value": 1000 }, { "source": 2, "target": 3, "value": 700 }, { "source": 2, "target": 4, "value": 300 }, { "source": 4, "target": 5, "value": 300 }] } } </oraInfoDisplay></pre>

Interactive widgets

For interactive widgets, use `ora.Invoke("actionCode", payload)`. This is the app-defined action path.

Enable Agent Teams

To enable agent teams to generate widgets, do these steps:

1. Go to AI Agent Studio and edit the workflow.
2. Go to the LLM or agent node that you want to generate widgets.
3. Edit the node and select the App Experience tab.
4. Select the options to enable widgets for actions and communications as needed.
5. Select the types of widget that are to be generated by the agent team.

Best Practices for Nodes in Workflows

Here are some best practices related to creating nodes for workflows, to be used in agentic apps:

- Start by deciding which app stages your workflow should handle.
- Branch early on `oraMessageHint` so each stage stays simple.
- For `InitDisplay`, return display output and optionally include actions or `oraComms` when that improves the startup experience. For `InitCommunications`, return `oraComms`. For `InitActions`, return actions.
- Use `oraAppContext` and `oraUserContext` whenever the app experience depends on who the user is or what object the app is currently showing.
- If a node needs the current question or a derived prompt, pass it explicitly instead of assuming a downstream node will reconstruct it for you.
- Test the exact stage you're writing before including that workflow into the app.

Build Agentic Apps Using App Builder

You can create agentic apps from the Apps tab in AI Agent Studio. To build agentic apps, do these steps:

1. Go to AI Agent Studio and open the Apps tab.
2.  Select **Add** and enter the name, code, and description for your agentic app.
3. Select **Update** to navigate to the App Builder.
4. Create an agentic app using one of these options:
 - a. Select one of the example agentic apps displayed and use its framework to edit further and build your agentic app.
 - b. Enter your app requirement in the **Ask Oracle** search bar. You can see the available apps and agent teams that can help you to create your agentic app.
 - c. Select **Start from scratch** to build your agentic app.
5. If you've chosen to use an example app or take help from Ask Oracle, you see the app's framework that you can further edit.
6. If you've chosen to build your app from scratch, you can start adding agent teams and other required artifacts to build your agentic app.
7. Enter the values in App Settings: **Name**, **Code**, and **Description**.

Add Agent Teams

Agent teams are one of the primary components of your agentic app.

1. Select **Add Section** and search for the agent team to include in your agentic app.
 **Note:** Only the published workflow agent teams that are enabled for using in apps are available for selection.
2. You can also assign custom agent teams for **Ask Oracle** and **Summary** sections.
If you're selecting a custom agent team for Summary, enter the **Summary Prompt**.

Add Communications

Communications enable your app to send emails and messages according to predefined templates.

Prerequisite

Define the templates for communication.

1. Select  and add a new template.
2. Select the format for the template.

You can add templates of type PowerPoint, PDF, email, or text.

- o PowerPoint: Add a PowerPoint template here. You can also define each slide separately. While creating the slide, specify the title for each slide, or select **Allow agent to choose my title**.
- o Email: Select **Allow agent to generate recipients**, **Allow agent to generate the subject** and add sections which can be filled by the agent when you select **Allow agent to fill in this section**. You can also include additional info in the email template.
- o PDF: Select **Allow agent to generate the title** and **Allow agent to fill in this section** to add the title and content for the PDF template.
- o Text: Select **Allow agent to fill in this section** to add content for the template of type Text.

Add the communication

1. Select **Add Communication** and add the title and description for the communication.
2. Select the agent the communication is applicable for.
 - a. Select the template that the agent must use when sending messages.
 - b. Enter the action to be performed in **Action Text**. For example, Send Email, Send Alert, Send to Agent.

Optionally, you can also select an agent to send the communication to in **Target Agent**.

Add Actions

Create actions that are to be processed in the course of app flow, including requiring human approval for any step. Actions and commands work in a simple sequence.

1. A widget or another UI element triggers a command
2. The command identifies which action to run
3. The action starts
4. The action runs its steps in order
5. Those steps perform the real behavior

Here, the command is the trigger, action is the specified workflow, and the steps are the ordered instructions within the workflow.

1. Select  and add a new action.
2. Enter the action code, display name, and description.

3. Select **Add Step** and choose the action.

Step Name	What it's for
Keep Action Available in UI	Keeps the original action available on UI even after it runs, instead of treating it as a one-time action.
Navigate to App	Opens another app. - Enter the code of the app to navigate to. - Select Pass Payload to Context to send the payload. - Enter the context to send the payload to.
Send Agent Command	Sends follow-up work to an agent. Use this when the action should ask an agent to do something next. - Select Pass Payload to Context to send the payload. - Enter the command.
Refresh Agents	Refreshes one or more agent displays. Use this when the user should immediately see updated content. Select the agent that's to be refreshed.
Switch App Context	Changes the current app context, optionally refreshing the app afterwards. - Select Use Payload as Context if needed. - Enter the context. - Select Reload App to refresh.

Publish the Agentic App

After adding the relevant artifacts and testing your app, publish your app. Users can view the published agents from the AI Agents page. To access this page, go to Me → Quick Actions → Show More → AI Agent Studio → AI Agents. If you're using **Home with Ask Oracle** as your home page layout, search for the **AI Agents** page.

Note: Make sure that your users have access to interact with the AI agents. For information, see [How can I give users access to AI agents?](#)

13 AI Agent Studio FAQs

When do I need a Custom AI Agent Subscription for AI Agent Studio?

AI Agent Studio is included with your Oracle Fusion SaaS Cloud subscription and comes with a variety of ready-to-use agent templates at no additional cost. You may use and make minor changes to these templates, such as uploading documents, adjusting display fields, or editing prompts, without requiring a Custom AI Agent subscription. However, a subscription is required if you want to create entirely new agents, significantly modify existing templates (for example, by adding new integrations, actions, or changing the agent's overall purpose), use third-party or marketplace agents, or select large language models (LLMs) that aren't provided by Oracle. Additional fees may also apply if your usage of premium LLMs exceeds your default token allocation. For more information, see [Where can I get information about premium subscription?](#) and [Fusion Custom AI Agents - Pricing FAQ \(KB864473\)](#).

Where can I get information about premium subscription?

In AI Agent Studio, locate the Recommendations panel, find **Unlock the full potential of AI Agent Studio**, and select **Learn More** to view premium subscription details.

The screenshot displays the AI Agent Studio web interface. At the top, there's a search bar labeled 'Search by agent template name ...' and two filter buttons: 'Family' and 'Product'. Below this is a yellow banner with the text 'Have questions about building better AI agents? Get expert guidance on designing, optimizing, and scaling high-performance AI agents.' The main content area is divided into two sections: 'AI Agents' and 'Marketplace'. The 'AI Agents' section shows a grid of agent templates, each with a 'New' badge, a title, a description, and a 'Copy Template' button. The templates include: 'Quote to Purchase Requisition Chat Assistant' (PRC - Self Service Procurement), 'Employment Lifecycle Policy Advisor' (HCM - Global Human Resources), 'Goods Delivery Advisor' (SCM - Inventory Management), 'Material Handling Advisor' (SCM - Inventory Management), 'Outbound Compliance Advisor' (SCM - Inventory Management), and 'Service Request Creation Assistant[Obsolete]' (CX - Service). The 'Marketplace' section shows a count of 23 items. On the right side, there's a 'Recommendations' section with a blue banner that says 'Unlock the full potential of AI Agents' and a 'Learn More' button. Below this is a 'Recently created' section showing agent teams like 'Deal Advisor Agent Team' and 'Incentive Administration Agent Team'. At the bottom, there's a navigation bar with icons and labels for 'AI Agent Studio', 'Apps', 'Agent Teams', 'Agents', 'Tools', 'Topics', 'Deep Link', 'Business Object', 'Credentials', and 'Monitoring and Evaluation'.

For more information, see [Fusion Custom AI Agents - Pricing FAQ \(KB864473\)](#).

Which large language models (LLMs) are currently used or supported for AI agents?

Oracle Fusion Cloud Applications support several LLMs from providers such as, Cohere, Meta, and OpenAI, for powering AI features including content authoring, summarization, and interactive Q&A. For the most current list of supported models, see [What Large Language Models \(LLM\) are in use or supported by Fusion SaaS Applications? \(KB796229\)](#).

Which scheduled processes do I need to run to make sure the features of AI Agent Studio work as expected?

Here are the scheduled processes required to ensure AI Agent Studio features work as expected.

Feature	Scheduled Process	Purpose	More Details
Access Requirements for AI Agent Studio	Import Resource Application Security Data	Imports resources and data from LDAP, and transfers the necessary information into the security tables of Fusion Applications.	Access Requirements for AI Agent Studio
	Import User and Role Application Security Data		
AI Agent Studio Assistant	Index AI Agent Studio Assistant Documents	Helps you get answers to questions about existing agents, search for agents, tools, and topics using natural language, and receive AI-powered suggestions for relevant resources to use in your agents.	Access Requirements for AI Agent Studio
	Index AI Agent Studio Assistant Objects and Attributes		
Monitor and Evaluate AI Agents	Aggregate AI Agent Usage and Metrics	Aggregates the metrics that are displayed in the Monitoring and Evaluation tab of AI Agent Studio.	Monitor and Evaluate AI Agents
Connectors	Content Intelligence SharePoint Connector	Connects your agents to accurate, context-aware, up-to-date source information. For example, using a SharePoint connector, you can import SharePoint files into articles as attachments, or a WebCrawl connector to contact selected sites to update the search index.	Schedule Content Intelligence Synchronization Process

Can I use any agent in my agent team or workflow?

No, you can't use just any agent in your agent team or workflow. You can use predefined agents and reuse agents created from the Agents tab in AI Agent Studio in other agent teams and workflows. However, if you create an agent directly within an agent team or workflow, that agent can't be reused and will only be available in the specific agent team or workflow where it was created.

When should I use a single-agent flow vs a multiagent process?

- Use a single-agent process for straightforward business requirements, such as retrieving answers from a single policy document.
- Opt for a multiagent process when dealing with more complex scenarios. For example, when multiple documents need to be parsed, or when the answers vary based on the user asking the question. In these cases, multiple agents are used, and the user's question is directed to the appropriate worker agent for processing.

What's the difference between the Copy Template and Use Template options in AI Agent Studio?

In AI Agent Studio, you can create agents from preconfigured templates using **Copy Template** or **Use Template**. The main difference between the two options is the order in which you rename the agent team, and update editable artifacts.

- **Copy Template:** Before creating the agent team, you can add a suffix to its name. The same suffix is also added to other editable artifacts in the agent team. After you add the suffix, the canvas opens with the agent team components, where you can modify editable artifacts. You can also edit the agent team details using **Agent Team Settings** ().
- **Use Template:** The Use Template option initially displays the general settings where you can rename the agent team and make other edits. After you create the agent team, the canvas opens with the agent team components, and you can modify the editable artifacts as needed.

For information about modifying artifacts in agents created from preconfigured templates, see [Can I edit a preconfigured agent team?](#)

Can I edit a preconfigured agent team?

You can't directly modify the components in a preconfigured agent team template. However, you can create copies of individual components (such as agents, tools, and topics), edit them, and then add the edited versions to your agent team after removing the original components. For example, to edit a topic that's part of a preconfigured agent team template you're using, do these steps.

1. Go to **AI Agent Studio**, and select the preconfigured agent team template you need, either using **Copy Template** or **Use Template**.
2. From the selected template, remove the topic you want to edit.
3. Go to the Topics tab, and search for the topic you removed.

Note: If you've removed an agent, go to the Agents tab. Similarly, for tools, use the Tools tab.

4. Select **Duplicate** () and create a copy of this topic.
5. Edit the copied topic as needed and add it back to your agent team.
6. Save and publish your agent team.

Can I migrate any agent team from one environment to another?

Yes. As an administrator, you can migrate (import) any individual agent team from one environment to another. However, if you've imported an agent team that belongs to a product family you don't have access to, you can't view the imported agent team in the new environment. For example, you're an Oracle Fusion Cloud Human Capital Management (HCM) administrator and you import an agent team that belongs to Oracle Fusion Cloud Supply Chain & Manufacturing (SCM). If your user doesn't have access to SCM product family in AI Agent Studio, you won't be able to view the imported agent team in the target environment.

For more information about migrating agent teams, see [Migrate AI Agents from One Instance to Another](#).

How do I display a welcome message in a chatbot agent?

You can configure your welcome message and any interactive starter questions for your chatbot agent in AI Agent Studio. These messages are displayed during the user's first interaction. You can also define follow-up questions, which are perfect for keeping the conversation going by suggesting logical next steps after an initial inquiry is resolved.

These settings can be configured while creating an agent, or editing it later.

- If you're creating a new agent, add the agent team from the **Agent Teams** tab, and go to the **Questions** subtab to configure the welcome message. For more information, see [Create Custom AI Agents of Type Supervisor](#) and [Create Custom AI Agents of Type Workflow](#).
- If you're editing an existing agent team, go to the **Agent Teams** tab and select **Edit**  to modify the agent team. Select **Agent Team Settings**  and then open the **Questions** subtab to configure the welcome message. For more information, see [Edit an Agent Team](#)

Note: If you see **View**  instead of **Edit**  , it indicates that the agent team is a preconfigured one, which can't be edited or directly modified. For more information, see [Can I edit a preconfigured agent team?](#)

How do I make agents respond faster?

AI agent response time is directly linked to the combined length of input and output text, measured in tokens for each transaction. For example, 100 tokens is roughly 75 words, though this ratio varies by model, writing style, and language. By being deliberate with your instructions, you ensure that your agents reason clearly and avoid unnecessary processing.

Here are some recommendations to help your agents respond faster.

- Edit summarization prompts to include only essential instructions relevant to your specific use case. The default prompts for supervisor and workflow agents are broadly written to cover many scenarios, so remove any general or redundant sections to streamline processing and improve response speed.
- Minimize the use of input and output tokens with concise prompts. Use retrieval-augmented generation (RAG) instead of overloading the context window, and set specific output length limits.
- Specify response length to avoid overly verbose answers.
- Use smaller, specialized agents that work in parallel and cache static instructions.
- Include only relevant information. Remove unnecessary details from context and documents to streamline processing.
- Consider a multi-agent approach, where specialized agents handle distinct tasks, such as research, coding, or Q&A, instead of loading all information into a single prompt.

By carefully selecting relevant context and breaking complex tasks into smaller, focused agents, you can improve both the speed and efficiency of your AI agents.

How do I track the usage of AI Agents created in AI Agent Studio?

You can use the Monitoring and Evaluation tab within the AI Agent Studio to track real-time interactions, analyze usage patterns, and identify errors to improve agent accuracy and performance. For more information, see [Monitor and Evaluate AI Agents](#).

How can I best reduce the risk of my agent showing hallucinations?

Large language models (LLMs) can unintentionally introduce extra details when they lack sufficient information.

To mitigate this issue, consider these strategies:

- Supply the necessary knowledge: Integrate relevant sources such as business objects, REST APIs, or RAG documents, ensuring the LLM has access to accurate and up-to-date information.
- Use explicit prompt instructions: Clearly instruct the LLM not to add or infer information beyond what's provided, reducing the risk of embellishment or fabrication.

How do I hide or restrict access to AI Agent Studio for specific users?

To restrict a user's access to AI Agent Studio, remove any related custom roles from the user's account.

1. Go to the Security Console, open the Users tab and search for the specific user.
2. From the search results, select the user, and edit the user account details.
3. In the Roles section of the Edit User Account page, select the custom roles associated with AI Agent Studio, and remove them using the  icon. These roles were added to the user account when access was provided. For a list of these roles, see [Provide Access to Configure AI Agents in all Products](#).
4. Save your changes.

Can I rename the AI chat label to something other than Ask Oracle, or change the design of the chat panel?

No, you can't change the background image or rename 'Ask Oracle' within the active chat window or AI Agent Studio preview.

What's the difference between editing a topic versus editing a system prompt?

When you update a topic, the changes are available across agents because you can reuse topics across agents. When you're updating a system prompt, the changes are available only for the agent being edited.

Tip: We recommend updating topics instead of prompts, so that you can reuse the topics.

What types of files can I upload as part of document tools in AI Agent Studio?

You can upload these types of files:

- PDF-TAGGED / SCANNED
- HTML
- JSON

- MARKDOWN
- XML
- TXT
- DOCX
- DOCX-MULTIMODAL
- XLSX
- CSV
- PPTX
- JPG
- PNG
- ZIP

For more information, see [Create Documents to Add to Document Tool](#).

Why does the external REST tool only support HTTPS transactions?

The tool is limited to only HTTPS to maintain data privacy in transit.

Why is the Provider list in the LLM subtab empty?

If the Provider list is empty when creating an agent team, it's likely due to missing permissions. To resolve this, make sure the required permission groups with the **AllRowsAllFields** security view are added to the custom role assigned to the user, and run the necessary scheduled processes to synchronize security data. For more information, see [LLM Provider LOV Empty when Creating a new Agent in Ai Agent Studio \(KB863374\)](#).

Why can't I see all agent teams in the Agent Teams tab of AI Agent Studio?

That's because only published agent teams are displayed by default. Use the **Draft** button to view agents that aren't published yet.

Why is my agent not fetching data from the document added to it?

Check if you've marked your document as published. If not, publish your document. To do so, from the Tools tab in AI Agent Studio, set the status of your document as **Ready to Publish**. Changing the status initiates the **Process Agent Documents** scheduled process. After the scheduled process completes, your document status is automatically changed to **Published**.

If your document status hasn't changed to **Published** even after five minutes, that means the scheduled process might have failed to trigger or complete. To resolve, do these steps:

1. Make sure your user has the **oraCommonFusionAISecurityAuthorization_create_Rag_OraResource** permission assigned.
2. From the Tools tab in AI Agent Studio, edit your document tool and change the status of your document from **Ready to Publish** to **Draft**.
3. Save the document tool.
4. Now, again change the status of your document back to **Ready to Publish**.
5. Save again to trigger the scheduled process.

For information about creating documents to be uploaded to the Document tool, see [Create Documents to Add to Document Tool](#).

Why can't I see the imported agent teams in the target environment?

You might not have access to the product family the imported agent team belongs to. For more information, see [Provide Access to Configure AI Agents in all Products](#).

How can I move the AI Agents link to the top of the Quick Actions list?

For information about reordering links in Quick Actions, see [Reorder Navigation Entries](#).

Why is the Roles list in the Security tab empty?

That's probably because permission groups aren't enabled for your duty role. Security administrators can enable permission groups for the duty role assigned to you, in Security Console. For details, see [Access Requirements for AI Agent Studio](#).